Qualifying Exam notes

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Abstract

We begin preparation for the qualifying exam.

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1 Algebraic geometry

The primary reference is Hartshorne's *Algebraic Geometry*, and the secondary reference is Vakil's *The Rising Sea*.

1.1 Sheaves

1.1.1 Presheaves

Definition 1.1. *A **presheaf** of abelian groups $\mathcal{F}$ on a space X is a contravariant functor from the poset category of open sets to the category of abelian groups.*

$$\mathcal{F} : \mathrm{Open}^{\mathrm{op}}(X) \to \mathrm{Ab}$$

Definition 1.2. *A presheaf $\mathcal{F}$ on a topological space X is a **sheaf** if it satisfies the following supplementary conditions:*

Separated *If U is an open set, if $\{V_i\}$ is an open covering of U, and if $s \in \mathcal{F}(U)$ is an element such that $s|_{V_i} = 0$ for all i, then $s = 0$;*

Gluing *If U is an open set, if $\{V_i\}$ is an open covering of U, and if we have elements $s_i \in \mathcal{F}(V_i)$ for each i, with the property that for each i, j, $s_i|_{V_i \cap V_j} = s_j|_{V_i \cap V_j}$, then there is an element $s \in \mathcal{F}(U)$ such that $s|_{V_i} = s_i$ for each i.*

Example 1.3 (Presheaf which is not a sheaf)**.** *Let X be a topological space and A an abelian group. Let $\mathcal{F}(U) = A$ for all open sets A. Then $\mathcal{F}$ is a presheaf with the obvious restriction maps, however $\mathcal{F}$ is not a sheaf because if U, V are disjoint open sets then $A = \mathcal{F}(U \cup V) = \mathcal{F}(U) \times \mathcal{F}(V) = A \times A$.*

Example 1.4 (Sheaf of functions)**.** *Let X be a topological space and for every open set $U \subset X$ let $C(U)$ denote the continuous functions $U \to \mathbb{R}$. Then C is a sheaf because every continuous function on U is completely determined by its restriction to any open cover U_i, and because continuous functions on U_i which agree on overlaps U_{ij} glue to continuous functions on U.*

The sheafs axioms are laid out so that sheaves behave like continuous functions on a topological space. However, there are many examples of sheaves which are not the sheaves of functions on a space.

Example 1.5 (Skyscraper sheaf)**.** *Let X be a topological space. For $x \in X$ and A an abelian group, the **skyscraper sheaf** $i_x(A)$ has:*

$$i_x(A)(U) = \begin{cases} A & \text{if } x \in U \\ 0 & \text{otherwise} \end{cases}$$

Skyscraper sheaves probe stalks, in the sense that they satisfy the following universal property: for any sheaf $\mathcal{F}$ on X,

$$\mathrm{Hom}(\mathcal{F}, i_x(A)) \cong \mathrm{Hom}(\mathcal{F}_x, A)$$

Definition 1.6. *Let $\mathcal{F}$ and $\mathcal{G}$ be sheaves on a topological space X. A* ***morphism of sheaves*** *$\varphi : \mathcal{F} \to \mathcal{G}$ is a collection of morphisms*

$$\varphi_U : \mathcal{F}(U) \to \mathcal{G}(U)$$

for each open set $U \subseteq X$, such that:

1. *The following diagram commutes for every pair of open sets $V \subseteq U$:*

$$\begin{tikzcd} \mathcal{F}(U) \arrow[r, "\varphi_U"] \arrow[d, "\rho_V^U"'] & \mathcal{G}(U) \arrow[d, "\rho_V^U"] \\ \mathcal{F}(V) \arrow[r, "\varphi_V"'] & \mathcal{G}(V) \end{tikzcd}$$

   where ρ_V^U denotes the restriction maps of both sheaves.

2. *Each φ_U is a morphism of abelian groups.*

Equivalently, a sheaf morphism is a natural transformation between the sheaves viewed as functors from the category of open sets of X to some algebraic category.

Definition 1.7. *The stalk of a sheaf $\mathcal{F}$ at a point $x \in X$ is the colimit*

$$\mathcal{F}_x = \varinjlim_{x \in U} \mathcal{F}(U)$$

where the limit is taken over all open sets U containing x.

A morphism of sheaves $\varphi : \mathcal{F} \to \mathcal{G}$ induces a morphism on stalks $\varphi_x : \mathcal{F}_x \to \mathcal{G}_x$ for each $x \in X$.

1.1.2 Leaf space, exactness, and constant sheaves

One way to think about the category of sheaves of abelian groups is that it should be very similar to the category of abelian groups. In fact, one takes theorems about abelian groups and tries to prove them for sheaves, or in general for any abelian category. The first question one can ask is about exact sequences.

The naive guess which turns out to be wrong is to declare that a sequence of sheaves of abelian groups

$$0 \to A \to B \to C \to 0$$

is exact if when evaluated for each open set U, the corresponding sequence of abelian groups is exact. This definition works for left exactness but turns out to give the wrong notion for surjectivity.

Example 1.8. *Consider $X = S^1$ and $X_1 = X$ and $X_2 =$ nontrivial double cover of X. Let F_1 and F_2 denote the corresponding sheaves of sections. Then the map $X_2 \to X_1$ is surjective, but the map $F_2(X) \to F_1(X)$ is not surjective. Indeed $F_2(X) = \emptyset$ and $F_1(X) = \{\mathrm{id}_X\}$.*

Thus we encounter two notions of surjectivity. If we think about all sheaves as the sheaves of sections of some space living over X, which we will introduce as the leaf space (or etale space), then morphisms of sheaves correspond fully faithfully to morphisms of leaf spaces. Then there is a local notion of surjectivity, which corresponds to the morphism of leaf spaces being surjective, and a global notion of surjective, corresponding to evaluation on all open sets.

Definition 1.9. *Let $\mathcal{F}$ be a sheaf on a topological space X. The* ***leaf space*** *$L(\mathcal{F})$ is the topological space defined by:*

$$L(\mathcal{F}) = \bigsqcup_{x \in X} \mathcal{F}_x$$

where $\mathcal{F}_x$ denotes the stalk of $\mathcal{F}$ at x, equipped with the topology generated by the basis:

$$\{\langle s, U\rangle \mid U \subseteq X \text{ open}, s \in \mathcal{F}(U)\}$$

where $\langle s, U\rangle = \{s_x \in \mathcal{F}_x \mid x \in U\}$ and s_x is the germ of s at x.

Remark 1.10. *The projection map $\pi : L(\mathcal{F}) \to X$ sending each germ s_x to x is a local homeomorphism. The original sheaf $\mathcal{F}$ can be recovered as the sheaf of continuous sections of π.*

The right definition of surjectivity to use is the local notion. By the definition of a leaf space, surjectivity of sheaves is equivalent to surjectivity on all stalks. Another clue that this is the right notion of surjectivity is that the local notion agrees with the notion of epimorphism in the category of sheaves of abelian groups.

Leaf spaces look very strange. For example, you might say that if X is a manifold and $\mathcal{F}$ is the sheaf of smooth functions on X, then the leaf space of $\mathcal{F}$ is locally homeomorphic to X and therefore is also a manifold. However the issue with this is that the leaf space is non-Hausdorff.

Definition 1.11. *Let $\mathcal{F}$ be a presheaf on a topological space X. The* ***sheafification*** *of $\mathcal{F}$ is the sheaf of continuous sections of the leaf space $L(\mathcal{F})$. It comes equipped with a morphism of presheaves $\mathcal{F} \to \mathcal{F}^+$ such that for any sheaf $\mathcal{G}$ and any morphism of presheaves $\mathcal{F} \to \mathcal{G}$, there is a unique morphism of sheaves $\mathcal{F}^+ \to \mathcal{G}$ making the following diagram commute:*

$$\begin{tikzcd} \mathcal{F} \arrow[r] \arrow[dr, "\exists!"'] & \mathcal{F}^+ \arrow[d] \\ & \mathcal{G} \end{tikzcd}$$

Definition 1.12. *A* ***constant sheaf*** *on a topological space X with value in an abelian group A is the sheafification of the constant presheaf defined by $\mathcal{F}(U) = A$ for all open sets $U \subseteq X$.*

Exercise 1.13. *On a connected locally connected space, $\underline{A}(U)$ is the set of locally constant functions $U \to A$, not literally just A for every open U.*

Solution 1.14. *Let $\mathcal{L}_A$ be the sheaf given by*

$$\mathcal{L}_A(U) = \{f : U \to A \mid f \text{ is locally constant}\}.$$

There is a natural morphism from the constant presheaf $P(U) = A$ to $\mathcal{L}_A$, sending $a \in A$ to the constant function with value a. By the universal property of sheafification, this induces a morphism

$$\Phi : \underline{A} = P^{\#} \to \mathcal{L}_A.$$

We claim that Φ is an isomorphism. It suffices to check stalks. For any $x \in X$,

$$(\underline{A})_x \cong P_x \cong A,$$

since sheafification does not change stalks and the stalk of the constant presheaf is the colimit of the constant system A. Also,

$$(\mathcal{L}_A)_x \cong A,$$

because the germ of a locally constant function is determined by its value at x. Under these identifications, Φ_x is the identity on A, hence an isomorphism. Therefore Φ is an isomorphism of sheaves.

Thus

$$\underline{A}(U) \cong \{f : U \to A \mid f \text{ is locally constant}\}.$$

If X is locally connected, then each connected component of U is open, and a function $U \to A$ is locally constant if and only if it is constant on each connected component. Therefore

$$\underline{A}(U) \cong \prod_{C \in \pi_0(U)} A.$$

1.1.3 Hom, kernel, image, cokernel, f_* and f^{-1}

The category of sheaves of abelian groups on a topological space X is an abelian category. In particular, given a morphism of sheaves

$$\varphi : \mathcal{F} \to \mathcal{G},$$

one can form its kernel, image, cokernel, and coimage, and these satisfy the usual formal properties familiar from modules and abelian groups. The point is that these constructions must be made sheaf-theoretically rather than merely sectionwise.

Definition 1.15. *Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves of abelian groups on X.*

1. *The **kernel sheaf** $\ker(\varphi)$ is the sheaf defined by*

$$(\ker \varphi)(U) := \ker\Big(\varphi_U : \mathcal{F}(U) \to \mathcal{G}(U)\Big).$$

2. *The **image presheaf** $\mathrm{im}^{\mathrm{pre}}(\varphi)$ is the presheaf defined by*

$$\mathrm{im}^{\mathrm{pre}}(\varphi)(U) := \mathrm{im}\Big(\varphi_U : \mathcal{F}(U) \to \mathcal{G}(U)\Big).$$

*Its associated sheaf is called the **image sheaf** and is denoted $\mathrm{im}(\varphi)$.*

3. *The* ***cokernel sheaf*** $\operatorname{coker}(\varphi)$ *is the sheaf associated to the presheaf*

$$U \longmapsto \mathcal{G}(U)/\varphi_U(\mathcal{F}(U)).$$

Remark 1.16. *The kernel is already a sheaf, because the sheaf axioms are checked inside $\mathcal{F}$. By contrast, the naive image and cokernel need not be sheaves, so one must sheafify.*

Proposition 1.17. *Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves of abelian groups.*

1. *For every $x \in X$,*
$$(\ker \varphi)_x \cong \ker(\varphi_x : \mathcal{F}_x \to \mathcal{G}_x).$$

2. *For every $x \in X$,*
$$(\operatorname{im} \varphi)_x \cong \operatorname{im}(\varphi_x : \mathcal{F}_x \to \mathcal{G}_x).$$

3. *For every $x \in X$,*
$$(\operatorname{coker} \varphi)_x \cong \operatorname{coker}(\varphi_x : \mathcal{F}_x \to \mathcal{G}_x).$$

Proof. Stalks are filtered colimits over neighborhoods of x, and filtered colimits of abelian groups are exact. For the kernel this gives

$$(\ker \varphi)_x = \varinjlim_{x \in U} \ker(\varphi_U) \cong \ker\Big(\varinjlim_{x \in U} \mathcal{F}(U) \to \varinjlim_{x \in U} \mathcal{G}(U)\Big) = \ker(\varphi_x).$$

The statements for image and cokernel follow similarly, using exactness of filtered colimits and the fact that sheafification does not change stalks. $\square$

The previous proposition explains why exactness for sheaves is tested on stalks.

Definition 1.18. *A sequence of sheaves of abelian groups*

$$\mathcal{F}' \xrightarrow{\alpha} \mathcal{F} \xrightarrow{\beta} \mathcal{F}''$$

is ***exact at*** $\mathcal{F}$ *if*

$$\operatorname{im}(\alpha) = \ker(\beta)$$

as subsheaves of $\mathcal{F}$. Equivalently, the sequence is exact at $\mathcal{F}$ if and only if for every point $x \in X$, the induced sequence on stalks

$$\mathcal{F}'_x \xrightarrow{\alpha_x} \mathcal{F}_x \xrightarrow{\beta_x} \mathcal{F}''_x$$

is exact.

Example 1.19 (Image presheaf need not be a sheaf)**.** *Let $\mathcal{O}$ denote the sheaf of holomorphic functions on $\mathbb{C}$, and consider the morphism*

$$d : \mathcal{O} \to \mathcal{O}, \qquad f \mapsto f'.$$

On a simply connected open set U, every holomorphic function has a primitive, so d_U is surjective. But on $U = \mathbb{C}^\times$, the function $1/z$ has no global primitive, so $1/z \notin \mathrm{im}(d_U)$. Thus the presheaf image is not determined by local data and is not a sheaf.

Its sheafification is the whole sheaf $\mathcal{O}$, because every holomorphic function is locally a derivative.

This example shows that a morphism of sheaves may be surjective on stalks without being surjective on sections over a given open set.

Definition 1.20. *A morphism of sheaves $\varphi : \mathcal{F} \to \mathcal{G}$ is called*

1. *a **monomorphism** if $\ker(\varphi) = 0$;*
2. *an **epimorphism** if $\mathrm{coker}(\varphi) = 0$;*
3. *an **isomorphism** if it is both a monomorphism and an epimorphism.*

Equivalently, φ is a monomorphism, epimorphism, or isomorphism if and only if each stalk map

$$\varphi_x : \mathcal{F}_x \to \mathcal{G}_x$$

has the corresponding property for every $x \in X$.

Proposition 1.21. *Let $f : X \to Y$ be a continuous map of topological spaces.*

1. *If $\mathcal{F}, \mathcal{G}$ are sheaves on X, then the assignment*

$$U \longmapsto \mathrm{Hom}_{\mathbb{S}_\approx(U)}(\mathcal{F}|_U, \mathcal{G}|_U)$$

   *defines a sheaf on X, called the **internal Hom sheaf** and denoted $\mathcal{H}om(\mathcal{F}, \mathcal{G})$*

2. *If $\mathcal{F}$ is a sheaf on X, then the direct image sheaf $f_*\mathcal{F}$ on Y is defined by*

$$(f_*\mathcal{F})(V) := \mathcal{F}(f^{-1}(V))$$

   for each open set $V \subseteq Y$.

3. *If $\mathcal{G}$ is a sheaf on Y, then the inverse image sheaf $f^{-1}\mathcal{G}$ on X is the sheafification of the presheaf*

$$U \longmapsto \varinjlim_{U \subseteq f^{-1}(V)} \mathcal{G}(V).$$

Remark 1.22. *For sheaves of abelian groups, f^{-1} is exact, while f_* is left exact but usually not right exact. This is the sheaf-theoretic origin of higher direct images $R^i f_*$.*

Remark 1.23. *If one works with sheaves of rings and modules, then one often uses the notation $f^*\mathcal{F} := \mathcal{O}_X \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}\mathcal{F}$ for the pullback of modules. For plain sheaves of abelian groups, the fundamental functor is f^{-1}.*

Exercise 1.24. *Show that a morphism of sheaves $\varphi : \mathcal{F} \to \mathcal{G}$ is an isomorphism if and only if it is an isomorphism on stalks, i.e. if and only if $\varphi_x : \mathcal{F}_x \to \mathcal{G}_x$ is an isomorphism for every $x \in X$.*

The forward direction is clear since stalks are functorial and an isomorphism of sheaves induces an isomorphism on stalks. **The converse direction is the nontrivial part, and it relies on the fact that if a sheaf has zero stalks everywhere, then it must be the zero sheaf.**

Solution 1.25. *Assume that for every $x \in X$, the stalk map*

$$\varphi_x : \mathcal{F}_x \to \mathcal{G}_x$$

is an isomorphism. Then for every x,

$$(\ker \varphi)_x \cong \ker(\varphi_x) = 0, \qquad (\operatorname{coker} \varphi)_x \cong \operatorname{coker}(\varphi_x) = 0.$$

Hence $\ker(\varphi)$ and $\operatorname{coker}(\varphi)$ have zero stalks everywhere.

We claim that if a sheaf $\mathcal{H}$ has $\mathcal{H}_x = 0$ for all $x \in X$, then $\mathcal{H} = 0$. Indeed, let $U \subseteq X$ be open and let $s \in \mathcal{H}(U)$. For every $x \in U$, the germ $s_x \in \mathcal{H}_x$ is zero, so there exists an open neighborhood $V_x \subseteq U$ of x such that $s|_{V_x} = 0$. Thus s restricts to zero on the open cover $\{V_x\}_{x \in U}$ of U, and by the separatedness axiom for sheaves we get $s = 0$. Therefore $\mathcal{H}(U) = 0$ for all open U, so $\mathcal{H} = 0$.

Applying this to $\ker(\varphi)$ and $\operatorname{coker}(\varphi)$, we get

$$\ker(\varphi) = 0, \qquad \operatorname{coker}(\varphi) = 0.$$

Therefore φ is both a monomorphism and an epimorphism, hence an isomorphism.

Exercise 1.26. *Construct a morphism of sheaves which is surjective on stalks but not surjective on sections over some open set.*

Note that the corresponding statement for injectivity is false: if a morphism of sheaves is injective on stalks, then it is injective on sections over every open set. This is because the separatedness axiom for sheaves implies that if a section restricts to zero for every stalk, then it must be zero.

Solution 1.27. *Consider the morphism of sheaves*

$$d : \mathcal{O} \to \mathcal{O}, \qquad f \mapsto f'$$

on the sheaf of holomorphic functions on $\mathbb{C}$. It is a morphism of sheaves because the derivative of a holomorphic function is holomorphic, and the restriction of a derivative is the derivative of the restriction. It is surjective on stalks because every germ of a holomorphic function has a primitive germ. However, it is not surjective on sections over $\mathbb{C}^\times$, since $1/z$ has no global primitive on $\mathbb{C}^\times$.

1.2 Schemes and Morphisms

1.2.1 Affine schemes and gluing

The basic objects in scheme theory are affine schemes.

Definition 1.28. *Let A be a commutative ring. The **affine scheme** associated to A is the locally ringed space*

$$\operatorname{Spec} A := (\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A})$$

where:

1. *the underlying topological space* $\operatorname{Spec} A$ *is the set of prime ideals of A;*
2. *the closed sets are of the form*

$$V(I) = \{\mathfrak{p} \in \operatorname{Spec} A : I \subseteq \mathfrak{p}\}$$

   for ideals $I \subseteq A$;

3. *the structure sheaf is characterized by*

$$\mathcal{O}_{\operatorname{Spec} A}(D(f)) = A_f$$

   on the basic distinguished opens

$$D(f) := \{\mathfrak{p} \in \operatorname{Spec} A : f \notin \mathfrak{p}\}.$$

The local ring at a point $\mathfrak{p} \in \operatorname{Spec} A$ is

$$\mathcal{O}_{\operatorname{Spec} A, \mathfrak{p}} \cong A_{\mathfrak{p}}.$$

In particular, points of an affine scheme carry local algebraic information.

Proposition 1.29. *The distinguished opens $D(f)$ form a basis for the Zariski topology on* $\operatorname{Spec} A$*, and for $f, g \in A$ one has*

$$D(f) \cap D(g) = D(fg).$$

Moreover,

$$D(f) \cong \operatorname{Spec} A_f$$

as locally ringed spaces.

Remark 1.30. *The last statement explains why localization corresponds geometrically to restricting to a distinguished open subset.*

Definition 1.31. *A **scheme** is a locally ringed space $(X, \mathcal{O}_X)$ such that every point $x \in X$ has an open neighborhood isomorphic, as a locally ringed space, to an affine scheme.*

Thus schemes are obtained by gluing affine schemes along open subsets.

Definition 1.32. *A **morphism of schemes** $f : X \to Y$ is a morphism of locally ringed spaces, i.e. a continuous map $f : X \to Y$ together with a morphism of sheaves of rings $f^\# : \mathcal{O}_Y \to f_*\mathcal{O}_X$ such that for every $x \in X$, the induced morphism on local rings*

$$f_x^\# : \mathcal{O}_{Y,f(x)} \to \mathcal{O}_{X,x}$$

is a local homomorphism of local rings.

The data of the morphism of sheaves $f^\#$ is saying how to pull back functions on Y to functions on X and in particular is extra data.

Remark 1.33. *A morphism between affine schemes $\operatorname{Spec} B \to \operatorname{Spec} A$ is precisely a ring homomorphism $A \to B$. Given a morphism of affine schemes, the map on global sections is the desired ring homomorphism. Conversely, given a ring homomorphism $\varphi : A \to B$, the corresponding morphism of affine schemes is given by the induced map on prime ideals and the morphism of structure sheaves*

$$f^\# : \mathcal{O}_{\operatorname{Spec} A} \to f_*\mathcal{O}_{\operatorname{Spec} B}$$

is defined on distinguished opens by

$$f^\#_{D(f)} : A_f \longrightarrow B_{\varphi(f)}, \qquad \frac{a}{f^n} \mapsto \frac{\varphi(a)}{\varphi(f)^n},$$

Proposition 1.34 (Gluing of schemes)**.** *Let $\{X_i\}_{i\in I}$ be schemes, and suppose for each i, j there are open subsets $U_{ij} \subseteq X_i$ together with isomorphisms*

$$\varphi_{ij} : U_{ij} \xrightarrow{\sim} U_{ji}$$

satisfying:

1. $\varphi_{ii} = \mathrm{id}$;
2. $\varphi_{ji} = \varphi_{ij}^{-1}$;
3. $\varphi_{jk} \circ \varphi_{ij} = \varphi_{ik}$ *on triple overlaps*

Then the X_i glue to a scheme X containing each X_i as an open subscheme.

1.2.2 Morphisms of finite type and finite morphisms

Definition 1.35. *A morphism of affine schemes*

$$\operatorname{Spec} B \to \operatorname{Spec} A$$

*is of **finite type** if B is a finitely generated A-algebra.*

Definition 1.36. *A morphism of schemes $f : X \to Y$ is* ***locally of finite type*** *if every point of X has an affine neighborhood $\operatorname{Spec} B$ mapping into an affine open $\operatorname{Spec} A \subseteq Y$ such that B is a finitely generated A-algebra. It is* ***of finite type*** *if in addition f is quasi-compact.*

Example 1.37. *The morphism*

$$\mathbb{A}^n_A = \operatorname{Spec} A[x_1, \ldots, x_n] \to \operatorname{Spec} A$$

is of finite type, and so is

$$\mathbb{P}^n_A \to \operatorname{Spec} A.$$

Exercise 1.38.

(a) Show that every open embedding is locally of finite type, and hence that every quasi-compact open embedding is of finite type. Show that every open embedding into a locally Noetherian scheme is of finite type.

(b) Show that the composition of two morphisms locally of finite type is locally of finite type. Deduce that the composition of two morphisms of finite type is of finite type.

(c) Suppose $\pi : X \to Y$ is locally of finite type, and Y is locally Noetherian. Show that X is also locally Noetherian. If $\pi : X \to Y$ is of finite type and Y is Noetherian, show that X is Noetherian.

Solution 1.39.

(a) *Let $j : U \hookrightarrow X$ be an open immersion. This is local on X, so let $\operatorname{Spec} A \subset X$ be an affine open subset. Then*

$$j^{-1}(\operatorname{Spec} A) = U \cap \operatorname{Spec} A$$

is an open subset of $\operatorname{Spec} A$, hence can be written as a union of distinguished opens:

$$U \cap \operatorname{Spec} A = \bigcup_i D(f_i).$$

Each $D(f_i) \cong \operatorname{Spec} A_{f_i}$, and $A_{f_i} = A[f_i^{-1}]$ is a finitely generated A-algebra. Thus j is locally of finite type.

If j is quasi-compact, then for each affine open $\operatorname{Spec} A \subset X$, the open subset $U \cap \operatorname{Spec} A$ is quasi-compact, so from the cover

$$U \cap \operatorname{Spec} A = \bigcup_i D(f_i)$$

we may extract a finite subcover

$$U \cap \operatorname{Spec} A = \bigcup_{m=1}^{r} D(f_{i_m}).$$

Hence $j^{-1}(\operatorname{Spec} A)$ is quasi-compact. Therefore j is quasi-compact and locally of finite type, so j is of finite type.

If X is locally Noetherian, then every affine open $\operatorname{Spec} A \subset X$ *has A Noetherian, and hence every open subset of* $\operatorname{Spec} A$ *is quasi-compact. Therefore every open immersion into X is quasi-compact. Since every open immersion is locally of finite type, it follows that every open immersion into a locally Noetherian scheme is of finite type.*

(b) *Let $X \xrightarrow{f} Y \xrightarrow{g} Z$ be morphisms locally of finite type. This is local on Z, so take an affine open* $\operatorname{Spec} C \subset Z$*, and let* $\operatorname{Spec} B \subset Y$ *lie over it, and* $\operatorname{Spec} A \subset X$ *lie over* $\operatorname{Spec} B$*.*

Then B is a finitely generated C-algebra, and A is a finitely generated B-algebra. Hence

$$A \cong B[x_1, \ldots, x_m]/I \cong C[y_1, \ldots, y_n, x_1, \ldots, x_m]/J,$$

so A is finitely generated over C. Thus $g \circ f$ is locally of finite type.

If f and g are of finite type, then they are also quasi-compact, and compositions of quasi-compact morphisms are quasi-compact. Hence $g \circ f$ is of finite type.

(c) *Suppose $\pi : X \to Y$ is locally of finite type and Y is locally Noetherian. This is local on Y, so assume $Y =$* $\operatorname{Spec} A$ *with A Noetherian. Let* $\operatorname{Spec} B \subset X$ *be an affine open mapping into* $\operatorname{Spec} A$*. Then B is a finitely generated A-algebra, hence Noetherian. Thus X is locally Noetherian.*

If π is of finite type and Y is Noetherian, then X is quasi-compact and locally Noetherian. A quasi-compact locally Noetherian scheme is Noetherian. Hence X is Noetherian.

Theorem 1.40 (Chevalley)**.** *If $f : X \to Y$ is a morphism of finite type between Noetherian schemes, then the image of any constructible subset of X is constructible in Y.*

Remark 1.41. *In particular, $f(X)$ is a constructible subset of Y. Thus if f is dominant (i.e. the image of f is dense in Y), its image contains a dense open subset of Y. This is saying something nontrivial because the target Y may have dense subsets which are not open (e.g. $\mathbb{Q} \subset \mathbb{R}$ is dense but not open), but Chevalley's theorem says that the image of a morphism of finite type cannot be one of those.*

Definition 1.42. *A ring map $A \to B$ is* ***finite*** *if B is finitely generated as an A-module. A morphism of schemes $f : X \to Y$ is finite if it is affine and for every affine open* $\operatorname{Spec} A \subseteq Y$*, the inverse image $f^{-1}(\operatorname{Spec} A)$ is affine, say* $\operatorname{Spec} B$*, with B finite as an A-module.*

Example 1.43. *Closed immersions are finite. Working locally, we are thinking about a morphism* $\operatorname{Spec} A/I \to \operatorname{Spec} A$ *for some ideal $I \subseteq A$. The ring A/I is generated as an A-module by the single element* 1*, so the morphism is finite.*

The morphism $k[t] \to k[u]$ given by $t \mapsto p(u)$ for some polynomial p of degree n is finite, since $k[u]$ is generated as a $k[t]$-module by $1, u, \ldots, u^{n-1}$.

Finite morphisms are precisely the morphisms which are affine and projective.

Example 1.44. *The open embedding $\mathbb{A}^2_k \setminus \{(0,0)\} \to \mathbb{A}^2_k$ has finite fibers, but is not affine (as $\mathbb{A}^2_k \setminus \{(0,0)\}$ is not affine) and hence not finite. Since the complement has codimension 2 and $\mathbb{A}^2_k$ is normal, every regular function on U extends uniquely to $\mathbb{A}^2_k$. Thus*

$$\Gamma(U, \mathcal{O}_U) = k[x, y].$$

If U were affine, then

$$U \cong \operatorname{Spec} \Gamma(U, \mathcal{O}_U) = \operatorname{Spec} k[x, y] = \mathbb{A}^2_k,$$

which is impossible since $U \neq \mathbb{A}^2_k$.

In general, open immersions are not finite because they are not affine, but even when they are affine, they need not be finite. In fact, an open immersion is finite exactly when it is also a closed immersion onto an open-and-closed subset. So for connected schemes, the only finite open immersions are isomorphisms onto connected components.

Exercise 1.45. *Show that the open embedding $\mathbb{A}^1_{\mathbb{C}} \setminus \{0\} \to \mathbb{A}^1_{\mathbb{C}}$ has finite fibers and is affine, but is not finite.*

Solution 1.46. *It is clear that the morphism is affine, since $\mathbb{A}^1_{\mathbb{C}} \setminus \{0\}$ is affine, given by the nonvanishing of x. It has finite fibers, since the fiber over 0 is empty and the fiber over any nonzero point is a single point. However, it is not finite, since the corresponding ring map $\mathbb{C}[x] \to \mathbb{C}[x, x^{-1}]$ is not finite, as x^{-1} is not integral over $\mathbb{C}[x]$.*

Exercise 1.47. *Show that if $f : X \to \operatorname{Spec} k$ is a finite morphism, then X is a finite union of discrete points, each with residual field a finite extension of k.*

Solution 1.48. *If $f : X \to \operatorname{Spec} k$ is finite, then X is affine, say*

$$X = \operatorname{Spec} A$$

with A a finite k-algebra. Since A is finite-dimensional as a k-vector space, every prime quotient $A/\mathfrak{p}$ is a finite-dimensional domain over k, hence a field. Thus every prime ideal of A is maximal.

Now A can only have finitely many maximal ideals, since if $\mathfrak{m}_1, \ldots, \mathfrak{m}_n$ are distinct maximal ideals, then the they are pairwise coprime, so the Chinese remainder theorem gives an surjection

$$A \to \prod_{i=1}^{n} A/\mathfrak{m}_i$$

so the dimension of A as a k-vector space is at least n.

Hence X is a finite set of closed points. For each point $x \in X$, the residue field is

$$\kappa(x) = A/\mathfrak{m}_x,$$

which is a finite extension of k. Thus X is a finite union of discrete points, each with residue field, finite over k again by the finite-dimensionality of A as a k-vector space.

Exercise 1.49. *Show that finite morphisms have finite fibers, i.e. for every point $y \in Y$, the fiber X_y is a finite scheme over $\kappa(y)$.*

Solution 1.50. *Let $f : X \to Y$ be a finite morphism, and let $y \in Y$. Since the question is local on Y, we may assume*

$$Y = \operatorname{Spec} B, \qquad X = \operatorname{Spec} A,$$

where A is a finite B-algebra. Let y correspond to a prime ideal $\mathfrak{q} \subset B$.

The fiber over y is

$$X_y \cong \operatorname{Spec}\Big(A \otimes_B \kappa(\mathfrak{q})\Big).$$

Note that prime ideals of $A \otimes_B \kappa(\mathfrak{q})$ correspond exactly to primes $\mathfrak{p} \subset A$ such that $\mathfrak{p} \cap B = \mathfrak{q}$.

Since A is finite as a B-module, the ring $A \otimes_B \kappa(\mathfrak{q})$ is a finite-dimensional $\kappa(\mathfrak{q})$-vector space. Thus X_y has only finitely many points. Thus finite morphisms have finite fibers.

Remark 1.51. *Let $\varphi : B \to A$ and $\mathfrak{q} \in \operatorname{Spec} B$. Set $S := B \setminus \mathfrak{q}$. Then $\kappa(\mathfrak{q}) = S^{-1}(B/\mathfrak{q})$, and*

$$A \otimes_B \kappa(\mathfrak{q}) \cong (A/\mathfrak{q}A) \otimes_{B/\mathfrak{q}} S^{-1}(B/\mathfrak{q}) \cong S^{-1}(A/\mathfrak{q}A)$$

Recall that $\operatorname{Spec}(S^{-1}R)$ has a point for every $\mathfrak{p} \in \operatorname{Spec} R : \ \mathfrak{p} \cap S = \varnothing$ via $\mathfrak{p} \mapsto S^{-1}\mathfrak{p}$.

Apply this with $R = A/\mathfrak{q}A$. Thus primes of $A \otimes_B \kappa(\mathfrak{q}) \cong S^{-1}(A/\mathfrak{q}A)$ correspond to primes $\mathfrak{p}' \subset A/\mathfrak{q}A$ such that $\mathfrak{p}' \cap S = \varnothing$.

Primes $\mathfrak{p}' \subset A/\mathfrak{q}A$ correspond to primes $\mathfrak{p} \subset A$ with $\mathfrak{q}A \subset \mathfrak{p}$ via $\mathfrak{p} \mapsto \mathfrak{p}/\mathfrak{q}A$. The condition $\mathfrak{p}' \cap S = \varnothing$ becomes $\varphi(B \setminus \mathfrak{q}) \cap \mathfrak{p} = \varnothing$.

The condition $\varphi(B \setminus \mathfrak{q}) \cap \mathfrak{p} = \varnothing$ is equivalent to $\varphi^{-1}(\mathfrak{p}) \subseteq \mathfrak{q}$. Combined with $\mathfrak{q}A \subseteq \mathfrak{p}$, we get $\mathfrak{q} \subseteq \varphi^{-1}(\mathfrak{p})$. Hence $\varphi^{-1}(\mathfrak{p}) = \mathfrak{q}$.

Exercise 1.52. *Show that the base change $f' : X \times_Y Y' \to Y'$ of a finite morphism $f : X \to Y$ is finite.*

Solution 1.53. *This is local on Y and Y', so we may assume $Y = \operatorname{Spec} A, \qquad X = \operatorname{Spec} B, \qquad Y' =$ Since f is finite, B is a finite A-algebra. Now*

$$X' = X \times_Y Y' \cong \operatorname{Spec}(B \otimes_A A').$$

So it is enough to show that $B \otimes_A A'$ is a finite A'-module. But if $b_1, \ldots, b_n$ generate B as an A-module, then the elements $b_i \otimes 1$ generate $B \otimes_A A'$ as an A'-module, by A-linearity of the tensor product.

1.2.3 Integrality and some commutative algebra

Definition 1.54. *A ring map $B \to A$ is **integral** if every element $a \in A$ satisfies a monic polynomial with coefficients in B. An integral ring morphism is an **integral extension** if it is injective.*

A morphism of schemes is ***integral*** *if it is affine and for every affine open* $\operatorname{Spec} A \subseteq Y$*, the inverse image* $f^{-1}(\operatorname{Spec} A)$ *is affine, say* $\operatorname{Spec} B$*, with* B *integral over* A*.*

The key point is topological. In particular, if $A \to B$ is integral, then $\operatorname{Spec} B \to \operatorname{Spec} A$ is closed and surjective. This geometry is a consequence of the algebraic fact known as the **lying over** and **going up** theorems.

Note that being an integral morphism is preserved by quotienting in the source and in the target, and localizing in the source. It is not preserved by localizing in the target.

Also note that if b_1, b_2 are integral over A, then so are $b_1 + b_2$ and $b_1 b_2$. Thus the set of elements of B that are integral over A form a subalgebra of B, called the **integral closure** of A in B. If A is integrally closed in its field of fractions, then we say that A is **integrally closed**. Also note that integrality is transitive: if $A \to B$ and $B \to C$ are integral, then so is $A \to C$.

Lemma 1.55. *Suppose* $\varphi : B \to A$ *is a ring morphism. Then* $a \in A$ *is integral over* B *if and only if it is contained in a subalgebra of* A *that is a finitely generated* B*-module.*

Proof. If a satisfies a monic polynomial of degree n, then the B-submodule of A generated by $1, a, \ldots, a^{n-1}$ is closed under multiplication, and hence is a subalgebra of A.

Conversely, assume that a is contained in a subalgebra $A' \subset A$ that is a finitely generated B-module. Choose a finite generating set $m_1, \ldots, m_n$ of A' (as a B-module). Then

$$am_i = \sum_j b_{ij} m_j \quad \text{for some } b_{ij} \in B.$$

Thus

$$(aI_n - [b_{ij}]_{ij}) \begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}, \tag{1}$$

where I_n is the $n \times n$ identity matrix. Recall that any $n \times n$ matrix M has an adjugate matrix $\operatorname{adj}(M)$ such that

$$\operatorname{adj}(M)\, M = \det(M)\, I_n.$$

Multiplying (1) by $\operatorname{adj}(aI_n - [b_{ij}])$, we obtain

$$\det(aI_n - [b_{ij}]) \begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Thus $\det(aI_n - [b_{ij}])$ annihilates each m_i, hence every element of A'. In particular, it annihilates 1 so

$$\det(aI_n - [b_{ij}]) = 0 \quad \text{in } A.$$

Expanding the determinant gives a monic polynomial in a with coefficients in B, so a is integral over B. $\square$

Proposition 1.56. *If B is finite as an A-module, then B is integral over A.*

Theorem 1.57 (Lying Over)**.** *Suppose $\varphi : B \to A$ is an integral extension. Then for any prime ideal $\mathfrak{q} \subset B$, there is a prime ideal $\mathfrak{p} \subset A$ such that*

$$\mathfrak{p} \cap B = \mathfrak{q}.$$

Note that B need not be Noetherian, and A need not be finitely generated over B.

Exercise 1.58. *Show that the special case where A is a field translates to: if $B \subset A$ and A is integral over B, then B is a field.*

Solution 1.59. *Let $b \in B$. Then $b^{-1} \in A$ is integral over B, so there is a monic polynomial*

$$(b^{-1})^n + a_{n-1}(b^{-1})^{n-1} + \cdots + a_0 = 0$$

with $a_i \in B$. Multiplying by b^n gives

$$1 + a_{n-1}b + \cdots + a_0 b^n = 0.$$

Thus $b^{-1} \in B$ so B is a field.

Proof. We first reduce to the case where $(B, \mathfrak{q})$ is a local ring. By localizing at $\mathfrak{q}$ (which preserves integrality), we may assume this.

Let $\mathfrak{p}$ be any maximal ideal of A. Consider the diagram

$$\begin{tikzcd} A \arrow[r] \arrow[d] & A/\mathfrak{p} \arrow[d] \\ B \arrow[r] & B/(\mathfrak{p} \cap B) \end{tikzcd}$$

The right vertical map is an integral extension. Since $A/\mathfrak{p}$ is a field, it follows that $B/(\mathfrak{p} \cap B)$ is also a field. Hence $\mathfrak{p} \cap B$ is maximal, and therefore equals $\mathfrak{q}$. Using the correspondence between prime ideals of A and prime ideals of $A_{\mathfrak{q}}$, we can contract the prime ideal $\mathfrak{p}$ back to get the desired prime ideal $\mathfrak{p}$ of A. $\square$

The same adjugate trick can be used to prove Nakayama's lemma, so we will do this now. They seem to have no connection conceptually.

Lemma 1.60 (Nakayama's lemma)**.** *Let I be an ideal of a ring A, and let M be a finitely generated A-module such that $IM = M$. Then there is an element $a \in A$ with $a \equiv 1 \mod I$ such that $aM = 0$.*

Proof. Say M is generated by $m_1, \ldots, m_n$. Then as $M = IM$, we have $m_i = \sum_j a_{ij} m_j$ for some $a_{ij} \in I$. Thus

$$(I_n - Z) \begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

where $Z = (a_{ij})$. Multiplying both sides on the left by $\operatorname{adj}(I_n - Z)$, we obtain

$$\det(I_n - Z)\begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Expanding the determinant gives something that is $1 \pmod{I}$, since Z has entries in I. $\square$

Suppose I is contained in all of the maximal ideals of A. Then $a \equiv 1 \mod I$ is a unit. Otherwise, (a) is not the whole ring and therefore is contained in some maximal ideal $\mathfrak{m}$. But then $a \cong 1 \mod \mathfrak{m}$, contradiction.

Thus an application of Nakayama's lemma says that $aM = 0$ for some unit a, so $M = 0$.

Lemma 1.61 (Nakayama's lemma)**.** *Suppose A is a ring, I is an ideal of A contained in all maximal ideals, and M is a finitely generated A-module. Suppose $M = IM$. Then $M = 0$.*

The most interesting case is when A is a local ring, and I is the maximal ideal.

Exercise 1.62 (Nakayama's lemma)**.** *Suppose $(A, \mathfrak{m})$ is a local ring, and let M be a finitely generated A-module. Suppose $f_1, \ldots, f_n \in M$ are such that their images generate $M/\mathfrak{m}M$. Then $f_1, \ldots, f_n$ generate M.*

Solution 1.63. *Let $N \subseteq M$ be the submodule generated by $f_1, \ldots, f_n$. Since the images of $f_1, \ldots, f_n$ generate $M/\mathfrak{m}M$, every class in $M/\mathfrak{m}M$ is represented by an element of N. Equivalently,*

$$M = N + \mathfrak{m}M.$$

Passing to the quotient by N, we obtain

$$\begin{aligned} M/N &= (N + \mathfrak{m}M)/N \\ &= \mathfrak{m}M/N \end{aligned}$$

Now M/N is a finitely generated A-module, because M is finitely generated. Since $(A, \mathfrak{m})$ is local, the ideal $\mathfrak{m}$ is contained in every maximal ideal of A; in fact it is the unique maximal ideal. Therefore Nakayama's lemma implies that $M/N = 0$, so $f_1, \ldots, f_n$ generate M.

Example 1.64. *Finite morphisms are integral, but integral morphisms need not have finite fibers. Let k be a field,*

$$B = k[x], \qquad A = \bigcup_{n \geq 1} k[x^{1/n}] \subset k(x^{1/\infty}).$$

Every element of A is integral over B, so $\operatorname{Spec} A \to \operatorname{Spec} B$ is integral. But A is not finite over B.

Look at the fiber over $(x) \in \operatorname{Spec} B$*. The primes of* A *lying over* (x) *correspond to primes of*

$$A \otimes_B \kappa((x)) \cong A/(x).$$

where $\kappa((x))$ *is the residue field of* B *at* (x)*, which is* k*. Now*

$$A/(x) \cong k[t_1, t_2, \dots]/(t_1, t_2^2, t_3^3, \dots)$$

where t_n *is the image of* $x^{1/n}$*. This ring has infinitely many prime ideals, so the fiber is infinite.*

Note that this counterexample is not of finite type. We can ask whether or not this is the only way for an integral morphism to fail to have finite fibers. The answer is indeed yes.

Proposition 1.65. *An integral morphism which is locally of finite type is finite.*

Proof. This is local on Y, so take

$$Y = \operatorname{Spec} A, \quad X = \operatorname{Spec} B.$$

By assumption we can write

$$B = A[b_1, \dots, b_n].$$

with each b_i is integral over A, so satisfies a monic polynomial:

$$b_i^{d_i} + a_{i,1} b_i^{d_i - 1} + \cdots + a_{i,d_i} = 0, \quad a_{i,j} \in A.$$

From these relations, every power b_i^m with $m \geq d_i$ can be expressed as an A-linear combination of $1, b_i, \dots, b_i^{d_i - 1}$.

Thus any monomial in the b_i reduces to an A-linear combination of finitely many monomials:

$$b_1^{e_1} \cdots b_n^{e_n}, \quad 0 \leq e_i < d_i.$$

There are finitely many such monomials and they generate B as an A-module. □

It is also useful to know that integral and finite morphisms are preserved by base change and composition.

1.2.4 Quasicompact, quasiseperated, and proper morphisms

Let $f : X \to Y$ be a morphism of schemes. The correct analogue of compactness is quasicompactness.

Definition 1.66. *The morphism* $f : X \to Y$ *is* ***quasicompact*** *if the inverse image of every affine open in* Y *is quasi-compact, i.e. if it can be covered by finitely many affine opens. A scheme* X *is* ***quasicompact*** *if the morphism* $X \to \operatorname{Spec} \mathbb{Z}$ *is quasicompact, i.e. if* X *can be covered by finitely many affine opens.*

Example 1.67. *Every affine scheme* $\operatorname{Spec} A \to \operatorname{Spec} \mathbb{Z}$ *is quasicompact, since* $\operatorname{Spec} A$ *can be covered by the single affine open* $\operatorname{Spec} A$.

Example 1.68. *The morphism* $\bigsqcup_{i\in\mathbb{N}} \operatorname{Spec} k \to \operatorname{Spec} k$ *is not quasi-compact, since the preimage of* $\operatorname{Spec} k$ *is an infinite disjoint union of affine schemes, which is not quasi-compact.*

Example 1.69. *Let* $A = k[x_1, x_2, \dots]$. *Then* $\operatorname{Spec} A \to \operatorname{Spec} k$ *is quasi-compact but not of finite type.*

Let $f : X \to Y$ be a morphism of schemes. The correct analogue of the Hausdorff condition is separatedness.

Definition 1.70. *The* ***diagonal morphism*** *of* f *is*

$$\Delta_f : X \to X \times_Y X, \qquad x \mapsto (x, x).$$

The morphism f *is called* ***quasi-separated*** *if* Δ_f *is quasi-compact, i.e. if the inverse image of every affine open in* $X \times_Y X$ *is quasi-compact. The morphism* f *is called* ***separated*** *if* Δ_f *is a closed immersion.*

Remark 1.71. *A scheme* X *is quasi-separated iff the intersection of any two affine open subsets is quasi-compact. A scheme* X *is separated iff the intersection of any two affine open subsets is affine.*

Example 1.72. *Every affine morphism is separated. In particular the diagonal morphism of an affine morphism corresponds to the multiplication map of rings and being surjective is therefore a closed immersion. In particular every affine scheme is separated over* $\operatorname{Spec} \mathbb{Z}$.

Example 1.73. *Projective space* $\mathbb{P}^n_Y \to Y$ *is separated. Cover* $\mathbb{P}^n_Y \times_Y \mathbb{P}^n_Y$ *by the opens* $U_i \times_Y U_j$ *where* U_i *is the standard affine open in* $\mathbb{P}^n_Y$ *where the* i*-th homogeneous coordinate is nonzero. The intersection of* $U_i \times_Y U_j$ *with the diagonal is the vanishing of* $x_i/x_j \subset U_i$. *A distinguished open in an affine scheme is affine, so the intersection of two affine opens in* $\mathbb{P}^n_Y$ *is affine, thus* $\mathbb{P}^n_Y$ *is separated over* Y.

Example 1.74 (Affine line with doubled origin)**.** ***The line with doubled origin is not separated.*** *Intuitively, the two origins cannot be distinguished by disjoint neighborhoods, and scheme-theoretically this failure is measured by the diagonal not being closed.*

The line with doubled origin is obtained by gluing two copies of $\mathbb{A}^1_k$ *along the open* $D(t)$ *via the identity map. The diagonal morphism is not closed. To test whether* $\Delta_X : X \to X \times X$ *is a closed immersion, it is enough to check on an open cover of* $X \times X$. *In particular, look at the open subset* $U_1 \times U_2 \subset X \times X$ *where* U_1 *and* U_2 *are the two copies of* $\mathbb{A}^1_k \setminus \{0\}$. *The inverse image of* $U_1 \times U_2$ *under the diagonal is exactly the set of points of* X *lying*

in both U_1 and U_2, namely the overlap $\Delta_X^{-1}(U_1 \times U_2) = U_1 \cap U_2 = D(t)$

So on this chart the diagonal restricts to the morphism

$$D(t) \longrightarrow U_1 \times U_2 \cong \operatorname{Spec} k[t_1, t_2]$$

given by $t_1 \mapsto t$ and $t_2 \mapsto t$. Its image is therefore $\{(a, a) : a \neq 0\}$, which is not closed in $U_1 \times U_2$ because it does not contain the point $(0, 0)$.

It is however quasi-separated. *Cover X by the two affine opens$X = X_1 \cup X_2$, with $X_i \cong \operatorname{Spec} k[t]$.*

Then $X \times X$ is covered by the four affine opens $X_i \times X_j$ ($i, j \in \{1, 2\}$). It suffices to show that for each i, j, the inverse image $\Delta_X^{-1}(X_i \times X_j)$ is quasi-compact.

- *If $i = j$, then $\Delta_X^{-1}(X_i \times X_i) = X_i$, which is affine, hence quasi-compact.*
- *If $i \neq j$, then $\Delta_X^{-1}(X_i \times X_j) = X_i \cap X_j = D(t)$, which is affine, hence quasi-compact.*

Therefore Δ_X is quasi-compact, so X is quasi-separated. Since each of these intersections is affine, it appears that we have just shown that X is separated, but this is not the case because there exist other affine opens $U, V \subset X$ whose intersection is not affine. The subtle point is that it is not enough to just check the affine opens in the given cover; we must check all affine opens. This is a contrast to quasi-separatedness or quasicompactness, where it suffices to check the affine opens in a given cover.

Example 1.75 (Affine line with infinitely many origins is not quasi-separated)**.** *Let X be obtained by gluing countably many copies*

$$X_i \cong \operatorname{Spec} k[t]$$

along the common open subset $D(t) = \operatorname{Spec} k[t, t^{-1}]$ via the identity.

This defines a scheme with infinitely many distinct origins $\{0_i\}$.

Let t denote the global function induced by the coordinate on each chart. For any subset $I \subset \mathbb{N}$, define

$$f_I := \prod_{i \in I} t.$$

Then f_I defines a global function on X, and

$$D(f_I)$$

is an affine open subset.

Choose a partition $\mathbb{N} = I \sqcup J$ into two infinite subsets, and set

$$U := D(f_I), \qquad V := D(f_J).$$

Then U and V are affine open subsets of X.

Their intersection is

$$U \cap V = D(f_I f_J) = D\Big(\prod_{n\in\mathbb{N}} t\Big),$$

which is the complement of all origins. Hence

$$U \cap V \cong \bigsqcup_{n\in\mathbb{N}} D(t).$$

This is an infinite disjoint union of affine schemes, hence not quasi-compact. Therefore X is not quasi-separated.

Definition 1.76. *A morphism $f : X \to Y$ is called* ***universally closed*** *if for every morphism $Y' \to Y$, the base change*

$$f' : X \times_Y Y' \to Y'$$

is a closed map on underlying topological spaces.

Exercise 1.77. *Show that $\pi : \mathbb{A}^1_k \to \operatorname{Spec} k$ is closed but not universally closed. Thus π is not proper.*

Recall that we saw that if $B \to A$ is integral, then $\operatorname{Spec} A \to \operatorname{Spec} B$ is closed. Since being integral is preserved by base change, it follows that integral morphisms are universally closed.

Proposition 1.78. *A morphism of schemes is integral if and only if it is affine and universally closed.*

Definition 1.79. *A morphism $f : X \to Y$ is called* ***proper*** *if it is of finite type, separated, and universally closed.*

This is the algebro-geometric replacement for compactness.

Example 1.80. *A closed immersion is proper. On affines, $A \to A/I$ is finitely generated so the morphism is of finite type. Closed immersions are affine morphisms and thus separated. A closed immersion is a homeomorphism onto a closed subset. Closed sets remain closed under any base change.*

Remark 1.81. *Proper morphisms are stable under composition and base change.*

Theorem 1.82. *Projective A-schemes are proper over A.*

Proof. Let $X \to \operatorname{Spec} A$ be a projective A-scheme. Then by definition the structure morphism factors as

$$X \hookrightarrow \mathbb{P}^n_A \to \operatorname{Spec} A$$

for some n, where the first map is a closed immersion. Since closed immersions are proper, and compositions of proper morphisms are proper, it suffices to show that

$$\mathbb{P}^n_A \to \operatorname{Spec} A$$

is proper.

We have already seen that this morphism is of finite type and separated. Therefore it remains to show that it is universally closed.

Now

$$\mathbb{P}^n_A = \mathbb{P}^n_{\mathbb{Z}} \times_{\operatorname{Spec}\mathbb{Z}} \operatorname{Spec} A.$$

So to prove that $\mathbb{P}^n_A \to \operatorname{Spec} A$ is universally closed, it suffices to show that for every scheme X, the projection

$$\mathbb{P}^n_X := \mathbb{P}^n_{\mathbb{Z}} \times_{\operatorname{Spec}\mathbb{Z}} X \to X$$

is a closed map.

Being a closed map is local on the target. Hence, after covering X by affine open subschemes, it suffices to prove that

$$\mathbb{P}^n_B \to \operatorname{Spec} B$$

is closed for every ring B.

But this is exactly the Fundamental Theorem of Elimination Theory, which says that given a system of homogeneous polynomial equations with coefficients in B, the set of $b \in \operatorname{Spec} B$ such that the system has a solution over $\kappa(b)$ is Zariski closed in $\operatorname{Spec} B$. □

Example 1.83. *Finite morphisms are proper. Indeed, a finite morphism is affine and of finite type, hence separated. It is also universally closed since it is integral.*

1.2.5 Valuative criterion of properness

The most efficient way to test properness in practice is often via valuation rings.

Definition 1.84. *A valuation ring is an integral domain R with fraction field K such that for every $x \in K^\times$, either $x \in R$ or $x^{-1} \in R$.*

Theorem 1.85 (Valuative criterion for separatedness)**.** *Let $f : X \to Y$ be a morphism of finite type. Then f is separated if and only if for every valuation ring R with fraction field K, every commutative diagram*

$$\begin{tikzcd} \operatorname{Spec} K \arrow[r] \arrow[d] & X \arrow[d, "f"] \\ \operatorname{Spec} R \arrow[r] & Y \end{tikzcd}$$

admits at most one dotted lift

$$\begin{tikzcd} \operatorname{Spec} K \arrow[r] \arrow[d] & X \arrow[d, "f"] \\ \operatorname{Spec} R \arrow[r] \arrow[ur, dashed] & Y. \end{tikzcd}$$

Theorem 1.86 (Valuative criterion for properness)**.** *Let $f : X \to Y$ be a morphism of finite type. Then f is proper if and only if for every valuation ring R with fraction field K, every commutative diagram*

$$\begin{tikzcd} \operatorname{Spec} K \arrow[r] \arrow[d] & X \arrow[d, "f"] \\ \operatorname{Spec} R \arrow[r] & Y \end{tikzcd}$$

admits a unique dotted lift

$$\begin{tikzcd} \operatorname{Spec} K \arrow[r] \arrow[d] & X \arrow[d, "f"] \\ \operatorname{Spec} R \arrow[r] \arrow[ur, dashed] & Y. \end{tikzcd}$$

The geometric meaning is that a map from the generic point of a trait extends uniquely across the special point.

Example 1.87. *The valuative criterion shows that $\mathbb{P}^n_k$ is proper over k: a morphism*

$$\operatorname{Spec} K \to \mathbb{P}^n_k$$

is given by homogeneous coordinates $[x_0 : \cdots : x_n]$ with $x_i \in K$, and after scaling we may assume all $x_i \in R$ and at least one is a unit, giving an extension

$$\operatorname{Spec} R \to \mathbb{P}^n_k.$$

1.2.6 Flat, smooth, and étale morphisms

These three notions measure how a family varies over the base.

Definition 1.88. *Let $A \to B$ be a ring map. We say that B is **flat over** A if the functor*

$$- \otimes_A B$$

is exact on the category of A-modules. Equivalently, B is flat over A if $\operatorname{Tor}_i^A(B, M) = 0$ for all $i > 0$ and all A-modules M.

Definition 1.89. *A morphism of schemes $f : X \to Y$ is flat if for every point $x \in X$, the local ring $\mathcal{O}_{X,x}$ is flat over $\mathcal{O}_{Y,f(x)}$.*

Flatness says that fibers vary without sudden jumps caused by torsion. It is a subtle but central notion.

Theorem 1.90 (Local criterion for flatness)**.** *Let $(A, \mathfrak{m}) \to (B, \mathfrak{n})$ be a local homomorphism of Noetherian local rings, and let M be a finitely generated B-module. Then M is flat over A if and only if*

$$\mathrm{Tor}_1^A(M, A/\mathfrak{m}) = 0.$$

Example 1.91. *The localization map of any multiplicative set is flat. This is because localization is an exact functor on modules. Free modules are flat, and being direct summands of free modules, projective modules are flat. This is because given an exact sequence of A-modules*

$$0 \to M' \otimes (P \oplus Q) \to M \otimes (P \oplus Q) \to M'' \otimes (P \oplus Q) \to 0$$

we can split it into two exact sequences because the maps are block diagonal.

Example 1.92. *If A is a ring, then $A[x_1, \ldots, x_n]$ is flat over A. This is because they are free, albeit of infinite rank. Flatness is preserved by base change, composition, and localization in the source and target.*

Exercise 1.93. *Show that if $\mathcal{F}$ is a locally free sheaf on a scheme X, then $\mathbb{P}\mathcal{F} \to X$ is flat.*

Solution 1.94. *$\mathbb{P}(\mathcal{F})$ is locally on X just $\mathbb{P}^{r-1}_X$, and projective space over a base is flat because it is covered by affine spaces over that base. On each such chart, the morphism to U is induced by a polynomial ring over $\mathcal{O}_U$. Polynomial rings are free, hence flat, over the base ring. Therefore each $D_+(x_i) \to U$ is flat.*

Example 1.95. *Open embeddings are flat. This is because they are localizations on the level of rings, and localization is an exact functor on modules.*

Exercise 1.96. *Show that a flat morphism induces an exact functor on quasi-coherent sheaves via pullback.*

Solution 1.97. *Recall that for any $\mathcal{O}_Y$-module $\mathcal{F}$,*

$$f^*\mathcal{F} = f^{-1}\mathcal{F} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X.$$

Now f^{-1} is exact on sheaves of abelian groups. This is because $(f^{-1}\mathcal{F})_x \cong \mathcal{F}_{f(x)}$ and exactness can be checked on stalks. So if

$$0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$$

is a short exact sequence of quasi-coherent sheaves on Y, then

$$0 \to f^{-1}\mathcal{F}' \to f^{-1}\mathcal{F} \to f^{-1}\mathcal{F}'' \to 0$$

is still exact. So it remains to check that tensoring with $\mathcal{O}_X$ over $f^{-1}\mathcal{O}_Y$ preserves exactness. This is exactly flatness, checked on stalks. For any $x \in X$ with $y = f(x)$,

$$(f^*\mathcal{F})_x \cong \mathcal{F}_y \otimes_{\mathcal{O}_{Y,y}} \mathcal{O}_{X,x}.$$

Since f is flat, $\mathcal{O}_{X,x}$ is a flat $\mathcal{O}_{Y,y}$-module. Therefore tensoring with $\mathcal{O}_{X,x}$ is exact, so

$$0 \to \mathcal{F}'_y \otimes_{\mathcal{O}_{Y,y}} \mathcal{O}_{X,x} \to \mathcal{F}_y \otimes_{\mathcal{O}_{Y,y}} \mathcal{O}_{X,x} \to \mathcal{F}''_y \otimes_{\mathcal{O}_{Y,y}} \mathcal{O}_{X,x} \to 0$$

is exact for every $x \in X$.

Hence

$$0 \to f^*\mathcal{F}' \to f^*\mathcal{F} \to f^*\mathcal{F}'' \to 0$$

is exact on stalks, and therefore exact as a sequence of sheaves.

Example 1.98. *A nonexample of a flat map is a quotient by a nonzerodivisor. For instance, if $A = k[x]$ and $B = A/(x)$, then B is not flat over A because $\mathrm{Tor}_A^1(B, k) \neq 0$.*

To see this, use the free resolution of B:

$$0 \to F_1 = A \xrightarrow{\cdot x} F_0 = A \to B \to 0.$$

Tensoring with $k = A/(x)$ over A:

$$0 \to k \xrightarrow{\cdot x} k \to B \otimes_A k \to 0.$$

Since $x = 0$ in k, the first map is zero:

$$0 \to k \xrightarrow{0} k \to B \otimes_A k \to 0.$$

Taking homology at $F_1 \otimes_A k$ gives

$$\mathrm{Tor}_A^1(B, k) = \ker(0)/\operatorname{im}(0) = k \neq 0.$$

Thus B is not flat over A. Geometrically, the map $\operatorname{Spec} B \to \operatorname{Spec} A$ is the inclusion of the origin into the affine line, which has generic fiber $B \otimes_A k(t) = (A/(t)) \otimes_A k(t) = 0$ but special fiber $B \otimes_A k = k$, so the fiber jumps in dimension.

Cohomology commutes with flat base change.

Theorem 1.99. *Suppose*

$$\begin{tikzcd} X' \arrow[r, "\rho'"] \arrow[d, "\pi'"'] & X \arrow[d, "\pi"] \\ Y' \arrow[r, "\rho"'] & Y \end{tikzcd}$$

is a Cartesian diagram, and π (and thus π') is quasicompact and separated (so higher pushforwards of quasicoherent sheaves by π and π' exist). Suppose also that ρ is flat, and $\mathcal{F}$ is a quasicoherent sheaf on X. Then the natural "push-pull" morphisms

$$\rho^* R^i \pi_* \mathcal{F} \cong R^i \pi'_* (\rho'^* \mathcal{F})$$

are isomorphisms. In particular, on the level of fibers, we have $(R^i\pi_*\mathcal{F}) \otimes \kappa(y) \cong H^i(X_y, \mathcal{F}_y)$ for every $y \in Y$.

Definition 1.100. *A morphism $f : X \to Y$ is* ***smooth of relative dimension*** *n if it is locally of finite type, flat, and for every geometric point $x \in X$, the fiber is regular of dimension n near x.*

A more practical criterion uses differentials.

Theorem 1.101 (Smoothness Criterion)**.** *Let $f : X \to Y$ be locally of finite presentation. The following are equivalent:*

1. *f is smooth of relative dimension n.*
2. *f is flat, $\Omega_{X/Y}$ is locally free of rank n, and for every $x \in X$, the natural map $\mathfrak{m}_x/\mathfrak{m}_x^2 \longrightarrow \Omega_{X/Y} \otimes \kappa(x)$ is surjective.*

Theorem 1.102 (Affine Jacobian Criterion)**.** *Suppose*

$$X = \operatorname{Spec} B = \operatorname{Spec} A[x_1, \ldots, x_m]/(f_1, \ldots, f_r), \quad Y = \operatorname{Spec} A.$$

Then f is smooth of relative dimension $m - r$ at a point if and only if:

- *the matrix*
$$\left(\frac{\partial f_i}{\partial x_j}\right)$$
has rank r at that point, and
- *B is flat over A.*

Equivalently, some $r \times r$ minor of the Jacobian matrix is invertible in the local ring.

Proposition 1.103. *If $f : X \to Y$ is smooth, then all geometric fibers are smooth. In particular, this means that for every geometric point $\bar{y} \to Y$, the fiber $X_{\bar{y}} = X \times_Y \operatorname{Spec} \kappa(\bar{y})$ is a smooth variety over $\kappa(\bar{y})$.*

Example 1.104. *The projection*

$$\mathbb{A}^n_Y \to Y$$

is smooth of relative dimension n.

Example 1.105. *The morphism*

$$\operatorname{Spec} k[x]/(x^2) \to \operatorname{Spec} k$$

is not smooth because it has a nonreduced fiber, and smooth morphisms have regular fibers, in particular reduced. However, it is flat because every module over a field is flat.

Example 1.106. *Let $A = k[t]$ and $B = A[x]/(x^2 - t)$. As an A-module, every element of B can be written uniquely as $a + bx$ for $a, b \in A$, Therefore $B \cong A \oplus A$ as an A-module, so B is flat over A. However, the fibers are not all isomorphic:*

- *For $t \neq 0$: the fiber is $\operatorname{Spec} k[x]/(x^2 - t)$, which is reduced (two distinct points if k is algebraically closed).*
- *For $t = 0$: the fiber is $\operatorname{Spec} k[x]/(x^2)$, which is nonreduced.*

Moreover, the morphism $\operatorname{Spec} B \to \operatorname{Spec} A$ is not smooth, because the fiber over $t = 0$ is singular. At the fiber over $t = 0$ which looks like $\operatorname{Spec} k[x]/(x^2)$, the scheme is 0-dimensional but the cotangent space $\mathfrak{m}/\mathfrak{m}^2$ is 1-dimensional. Thus flatness does not imply reduced fibers or that the fibers are isomorphic, and flatness does not imply smoothness.

Example 1.107. *A smooth morphism need not be étale. For example, the projection $\mathbb{A}^n_Y \to Y$ is smooth of relative dimension n, but it is not étale unless $n = 0$.*

Exercise 1.108. *Give examples of morphisms that are:*

1. *unramified morphisms that are not étale*
2. *finite but not étale morphisms*

Definition 1.109. *A morphism $f : X \to Y$ is **étale** if it is smooth of relative dimension 0.*

Equivalently, étale means unramified (i.e. $\Omega_{X/Y} = 0$) and flat of finite presentation. Étale morphisms are the algebro-geometric analogue of local isomorphisms.

Example 1.110. *If k is a field and $a \in k^\times$, then*

$$\operatorname{Spec} k[t, t^{-1}] \to \operatorname{Spec} k[s, s^{-1}], \qquad s \mapsto t^n$$

is étale whenever n is invertible in k.

Example 1.111. *Recall that a field extension L/k is separable if every element of L has minimal polynomial over k with distinct roots, i.e. separable. This is a purely characteristic p phenomenon, since in characteristic zero every field extension is separable. $\operatorname{Spec} L \to \operatorname{Spec} k$ is étale if and only if L is a finite separable extension of k. If $\operatorname{Spec} L \to \operatorname{Spec} k$ is étale, then L is a finite extension of k because the morphism is of finite type, and L is separable over k because $\Omega_{L/k} = 0$. Conversely, if L is a finite separable extension of k, then $\operatorname{Spec} L \to \operatorname{Spec} k$ is étale. This is because by the primitive element theorem, we can write $L = k[x]/(f)$ where f is a separable polynomial, and the Jacobian criterion for étaleness says that this morphism is étale if and only if $f' \in L$ is invertible, which is equivalent to f being separable. Any repeated root of f would be a common root of f and f', so f' is invertible in L if and only if f has no repeated roots, i.e. f is separable.*

Example 1.112. *The map $\operatorname{Spec} A[x]/(f) \to \operatorname{Spec} A$ with f' invertible in A is étale.*

There is a standard exact sequence:

$$(f)/(f)^2 \longrightarrow \Omega_{A[x]/A} \otimes B \longrightarrow \Omega_{B/A} \to 0.$$

Now $\Omega_{A[x]/A} \cong A[x] \cdot dx$ *so tensoring with* B *and using the relation* $f = 0$ *gives the relation* $df = 0$. *Since* $df = f'(x)\,dx$, *we obtain*

$$\Omega_{B/A} \cong B\,dx/(f'(x)\,dx) \cong B/(f'(x)) \cdot dx.$$

Étale requires $\Omega_{B/A} = 0$ *which is equivalent to* $f'(x)$ *being invertible in* B.

Example 1.113. *Open immersions are etale.*

Exercise 1.114. *Show that* $\operatorname{Spec} k \to \operatorname{Spec} k[t]/(t^2)$ *is not an etale morphism. In general, classify the etale morphisms to* $\operatorname{Spec} k$ *and and to* $\operatorname{Spec} \overline{k}[t]/(t^2)$.

Solution 1.115. *The morphism*

$$\operatorname{Spec} k \to \operatorname{Spec} k[t]/(t^2)$$

corresponds to the quotient map

$$A := k[t]/(t^2) \to k = A/(t).$$

This is not étale because it is not flat: over the local ring A, *the module* k *is not free, hence not flat.*

More generally, an étale morphism to $\operatorname{Spec} k$ *is the same as a finite étale* k*-scheme. Being etale, we can concklude that it is dimension zero, so it is a finite disjoint union of discrete points, each with residue field which must be a finite separable extension of* k *by etaleness. Hence the morphism is of the form*

$$\operatorname{Spec}\Big(\prod_{i=1}^{r} L_i\Big) \to \operatorname{Spec} k$$

with each L_i/k *a finite separable field extension.*

If $A = \overline{k}[t]/(t^2)$, *then every étale* A*-algebra* B *satisfies* B/tB *is finite étale over* $\overline{k} = A/(t)$ *because étaleness is preserved by base change. Therefore*

$$B/tB \cong \overline{k}^r.$$

Write

$$\overline{k}^r = \overline{k}e_1 \times \cdots \times \overline{k}e_r,$$

where the e_i *are the standard orthogonal idempotents:*

$$e_i^2 = e_i, \qquad e_i e_j = 0 \ (i \neq j), \qquad \sum_i e_i = 1.$$

Since $(t) \subset A$ is nilpotent, the ideal $tB \subset B$ is also nilpotent. Idempotents lift uniquely across nilpotent ideals, so each $e_i \in B/tB$ lifts to an idempotent $\widetilde{e}_i \in B$. These lifts remain orthogonal and sum to 1. Therefore

$$B \cong \prod_{i=1}^{r} B_i, \qquad B_i := \widetilde{e}_i B.$$

So we reduce to the case where B is connected, meaning $B/tB \cong \overline{k}$. Assume now B is connected étale over A. Because B is finite étale over A, it is finite projective over A. Since A is local, finite projective A-modules are free. Thus

$$B \cong A^n$$

as an A-module for some rank n.

Reduce modulo t implies that $n = 1$.

$$B/tB \cong (A^n)/t(A^n) \cong (A/t)^n \cong \overline{k}^n$$

So B is free of rank 1 as an A-module, hence $B \cong A$ as an A-algebra. Therefore the only étale morphisms to $\operatorname{Spec} \overline{k}[t]/(t^2)$ are finite disjoint unions of copies of $\operatorname{Spec} A$, i.e. of the form

$$\bigsqcup_{i=1}^{r} \operatorname{Spec} A \to \operatorname{Spec} A.$$

Thus the étale morphisms to $\operatorname{Spec} \overline{k}[t]/(t^2)$ are exactly the finite disjoint unions

$$\bigsqcup_{i=1}^{r} \operatorname{Spec} A \to \operatorname{Spec} A.$$

Note that the same argument shows the same conclusion holds for any local ring whose maximal ideal is nilpotent and whose residue field is algebraically closed. The general result relevant here is for Henselian local rings. let $(R, \mathfrak{m}, \kappa)$ be a henselian local ring. The category of finite étale ring extensions $R \to S$ is equivalent to the category of finite étale κ-algebras via the functor $S \mapsto S/\mathfrak{m}S$.

The connection is that nilpotent is about finite order liftings whereas Henselian is about infinite order liftings. Note that local rings with nilpotent maximal ideal are $\mathfrak{m}$-adically complete, and thus Henselian.

Remark 1.116. *Smooth morphisms are stable under base change and composition, and the same is true for étale morphisms.*

1.3 Quasicoherent sheaves

1.3.1 Definitions and basic properties

Definition 1.117. *Let X be a scheme. An $\mathcal{O}_X$-module $\mathcal{F}$ is* ***quasi-coherent*** *if for every affine open subset $U = \operatorname{Spec} A \subset X$, there exists an A-module M such that*

$$\mathcal{F}|_U \cong \widetilde{M},$$

where $\widetilde{M}$ is the sheaf associated to M.

Definition 1.118. *If M is an A-module, the associated sheaf $\widetilde{M}$ on* $\operatorname{Spec} A$ *is defined by*

$$\widetilde{M}(D(f)) = M_f.$$

Proposition 1.119. *Quasi-coherence is a local property: an $\mathcal{O}_X$-module $\mathcal{F}$ is quasi-coherent if and only if there exists an affine open cover $\{U_i\}$ of X such that $\mathcal{F}|_{U_i}$ is quasi-coherent for all i.*

Proposition 1.120. *The category of quasi-coherent sheaves on a scheme X is an abelian subcategory of the category of all $\mathcal{O}_X$-modules. In particular, it is closed under kernels, cokernels, and extensions.*

Definition 1.121. *A quasi-coherent sheaf $\mathcal{F}$ on a scheme X is* ***coherent*** *if:*

1. *$\mathcal{F}$ is of finite type (i.e. locally finitely generated), and*
2. *for every open $U \subset X$ and every morphism $\mathcal{O}_X^{\oplus n}|_U \to \mathcal{F}|_U$, the kernel is of finite type.*

Proposition 1.122. *If $X = \operatorname{Spec} A$ is affine, then:*

- *quasi-coherent sheaves correspond to A-modules,*
- *coherent sheaves correspond to finitely generated A-modules (when A is Noetherian).*

Remark 1.123. *On a Noetherian scheme, a quasi-coherent sheaf is coherent if and only if it is locally finitely generated.*

Example 1.124. *On any scheme X, the structure sheaf $\mathcal{O}_X$ is quasi-coherent (and in fact coherent).*

Example 1.125. *If $X = \operatorname{Spec} A$ is an affine scheme, if $Y \subseteq X$ is the closed subscheme defined by an ideal $\mathfrak{a} \subseteq A$ and if $i : Y \to X$ is the inclusion morphism, then $i_*(\mathcal{O}_Y)$ is a quasi-coherent (in fact coherent) $\mathcal{O}_X$-module. Indeed, it is isomorphic to $(A/\mathfrak{a})^\sim$.*

Example 1.126. *If U is an open subscheme of a scheme X, with inclusion map $j : U \to X$, then the sheaf $j_!(\mathcal{O}_U)$ obtained by extending $\mathcal{O}_U$ by zero outside of U, i.e. the sheafification of the presheaf*

$$j_!(\mathcal{O}_U)(V) = \begin{cases} \mathcal{O}_U(V) & V \subseteq U, \\ 0 & \textit{otherwise} \end{cases}$$

is an $\mathcal{O}_X$-module, but it is not in general quasi-coherent.

For example, suppose X is integral, and $V = \operatorname{Spec} A$ is any open affine subset of X not contained in U. Then $j_!(\mathcal{O}_U)|_V$ has no global sections over V because a section in

$j_!(\mathcal{O}_U)(V)$ must live in $\mathcal{O}_U(V \cap U)$ and vanish near $V \setminus U$, meaning that there exists an open $W \subset V$ with $V \setminus U \subset W$ so that $s|_{W \cap U} = 0$. But $W \cap U$ is an open subset of $V \cap U$, and so s vanishes on a nonempty open subset of an integral scheme, meaning $s = 0$. However, $j_!(\mathcal{O}_U)|_V$ is not the zero sheaf, since it has nonzero stalks. Note that the stalks of $j_!(\mathcal{O}_U)$ are the same as the stalks of $\mathcal{O}_U$. Hence it cannot be of the form $\widetilde{M}$ for any A-module M.

Example 1.127. *If Y is a closed subscheme of a scheme X, then the sheaf $\mathcal{O}_X|_Y$ is not in general quasi-coherent on Y. In fact, it is not even an $\mathcal{O}_Y$-module in general.*

Example 1.128. *Let X be an integral noetherian scheme, and let $\mathcal{K}$ be the constant sheaf with group K equal to the function field of X. Then $\mathcal{K}$ is a quasi-coherent $\mathcal{O}_X$-module, but it is not coherent unless X is reduced to a point.*

Proposition 1.129. *Let X be a scheme. The kernel, cokernel, and image of any morphism of quasi-coherent sheaves are quasi-coherent. Any extension of quasi-coherent sheaves is quasi-coherent. If X is noetherian, the same is true for coherent sheaves.*

1.3.2 Modules on affine and projective schemes

Theorem 1.130. *Let $X = \operatorname{Spec} A$ be an affine scheme. The functor*

$$M \longmapsto \widetilde{M}$$

defines an equivalence of categories between the category of A-modules and the category of quasi-coherent sheaves on X. Under this equivalence:

- $\Gamma(X, \widetilde{M}) \cong M$,
- *morphisms of sheaves correspond to module homomorphisms.*
- *For $f \in A$,*
$$\widetilde{M}(D(f)) \cong M_f.$$

Definition 1.131. *Let S be a graded ring and let M be a graded S-module. We define the sheaf associated to M on $\operatorname{Proj} S$, denoted by $\widetilde{M}$, as follows. For each $\mathfrak{p} \in \operatorname{Proj} S$, let $M_{(\mathfrak{p})}$ be the group of elements of degree 0 in the localization $T^{-1}M$, where T is the multiplicative system of homogeneous elements of S not in $\mathfrak{p}$. For any open subset $U \subseteq \operatorname{Proj} S$ we define $\widetilde{M}(U)$ to be the set of functions s from U to $\coprod_{\mathfrak{p} \in U} M_{(\mathfrak{p})}$ which are locally fractions. This means that for every $\mathfrak{p} \in U$, there is a neighborhood V of $\mathfrak{p}$ in U, and homogeneous elements $m \in M$ and $f \in S$ of the same degree, such that for every $\mathfrak{q} \in V$, we have $f \notin \mathfrak{q}$, and $s(\mathfrak{q}) = m/f$ in $M_{(\mathfrak{q})}$. We make $\widetilde{M}$ into a sheaf with the obvious restriction maps.*

Proposition 1.132. *Let S be a graded ring, and M a graded S-module. Let $X = \operatorname{Proj} S$.*

(a) For any $\mathfrak{p} \in X$, the stalk $(\widetilde{M})_{\mathfrak{p}} = M_{(\mathfrak{p})}$.

(b) *For any homogeneous $f \in S_+$, we have*

$$\widetilde{M}|_{D_+(f)} \cong (M_{(f)})^\sim$$

via the isomorphism of $D_+(f)$ with $\operatorname{Spec} S_{(f)}$ *where $M_{(f)}$ denotes the group of elements of degree* 0 *in the localized module M_f.*

(c) *$\widetilde{M}$ is a quasi-coherent $\mathcal{O}_X$-module. If S is noetherian and M is finitely generated, then $\widetilde{M}$ is coherent.*

This constructions gives a functor from the category of graded S-modules to the category of quasi-coherent $\mathcal{O}_X$-modules. It is not an equivalence of categories, but it becomes one once we restrict to the full subcategory of graded S-modules which are **saturated**.

Definition 1.133. *Let $\mathcal{F}$ be a quasi-coherent sheaf on* $\operatorname{Proj} S$. *Then we define the graded S-module $\Gamma_*(\mathcal{F})$ by*

$$\Gamma_*(\mathcal{F}) = \bigoplus_{n\in\mathbb{Z}} \Gamma(X, \mathcal{F}(n)).$$

*A module M is **saturated** if the natural map $M \to \Gamma_*(\widetilde{M})$, which takes $m \in M_n$ to the global section $s_m \in \Gamma(X, \widetilde{M}(n))$ which on a standard open $D_+(f)$, $s_m|_{D_+(f)} = \frac{m}{1} \in (M_f)_0$, is an isomorphism.*

Theorem 1.134. *The functor $M \mapsto \widetilde{M}$ defines an equivalence of categories between the category of saturated graded S-modules and the category of quasi-coherent $\mathcal{O}_X$-modules on $X = \operatorname{Proj} S$.*

It is important to remember that two graded S-modules can give rise to the same sheaf on $\operatorname{Proj} S$. However there is a best representative of the isomorphism class of a sheaf, which is the saturation of any module giving rise to that sheaf. Also note that two ideal sheaves on $\operatorname{Proj} S$ cut out the same closed subscheme if and only if they have the same saturation.

Example 1.135. *Let*

$$S = k[x_0, x_1], \qquad S_+ = (x_0, x_1), \qquad M = S/(x_0, x_1).$$

Then every element of M is killed by S_+, so M is entirely S_+-torsion. On any standard open $D_+(f)$ with $f \in S_+$ homogeneous, we have

$$\widetilde{M}(D_+(f)) = (M_f)_0 = 0,$$

because f acts by 0 *on M, so localizing at f kills the module. Hence*

$$\widetilde{M} = 0.$$

Consequently,

$$\Gamma_*(\widetilde{M}) = 0,$$

so the saturation of M is zero. Geometrically, M is supported at the irrelevant ideal S_+, which is invisible on $\operatorname{Proj} S$.

Definition 1.136. *Let S be a graded ring, and let $X = \operatorname{Proj} S$. For any $n \in \mathbb{Z}$, we define the sheaf $\mathcal{O}_X(n)$ to be $S(n)^\sim$. We call $\mathcal{O}_X(1)$* ***the twisting sheaf of Serre****. For any sheaf of $\mathcal{O}_X$-modules, $\mathcal{F}$, we denote by $\mathcal{F}(n)$ the twisted sheaf $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_X(n)$.*

Proposition 1.137. *Let S be a graded ring and let $X = \operatorname{Proj} S$. Assume that S is generated by S_1 as an S_0-algebra. The sheaf $\mathcal{O}_X(n)$ is locally free of rank 1 for every $n \in \mathbb{Z}$.*

We end this section with some special results about sheaves on a projective scheme over a noetherian ring.

Definition 1.138. *Let X be a scheme and let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$-modules. We say that $\mathcal{F}$ is* ***generated by global sections*** *if there is a family of global sections $\{s_i\}_{i\in I}$, $s_i \in \Gamma(X, \mathcal{F})$, such that for each $x \in X$, the images of s_i in the stalk $\mathcal{F}_x$ generate that stalk as an $\mathcal{O}_{X,x}$-module.*

Note that $\mathcal{F}$ is generated by global sections if and only if $\mathcal{F}$ can be written as a quotient of a free sheaf. Indeed, the generating sections $\{s_i\}_{i\in I}$ define a surjective morphism of sheaves

$$\bigoplus_{i\in I} \mathcal{O}_X \longrightarrow \mathcal{F},$$

and conversely.

Example 1.139. *Any quasi-coherent sheaf on an affine scheme is generated by global sections. Indeed, if $\mathcal{F} = \widetilde{M}$ on $\operatorname{Spec} A$, any set of generators for M as an A-module will do.*

Example 1.140. *Let $X = \operatorname{Proj} S$, where S is a graded ring which is generated by S_1 as an S_0-algebra. Then the elements of S_1 give global sections of $\mathcal{O}_X(1)$ which generate it.*

Theorem 1.141 (Serre)**.** *Let X be a projective scheme over a noetherian ring A, let $\mathcal{O}_X(1)$ be a very ample invertible sheaf on X, and let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module. Then there is an integer n_0 such that for all $n \geq n_0$, the sheaf $\mathcal{F}(n)$ can be generated by a finite number of global sections.*

Theorem 1.142. *Let k be a field, let A be a finitely generated k-algebra, let X be a projective scheme over A, and let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module. Then $\Gamma(X, \mathcal{F})$ is a finitely generated A-module. In particular, if $A = k$, $\Gamma(X, \mathcal{F})$ is a finite-dimensional k-vector space.*

1.3.3 Operations: tensor product, Hom, pullback, pushforward

Definition 1.143. *If $\mathcal{F}, \mathcal{G}$ are $\mathcal{O}_X$-modules, their tensor product is*

$$\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}.$$

If both are quasi-coherent, then so is the tensor product. The sheaf $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F},\mathcal{G})$ assigns to each open set U the group

$$\mathrm{Hom}_{\mathcal{O}_U}(\mathcal{F}|_U, \mathcal{G}|_U).$$

Definition 1.144. *Let $f : X \to Y$ be a morphism of schemes. The **pullback** of $\mathcal{F}$ is*

$$f^*\mathcal{F} = f^{-1}\mathcal{F} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X.$$

*The **pushforward** is*

$$f_*\mathcal{F}(U) = \mathcal{F}(f^{-1}(U)).$$

Proposition 1.145. *If $f : X \to Y$ is a morphism of schemes, then f^* preserves quasi-coherence. If f is quasi-compact and separated, then f_* preserves quasi-coherence. It is not true in general that f_* preserves coherence, even if X and Y are noetherian. However if f is proper and X and Y are noetherian, then f_* does preserve coherence.*

Example 1.146. *Quasicoherent pushforward can fail without the quasi-compact and separated hypothesis. Let $X_i \cong \operatorname{Spec}\mathbb{Z}$ for $i \in \mathbb{N}$, and set*

$$X = \bigsqcup_{i\in\mathbb{N}} X_i, \qquad S = \operatorname{Spec}\mathbb{Z}.$$

Let $f : X \to S$ be the morphism induced by the identities on each component. We show that $f_*\mathcal{O}_X$ is not quasi-coherent.

Recall that for a quasi-coherent sheaf $\mathcal{F}$ on S, the natural map

$$m_U : \mathcal{F}(S) \otimes_{\mathcal{O}_S(S)} \mathcal{O}_S(U) \to \mathcal{F}(U)$$

is an isomorphism for every affine open $U \subset S$.

Take $U = D(2) \subset S$. Then

$$f_*\mathcal{O}_X(S) = \prod_{i\in\mathbb{N}} \mathbb{Z}, \qquad f_*\mathcal{O}_X(U) = \prod_{i\in\mathbb{N}} \mathbb{Z}\left[\tfrac{1}{2}\right].$$

Thus the map becomes

$$\left(\prod_{i\in\mathbb{N}} \mathbb{Z}\right) \otimes_{\mathbb{Z}} \mathbb{Z}\left[\tfrac{1}{2}\right] \longrightarrow \prod_{i\in\mathbb{N}} \mathbb{Z}\left[\tfrac{1}{2}\right].$$

This map is not surjective: elements in the image have a uniform bound on the power of 2 in denominators. This is because $\mathbb{Z}[1/2]$ is a filtered colimit of free $\mathbb{Z}$-modules $\mathbb{Z}[1/2] = \varinjlim_n 2^{-n}\mathbb{Z}$, so any element comes from some fixed n. However elements in the codomain like

$$\left(\frac{1}{2^i}\right)_{i\in\mathbb{N}} \in \prod_{i\in\mathbb{N}} \mathbb{Z}\left[\tfrac{1}{2}\right]$$

do not. Hence m_U is not surjective, so $f_*\mathcal{O}_X$ is not quasi-coherent.

Proposition 1.147. *The functor on sheaves of abelian groups f^{-1} is exact. The functor on $\mathcal{O}_X$-modules f^* is exact if f is flat. The functor f_* is left exact, but not in general right exact.*

The failure of f_* to be right exact leads to the need for sheaf cohomology, which measures the failure of exactness of f_*.

Pushforward and pullback are adjoint in the following sense.

Proposition 1.148. *For any $\mathcal{O}_X$-module $\mathcal{F}$ and any $\mathcal{O}_Y$-module $\mathcal{G}$, there is a natural isomorphism*

$$\mathrm{Hom}_{\mathcal{O}_X}(f^*\mathcal{G}, \mathcal{F}) \cong \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{G}, f_*\mathcal{F})$$

Proof. On stalks we have

$$\mathrm{Hom}_{\mathcal{O}_{X,x}}\Big((f^*\mathcal{G})_x, \mathcal{F}_x\Big) \cong \mathrm{Hom}_{\mathcal{O}_{Y,y}}(\mathcal{G}_y, \mathcal{F}_x).$$

Note that the formula for the stalk of the pushforward is

$$(f_*\mathcal{F})_y \cong \varinjlim_{y\in U} \mathcal{F}(f^{-1}(U))$$

which in general only maps to $\mathcal{F}_x$ for $x \in f^{-1}(y)$, but since the statement is local on X we can just pass to $\mathcal{F}_x$.s Since

$$(f^*\mathcal{G})_x \cong \mathcal{G}_y \otimes_{\mathcal{O}_{Y,y}} \mathcal{O}_{X,x},$$

this becomes the familiar extension–restriction of scalars adjunction

$$\mathrm{Hom}_{\mathcal{O}_{X,x}}\Big(\mathcal{G}_y \otimes_{\mathcal{O}_{Y,y}} \mathcal{O}_{X,x}, \mathcal{F}_x\Big) \cong \mathrm{Hom}_{\mathcal{O}_{Y,y}}(\mathcal{G}_y, \mathcal{F}_x)$$

Then the statement becomes the familiar identification

$$\mathrm{Hom}_B(M \otimes_A B, N) \;\cong\; \mathrm{Hom}_A(M, N)$$

which is obviously true. □

1.3.4 Locally free sheaves and vector bundles

Definition 1.149. *An $\mathcal{O}_X$-module $\mathcal{F}$ is **locally free of rank** r if there exists an open cover $\{U_i\}$ such that*

$$\mathcal{F}|_{U_i} \cong \mathcal{O}_{U_i}^{\oplus r}.$$

*A **vector bundle** on X is a locally free sheaf of finite rank.*

If $\mathcal{F}$ and $\mathcal{G}$ are locally free of finite rank, then so is $\mathcal{H}om(\mathcal{F}, \mathcal{G})$, $\mathcal{F} \otimes \mathcal{G}$, and the dual $\mathcal{F}^\vee = \mathcal{H}om(\mathcal{F}, \mathcal{O}_X)$.

Proposition 1.150. *If $X = \operatorname{Spec} A$, then locally free sheaves correspond to projective A-modules.*

Proposition 1.151. *If A is a local ring, then every projective A-module is free. Hence locally free sheaves over* $\operatorname{Spec} A$ *are free.*

Exercise 1.152. *Let $f : X \to Y$ be a morphism of schemes.*

(a) Show by example that if $\mathcal{F}$ is coherent on X, then $f_*\mathcal{F}$ need not be coherent on Y, even if X and Y are varieties over a field k.

(b) If f is a finite morphism of noetherian schemes, and if $\mathcal{F}$ is coherent on X, then $f_*\mathcal{F}$ is coherent on Y.

Solution 1.153.

(a) Consider $\operatorname{Spec} k[t] \to \operatorname{Spec} k$ *and the coherent sheaf $\mathcal{O}_{\operatorname{Spec} k[t]}$ on* $\operatorname{Spec} k[t]$. *The pushforward of $\mathcal{O}_{\operatorname{Spec} k[t]}$ to* $\operatorname{Spec} k$ *is the quasi-coherent sheaf corresponding to the k-module $k[t]$. This is not a finite-dimensional k-vector space, hence not a coherent sheaf on* $\operatorname{Spec} k$.

(b) This is local on Y, so assume

$$Y = \operatorname{Spec} A, \qquad X = \operatorname{Spec} B,$$

where B is a finite A-algebra. Let $\mathcal{F}$ be coherent on X. Since X is noetherian affine, $\mathcal{F}$ is of the form

$$\mathcal{F} \cong \widetilde{M}$$

for some finitely generated B-module M.

Now for any A-module N, the associated quasi-coherent sheaf on Y is $\widetilde{N}$. In the affine case, pushforward along $\operatorname{Spec} B \to \operatorname{Spec} A$ *sends $\widetilde{M}$ to $\widetilde{M}$, where M is viewed as an A-module via $A \to B$. More precisely, for any distinguished open $D(f) \subseteq \operatorname{Spec} A$,*

$$f_*\widetilde{M}(D(f)) = \widetilde{M}(\operatorname{Spec} B \times_{\operatorname{Spec} A} D(f)) = \widetilde{M}(\operatorname{Spec} B_f) = M_f.$$

Thus

$$f_*\mathcal{F} \cong \widetilde{M}$$

as a quasi-coherent sheaf on $Y = \operatorname{Spec} A$.

It remains to show that M is finitely generated as an A-module. Since B is finite over A, choose $b_1, \ldots, b_r \in B$ generating B as an A-module. Since M is finitely generated as a B-module, choose $m_1, \ldots, m_s \in M$ generating M over B. Then every element of M is a B-linear combination of the m_j, hence an A-linear combination of the finitely many elements

$$b_i m_j \qquad (1 \le i \le r,\ 1 \le j \le s).$$

So M is finitely generated as an A-module. Therefore $f_*\mathcal{F} \cong \widetilde{M}$ is a coherent sheaf on Y.

One may ask when the pushforward of a vector bundle is a vector bundle. Knowing the morphism is proper is not enough, as the following example shows. The answer is Grauert's theorem.

Example 1.154. *Properness alone does not guarantee that the pushforward of a vector bundle is locally free. Let*

$$i : \operatorname{Spec} k \hookrightarrow \mathbb{A}^1_k$$

be the inclusion of the origin. This is a proper morphism, since every closed immersion is proper. The structure sheaf

$$\mathcal{O}_{\operatorname{Spec} k}$$

is a vector bundle of rank 1 *on* $\operatorname{Spec} k$*. However,*

$$i_* \mathcal{O}_{\operatorname{Spec} k}$$

is the skyscraper sheaf at the origin on $\mathbb{A}^1_k$*, which is not locally free.*

Theorem 1.155 (Grauert)**.** *Let* $\pi : X \to S$ *be a proper flat morphism, with* S *reduced and locally Noetherian, and let* $\mathcal{F}$ *be a coherent sheaf on* X *flat over* S*, meaning that for every* x*, the stalk* $\mathcal{F}_x$ *is a flat* $\mathcal{O}_{S,\pi(x)}$*-module via the structure morphism* $\mathcal{O}_{S,\pi(x)} \to \mathcal{O}_{X,x}$*.*

Suppose that for some $p \geq 0$ *the function*

$$s \longmapsto h^p(X_s, \mathcal{F}_s)$$

is locally constant on S*.*

Then:

1. $R^p\pi_*\mathcal{F}$ *is locally free on* S*;*
2. *for every morphism* $f : T \to S$*, the natural base change map*

$$f^*(R^p\pi_*\mathcal{F}) \longrightarrow R^p\pi'_*(\mathcal{F}_T)$$

is an isomorphism, where $\pi' : X_T := X \times_S T \to T$*.*

In particular, every morphism of smooth projective curves over a field is proper and flat, so the pushforward of a vector bundle is always a vector bundle. This could also be seen by a direct argument. Since the local rings of a smooth curve are DVRs, it suffices to show that $f_*\mathcal{E}$ is torsion-free. This follows from the constancy of the function

$$y \mapsto h^0(X_y, \mathcal{E}_y),$$

since any torsion would cause the fiber dimension to jump. Hence $f_*\mathcal{E}$ is locally free. This function is constant because we are in the curve case, so the fibers are zero-dimensional, and the Euler characteristic is constant in flat families. Also we have the vanishing of $R^1 f_*$ because f is affine.

Theorem 1.156 (Projection formula). *Let $f : X \to Y$ be a morphism of schemes, let $\mathcal{F}$ be a quasi-coherent sheaf on X, and let $\mathcal{E}$ be a locally free sheaf on Y. Then there is a natural isomorphism*

$$f_*(\mathcal{F} \otimes f^*\mathcal{E}) \cong f_*\mathcal{F} \otimes \mathcal{E}.$$

Example 1.157 (Projection formula on an affine chart). *Let $A \to B$ be a ring map, corresponding to a morphism*

$$f : \operatorname{Spec} B \to \operatorname{Spec} A.$$

If M is a B-module and N an A-module, then the projection formula says

$$f_*(\widetilde{M} \otimes f^*\widetilde{N}) \cong f_*\widetilde{M} \otimes \widetilde{N}.$$

On global sections this becomes an isomorphism of A-modules

$$\operatorname{Res}_A^B(M \otimes_B (B \otimes_A N)) \cong \operatorname{Res}_A^B(M) \otimes_A N$$

where the left hand side obviously simplifies to $\operatorname{Res}_A^B(M \otimes_A N)$. Hence the projection formula is just the statement that restriction of scalars commutes with tensor product.

1.4 Divisors

1.4.1 Weil divisors

Let X be an Noetherian integral separated scheme which is regular in codimension one, meaning that the local ring at every codimension one point is a regular local ring. We need the regularity assumption to ensure that the local ring at every codimension one point is a discrete valuation ring. Recall that the notion of a 1-dimensional regular local ring is precisely a discrete valuation ring.

The most important examples of such schemes are nonsingular varieties over a field and Noetherian normal schemes. Recall that a variety over a field is a separated integral (reduced and irreducible) scheme of finite type. On a nonsingular variety, the local ring at every closed point is regular, hence all local rings are regular, since they are localizations of local rings at closed points. In particular, every $x \in X$ lies in the closure of some closed point $y \in X$. Then $\mathcal{O}_{X,x} = (\mathcal{O}_{X,y})_{\mathfrak{p}_x}$ and now use the fact that regularity is preserved by localization. A Noetherian ring is **regular** if every localization at a prime ideal is a regular local ring. A Noetherian local ring $(R, \mathfrak{m})$ is **regular** if and only if $\dim R = \dim_{R/\mathfrak{m}} \mathfrak{m}/\mathfrak{m}^2$ where $\dim R$ is the Krull dimension of R. It is a theorem that a regular local ring is a UFD, so in particular integrally closed.

On a Noetherian normal scheme, any local ring of dimension one is an integrally closed domain, hence a discrete valuation ring, and therefore regular. Recall that a scheme is **normal** if all of its local rings are normal, i.e. integrally closed domains. A fact is that an affine variety X is normal if and only if its coordinate ring is a normal domain, i.e. a domain integrally closed in its field of fractions. For a Noetherian scheme, if x has codimension one, then $\dim \mathcal{O}_{X,x} = 1$ by a standard argument about chains of prime ideals. The key commutative algebra fact is that if A is a Noetherian local domain of dimension 1 and A is integrally closed, then A is a discrete valuation ring. A DVR has principal maximal ideal, so $\dim_{A/\mathfrak{m}} \mathfrak{m}/\mathfrak{m}^2 = 1$ is regular local ring.

Recall that an integral scheme has the very important property that if two rational functions agree on a nonempty open subset, then they are equal. This is the same thing as saying that $\mathcal{K}_X$ is constant or $K(U) = K(X)$ for all nonempty U. Take $f, g \in K(X)$. Assume $f|_U = g|_U$. Then $h := f - g$ restricts to 0 on U. Since X is integral, every rational function is represented on some nonempty affine open $\operatorname{Spec} A$ by an element of $\operatorname{Frac}(A)$, and if it vanishes on a nonempty open subset of $\operatorname{Spec} A$, then it is zero in $\operatorname{Frac}(A)$. Indeed, write

$$h = \frac{a}{b} \in \operatorname{Frac}(A), \qquad b \neq 0.$$

If h vanishes on a nonempty distinguished open $D(s) \subseteq U \cap \operatorname{Spec} A$, then $a/1 = 0$ in A_s, so for some n,

$$s^n a = 0.$$

Since A is a domain and $s \neq 0$, this implies $a = 0$, hence $h = 0$. Thus $f = g$.

Exercise 1.158. *Can normality be checked on an affine open cover? If X is a normal scheme and $\operatorname{Spec} A \subset X$ is an affine open, does it follow that A is a normal domain?*

Solution 1.159. *A scheme X is normal if and only if all local rings $\mathcal{O}_{X,x}$ are normal domains. Since this is a local condition on stalks, it can be checked on any open cover, in particular on an affine open cover.*

If $\operatorname{Spec} A \subseteq X$ is an open affine subscheme, then A is a normal ring. However, A need not be a domain unless U is irreducible, or unless X itself is integral.

Normality for a scheme usually means all local rings are normal domains, but the scheme need not be irreducible. A disconnected union of normal varieties is normal, and an affine open in it can have a ring that is normal but not a domain.

Definition 1.160. *A **Weil divisor** on X is a formal finite sum*

$$D = \sum_{Z \subset X} n_Z [Z],$$

*where the sum runs over irreducible closed subsets Z of codimension 1, and $n_Z \in \mathbb{Z}$. A Weil divisor D is **effective** if all $n_Z \geq 0$.*

Definition 1.161. *Let $K(X)$ be the function field of X. For $f \in K(X)^\times$, the **principal divisor** associated to f is*

$$\operatorname{div}(f) = \sum_Z \operatorname{ord}_Z(f)\,[Z],$$

*where $\operatorname{ord}_Z(f)$ is the valuation at the codimension 1 point corresponding to Z. Two Weil divisors D_1, D_2 are **linearly equivalent** if*

$$D_1 - D_2 = \operatorname{div}(f)$$

*for some $f \in K(X)^\times$. The **class group** $\operatorname{Cl}(X)$ is the group of Weil divisors modulo linear equivalence.*

Proposition 1.162. *Let A be a Noetherian domain. Then A is a unique factorization domain if and only if $X = \operatorname{Spec} A$ is normal and $\operatorname{Cl}(X) = 0$.*

Proof. In the forward direction, first we check that if A is a UFD, then A is integrally closed. Let A be a UFD, and let $K = \operatorname{Frac}(A)$. Suppose $x \in K$ is integral over A. Write

$$x = \frac{a}{b}$$

with $a, b \in A$ and $\gcd(a, b) = 1$. Since x is integral over A, it satisfies a monic polynomial

$$x^n + c_{n-1}x^{n-1} + \cdots + c_0 = 0 \qquad \text{with } c_i \in A.$$

Multiplying by b^n gives

$$a^n + c_{n-1}a^{n-1}b + \cdots + c_0 b^n = 0,$$

so

$$a^n = -b(c_{n-1}a^{n-1} + \cdots + c_0 b^{n-1}).$$

Thus $b \mid a^n$. Since A is a UFD and $\gcd(a, b) = 1$, this forces b to be a unit. Hence $x \in A$. Therefore A is integrally closed.

Assume R is a UFD and let $\mathfrak{p}$ be a height 1 prime ideal. Take $x \in \mathfrak{p}$ nonzero and let $x = a_1 \cdots a_n$ be a factorization into irreducibles. Since $\mathfrak{p}$ is prime we see that $a_i \in \mathfrak{p}$ for some i. The ideal (a_i) is prime. Since $\mathfrak{p}$ has height 1 we conclude that $(a_i) = \mathfrak{p}$.

Conversely, assume R is a domain so that every height 1 prime ideal is principal. Since R is Noetherian every nonzero nonunit element x has a factorization into irreducibles. Now recall that if R is a domain and every nonzero, nonunit factors into irreducibles, then R is a UFD if and only if every irreducible element is prime. So it suffices to show that every irreducible element is prime. Let $\mathfrak{p}$ be a prime minimal over (x). Then $\mathfrak{p}$ has height 1. This is because if $\mathfrak{p}$ is minimal over (x), then the only prime ideal of $R_\mathfrak{p}$ containing x is the maximal ideal $\mathfrak{p}R_\mathfrak{p}$. This is true because the primes of $R_\mathfrak{p}$ correspond 1-to-1 with the primes of R contained in $\mathfrak{p}$. Then $\dim(R_\mathfrak{p}) = 1$ if x is not nilpotent in $R_\mathfrak{p}$.

This is because if R is a local Noetherian ring of dimension 1, then $\dim R = 1$ if and only if there exists $x \in \mathfrak{m}$ such that x is not nilpotent and $V(x) = \{\mathfrak{q} \in \operatorname{Spec} R : x \in \mathfrak{q}\}$ is precisely the closed point $\{\mathfrak{m}\}$. Now apply this fact and the minimality of $\mathfrak{p}$, along with the fact that $\dim R_\mathfrak{p} = \operatorname{ht}(\mathfrak{p})$.

So we see that $\mathfrak{p}$ is height one and therefore by assumption $\mathfrak{p} = (y)$. Hence $x = yz$ and z is a unit as x is irreducible. Thus $(x) = (y)$ and we see that x is prime.

So what we must show is that if A is an integrally closed domain, then every prime ideal of height 1 is principal if and only if $\operatorname{Cl}(\operatorname{Spec} A) = 0$.

One direction is easy: if every prime ideal of height 1 is principal, consider a prime divisor $Y \subseteq X = \operatorname{Spec} A$. Then Y corresponds to a prime ideal $\mathfrak{p}$ of height 1. If $\mathfrak{p}$ is generated by an element $f \in A$, then clearly the divisor of f is

$$(f) = Y.$$

Thus every prime divisor is principal, so $\operatorname{Cl}(X) = 0$.

For the converse, suppose $\mathrm{Cl}(X) = 0$. Let $\mathfrak{p}$ be a prime ideal of height 1, and let Y be the corresponding prime divisor. Then there is an $f \in K$, the quotient field of A, with

$$(f) = Y.$$

We will show that in fact $f \in A$ and that f generates $\mathfrak{p}$. Since $v_Y(f) = 1$, we have $f \in A_{\mathfrak{p}}$, and f generates $\mathfrak{p}A_{\mathfrak{p}}$. If $\mathfrak{p}' \subset A$ is any other prime ideal of height 1, then $\mathfrak{p}'$ corresponds to a prime divisor $Y' \subset X$, and $v_{Y'}(f) = 0$, so $f \in A_{\mathfrak{p}'}$.

Now the algebraic result below implies that $f \in A$ In fact,

$$f \in A \cap \mathfrak{p}A_{\mathfrak{p}} = \mathfrak{p}$$

Now to show that f generates $\mathfrak{p}$, let g be any other element of $\mathfrak{p}$. Then $v_Y(g) \geq 1$ and $v_{Y'}(g) \geq 0$ for all $Y' \neq Y$. Hence

$$v_{Y'}(g/f) \geq 0$$

for all prime divisors Y' (including Y). Thus

$$g/f \in A_{\mathfrak{p}'}$$

for all $\mathfrak{p}'$ of height 1, so by the algebraic result again, $g/f \in A$ In other words, $g \in Af$, which shows that $\mathfrak{p}$ is a principal ideal, generated by f. $\square$

We have the following algebraic result which is often referred to as "algebraic Hartogs' lemma". It is saying that regularity can be checked at the generic points of all prime divisors.

Theorem 1.163. *Let A be an integrally closed Noetherian domain. Then*

$$A = \bigcap_{\mathrm{ht}(\mathfrak{p})=1} A_{\mathfrak{p}},$$

where the intersection is taken over all prime ideals of height 1. Moreover, if A is normal, then each $A_{\mathfrak{p}}$ is a discrete valuation ring.

Remark 1.164. *I was originally confused about what this lemma is really saying. If A is dimension 1, then the statement is completely obvious because every prime ideal is either zero or maximal. This is because if $\mathfrak{p}$ is a nonzero prime ideal, then $\mathfrak{p}$ is properly contained in a maximal ideal $\mathfrak{m}$, and the chain $0 \subset \mathfrak{p} \subseteq \mathfrak{m}$ would witness that $\dim A \geq 2$, a contradiction. Then in dimension 1, the theorem says that*

$$A = \bigcap_{\mathfrak{p} \text{ of height } 1} A_{\mathfrak{p}} = \bigcap_{\mathfrak{m} \text{ maximal}} A_{\mathfrak{m}}$$

which is true for all Noetherian domains, not just integrally closed ones. The reason is as follows. Let $x \in \bigcap_{\mathfrak{m}} A_{\mathfrak{m}}$. Define

$$I = \{a \in A : ax \in A\}.$$

This is an ideal of A, and $I \neq 0$ because $x \in K$, so if $x = b/c$ with $b, c \in A$, $c \neq 0$, then $c \in I$.

Now suppose $x \notin A$. Then $1 \notin I$, so I is a proper ideal. Hence $I \subseteq \mathfrak{m}$ for some maximal ideal $\mathfrak{m}$. But $x \in A_{\mathfrak{m}}$, so we can write

$$x = \frac{u}{s} \qquad \text{with } u \in A,\ s \notin \mathfrak{m}.$$

Then

$$sx = u \in A,$$

so by definition of I,

$$s \in I \subseteq \mathfrak{m}.$$

This contradicts $s \notin \mathfrak{m}$. Therefore I cannot be proper, so $I = A$. Hence $1 \in I$, which means that $x \in A$.

Remark 1.165. *In general, recall that we have the notion of a Dedekind domain, which is a Noetherian integrally closed domain of dimension 1. Let C be a nonsingular geometrically integral affine algebraic curve over a field k. Then the coordinate ring $k[C]$ of regular functions on C is a Dedekind domain, regardless if C is smooth.*

Another example of a Dedekind domain is the ring of integers $\mathcal{O}_K$ of a number field K. The prime ideals of $\mathcal{O}_K$ correspond to the closed points of $\operatorname{Spec} \mathcal{O}_K$, because every nonzero prime ideal is maximal. An interesting example here to study is $\mathbb{Z}[\sqrt{5}]$ which is not normal because $5 \cong 1 \pmod 4$ and the element $(1+\sqrt{5})/2$ is integral over $\mathbb{Z}[\sqrt{5}]$ but not in $\mathbb{Z}[\sqrt{5}]$. Even though it is not normal, the conclusion of the theorem still holds, because it is a Noetherian domain of dimension 1. Another funny example to study here is $\mathbb{Z}[\sqrt{-5}]$ which is normal but not a UFD, because the ideal $(2, 1+\sqrt{-5})$ is not principal. Here, we have a chain of ideals $(0) \subset (2) \subset (2, 1+\sqrt{-5}) \subset \mathcal{O}_K$, but the point is that $(2) = (2, 1+\sqrt{-5})^2$, is not prime, so the chain does not witness that $\dim \mathcal{O}_K \geq 2$.

Example 1.166. *If X is affine n-space $\mathbb{A}^n_k$ over a field k, then*

$$\operatorname{Cl}(X) = 0.$$

Indeed,

$$X = \operatorname{Spec} k[x_1, \ldots, x_n],$$

and the polynomial ring is a UFD.

Example 1.167. *If A is a Dedekind domain, then $\operatorname{Cl}(\operatorname{Spec} A)$ is just the ideal class group of A, as defined in algebraic number theory. Thus the first proposition generalizes the fact that A is a UFD if and only if its ideal class group is 0.*

Proposition 1.168. *Let X be the projective space $\mathbb{P}^n_k$ over a field k. For any divisor*

$$D = \sum n_i Y_i,$$

define the degree of D by

$$\deg D = \sum n_i \deg Y_i,$$

where $\deg Y_i$ is the degree of the hypersurface Y_i. Let H be the hyperplane $x_0 = 0$. Then:

(a) *if D is any divisor of degree d, then $D \sim dH$;*

(b) *for any $f \in K^\times$, $\deg(f) = 0$;*

(c) *the degree function gives an isomorphism*

$$\deg : \mathrm{Cl}(X) \xrightarrow{\sim} \mathbb{Z}.$$

Proof. Let $S = k[x_0, \ldots, x_n]$ be the homogeneous coordinate ring of X. If g is a homogeneous element of degree d, we can factor it into irreducible polynomials $g = g_1^{n_1} \cdots g_r^{n_r}$. Then g_i defines a hypersurface Y_i of degree $d_i = \deg g_i$, and we can define the divisor of g to be $(g) = \sum n_i Y_i$. Then $\deg(g) = d$.

Now a rational function f on X is a quotient g/h of homogeneous polynomials of the same degree. Clearly $(f) = (g) - (h)$, so we see that $\deg(f) = 0$, which proves (b).

If D is any divisor of degree d, we can write it as a difference $D_1 - D_2$ of effective divisors of degrees d_1, d_2 with $d_1 - d_2 = d$. Let $D_1 = (g_1)$ and $D_2 = (g_2)$. This is possible because an irreducible hypersurface in $\mathbb{P}^n$ corresponds to a homogeneous prime ideal of height 1 in S, which is principal. Taking power products we can get any effective divisor as (g) for some homogeneous g. Now $D - dH = (f)$ where $f = g_1/x_0^d g_2$ is a rational function on X. This proves (a). Statement (c) follows from (a), (b), and the fact that $\deg H = 1$. $\square$

Proposition 1.169. *Let X satisfy the hypotheses at the beginning of this section, and let Z be a proper closed subset of X, and let $U = X - Z$ be the complement. Then:*

(a) *there is a surjective homomorphism*

$$\mathrm{Cl}\, X \to \mathrm{Cl}\, U$$

defined by

$$D = \sum n_i Y_i \longmapsto \sum n_i (Y_i \cap U),$$

where we ignore those $Y_i \cap U$ which are empty;

(b) *if $\mathrm{codim}(Z, X) \geq 2$, then*

$$\mathrm{Cl}\, X \to \mathrm{Cl}\, U$$

is an isomorphism;

(c) *if Z is an irreducible subset of codimension 1, then there is an exact sequence*

$$\mathbb{Z} \longrightarrow \mathrm{Cl}\, X \longrightarrow \mathrm{Cl}\, U \longrightarrow 0,$$

where the first map is defined by

$$1 \longmapsto 1 \cdot Z.$$

Proof. (a) If Y is a prime divisor on X, then $Y \cap U$ is either empty or a prime divisor on U. If $f \in K^\times$, and

$$(f) = \sum n_i Y_i,$$

then considering f as a rational function on U, we have

$$(f)_U = \sum n_i(Y_i \cap U),$$

so indeed we have a homomorphism $\operatorname{Cl} X \to \operatorname{Cl} U$ It is surjective because every prime divisor of U is the restriction of its closure in X.

(b) The groups $\operatorname{Div} X$ and $\operatorname{Cl} X$ depend only on subsets of codimension 1, so removing a closed subset Z of codimension ≥ 2 does not change anything.

(c) The kernel of

$$\operatorname{Cl} X \to \operatorname{Cl} U$$

consists of divisors whose support is contained in Z. If Z is irreducible, the kernel is just the subgroup of $\operatorname{Cl} X$ generated by $1 \cdot Z$ $\square$

Example 1.170. *Let Y be an irreducible curve of degree d in $\mathbb{P}^2_k$. Then*

$$\operatorname{Cl}(\mathbb{P}^2 - Y) = \mathbb{Z}/d\mathbb{Z}.$$

Example 1.171. *Let k be a field, let*

$$A = k[x, y, z]/(xy - z^2),$$

and let

$$X = \operatorname{Spec} A.$$

Then X is an affine quadric cone in $\mathbb{A}^3_k$. We will show that

$$\operatorname{Cl}(X) \cong \mathbb{Z}/2\mathbb{Z},$$

and that it is generated by a ruling of the cone, say

$$Y: \ y = z = 0.$$

Note that Y is a prime divisor. To see this we check that X is integral so codimension-1 irreducible implies prime divisor, and that Y is irreducible closed of codimension 1. First, A is a domain since $xy - z^2$ is irreducible in $k[x, y, z]$. To see that Y is irreducible, we compute $A/\mathfrak{p} \cong k[x]$, which is a domain, so $\mathfrak{p}$ is prime and Y is irreducible. For codimension $\dim A = 2$ (hypersurface in $\mathbb{A}^3$) and $\dim A/\mathfrak{p} = 1$, so $\operatorname{ht}(\mathfrak{p}) = 1$ and Y has codimension 1.

We have an exact sequence

$$\mathbb{Z} \longrightarrow \operatorname{Cl}(X) \longrightarrow \operatorname{Cl}(X - Y) \longrightarrow 0,$$

where the first map sends $1 \mapsto 1 \cdot Y$. Now Y can be cut out set-theoretically by the function y. In fact, the divisor of y is

$$\operatorname{div}(y) = 2 \cdot Y,$$

because z generates the maximal ideal of the local ring at the generic point of Y. This is because $x \notin \mathfrak{p}$ so x becomes invertible in $A_\mathfrak{p}$ and so $y = z^2/x$ implies $\mathfrak{p}A_\mathfrak{p} = (y, z)A_\mathfrak{p} = zA_\mathfrak{p}$,

so z is a uniformizer at the generic point, and y has valuation 2 at the generic point. Since Y is set theoretically cut out by y, we have

$$X - Y = D(y) = \operatorname{Spec} A_y.$$

Now

$$A_y = k[x, y, y^{-1}, z]/(xy - z^2).$$

In this ring,

$$x = y^{-1}z^2,$$

so we can eliminate x, and find

$$A_y \cong k[y, y^{-1}, z].$$

This is a UFD and so $\mathrm{Cl}(X - Y) = 0$. Thus we see that $\mathrm{Cl}(X)$ is generated by Y, and that

$$2 \cdot Y = 0.$$

It remains to show that Y itself is not a principal divisor. Since A is integrally closed, it is equivalent to show that the prime ideal of Y, namely

$$\mathfrak{p} = (y, z),$$

is not principal. Here we are using the following fact to establish this equivalence. Let A be a Noetherian normal domain, and let $\mathfrak{p}$ be a height-1 prime. Then

$$Y = V(\mathfrak{p})$$

is a prime divisor. The claim is Y is a principal Weil divisor if and only if $\mathfrak{p}$ is a principal ideal. If $\mathfrak{p} = (f)$ for some $f \in A$, then $\mathrm{ord}_{\mathfrak{p}}(f) = 1$ and $\mathrm{ord}_{\mathfrak{p}'}(f) = 0$ for all other height-1 primes $\mathfrak{p}'$, so $\mathrm{div} f = Y$. Conversely, if Y is a principal Weil divisor, then $Y = (f)$ for some $f \in K(X)^\times$ and $\mathrm{ord}_{\mathfrak{p}}(f) = 1$ and $\mathrm{ord}_{\mathfrak{p}'}(f) = 0$ for all other height-1 primes $\mathfrak{p}'$, so $f \in A_{\mathfrak{p}}$ and f generates $\mathfrak{p}A_{\mathfrak{p}}$. Also f is a unit in $A_{\mathfrak{p}'}$ for all other height-1 primes $\mathfrak{p}'$.

Let $I = (f) \subset A$. Then for every height-1 prime $\mathfrak{q}$:

$$IA_{\mathfrak{q}} = \begin{cases} \mathfrak{p}A_{\mathfrak{p}} & \text{if } \mathfrak{q} = \mathfrak{p}, \\ A_{\mathfrak{q}} & \text{if } \mathfrak{q} \neq \mathfrak{p}. \end{cases}$$

But $\mathfrak{p}$ has the same property:

$$\mathfrak{p}A_{\mathfrak{q}} = \begin{cases} \mathfrak{p}A_{\mathfrak{p}} & \text{if } \mathfrak{q} = \mathfrak{p}, \\ A_{\mathfrak{q}} & \text{if } \mathfrak{q} \neq \mathfrak{p}. \end{cases}$$

In a normal domain,

$$I = \bigcap_{\mathrm{ht}(\mathfrak{q})=1} IA_{\mathfrak{q}}.$$

So:

$$(f) = \bigcap IA_{\mathfrak{q}} = \bigcap \mathfrak{p}A_{\mathfrak{q}} = \mathfrak{p}.$$

Let

$$\mathfrak{m} = (x, y, z),$$

and note that $\mathfrak{m}/\mathfrak{m}^2$ *is a* 3*-dimensional vector space over* k *generated by* $\bar{x}, \bar{y}, \bar{z}$*, the images of* x, y, z*. Now* $\mathfrak{p} \subseteq \mathfrak{m}$*, and the image of* $\mathfrak{p}$ *in* $\mathfrak{m}/\mathfrak{m}^2$ *contains* $\bar{y}$ *and* $\bar{z}$*. Hence* $\mathfrak{p}$ *cannot be a principal ideal.*

Example 1.172. *We compute* $\mathrm{Cl}(\mathbb{P}^1 \times \mathbb{P}^1) = \mathbb{Z} \oplus \mathbb{Z}$ *where the isomorphism is given by pulling back the hyperplane class from each factor. Let* $X = \mathbb{P}^1_k \times \mathbb{P}^1_k$*. Write* $H_1 = \{\infty\} \times \mathbb{P}^1$ *and* $H_2 = \mathbb{P}^1 \times \{\infty\}$*. Then* $U := X \setminus (H_1 \cup H_2) \cong \mathbb{A}^1 \times \mathbb{A}^1$*. Since* $k[x, y]$ *is a UFD, we have* $\mathrm{Cl}(U) = 0$*.*

Now use the standard exact sequence for removing codimension-1 components: if $Z \subset X$ *is a union of irreducible codimension-1 closed subsets, then*

$$\bigoplus_i \mathbb{Z}[Z_i] \longrightarrow \mathrm{Cl}(X) \longrightarrow \mathrm{Cl}(X \setminus Z) \longrightarrow 0$$

Applying this to $Z = H_1 \cup H_2$ *gives* $\mathbb{Z}[H_1] \oplus \mathbb{Z}[H_2] \longrightarrow \mathrm{Cl}(X) \longrightarrow 0$*, so* $\mathrm{Cl}(X)$ *is generated by* H_1 *and* H_2*. As for the injectivity, suppose that* $aH_1 + bH_2$ *is principal. Then since* X *is smooth, we can identify* $\mathrm{Cl}(X)$ *with the Picard group* $\mathrm{Pic}(X)$*, and the line bundle associated to* $aH_1 + bH_2$ *is* $\mathcal{O}_X(aH_1 + bH_2)$*. Then* $\mathcal{O}_X(aH_1 + bH_2)$ *is trivial, and pulling back along the inclusion* $\mathbb{P}^1 \times * \hookrightarrow X$ *gives a trivial line bundle on* $\mathbb{P}^1$*. Now recall that if* D *is a Cartier divisor on any scheme* X *and* $C \subset X$ *is a closed subscheme not contained in* $\mathrm{Supp}(D)$*, then*

$$\mathcal{O}_X(D)\Big|_C \cong \mathcal{O}_C(D|_C),$$

where $D|_C$ *is the induced Cartier divisor on* C*. The defining equation of* aH_1*, when restricted to* $\mathbb{P}^1 \times *$*, vanishes to order* a *at the point* ∞*. Therefore,* $\mathcal{O}_{\mathbb{P}^1}(aH_1|_{\mathbb{P}^1})$ *is trivial which implies* $a = 0$*. Similarly, pulling back along* $* \times \mathbb{P}^1$ *gives* $b = 0$*. Thus* H_1 *and* H_2 *are linearly independent in* $\mathrm{Cl}(X)$*, so* $\mathrm{Cl}(X) \cong \mathbb{Z} \oplus \mathbb{Z}$*.*

Example 1.173. *Consider a quadric surface* X *in* $\mathbb{P}^3$ *defined by a homogeneous degree 2 polynomial. We can identify this as the locus of singular matrices in* $\mathbb{P}^3$ *where we think of* $\mathbb{P}^3$ *as the projectivization of the space of* 2×2 *matrices. Then* X *is isomorphic to* $\mathbb{P}^1 \times \mathbb{P}^1$ *since every singular matrix has a unique 1-dimensional kernel and a unique 1-dimensional cokernel. The hyperplane class* $\mathcal{O}_{\mathbb{P}^3}(1)$ *restricts to* $\mathcal{O}_X(1, 1)$ *under this isomorphism, because the restriction to* $\mathbb{P}^1 \times *$ *is* $\mathcal{O}_{\mathbb{P}^1}(1)$*. This is saying that the composition* $\mathbb{P}^1 \hookrightarrow X \hookrightarrow \mathbb{P}^3$ *is a linear embedding. Equivalently, we have the Segre embedding* $\mathbb{P}^1 \times \mathbb{P}^1 \hookrightarrow \mathbb{P}^3$ *given by* $([x_0 : x_1], [y_0 : y_1]) \mapsto [x_0y_0 : x_0y_1 : x_1y_0 : x_1y_1]$ *which corresponds to the choice of line bundle* $\mathcal{O}_{\mathbb{P}^1}(1) \boxtimes \mathcal{O}_{\mathbb{P}^1}(1)$ *on* $\mathbb{P}^1 \times \mathbb{P}^1$*. This by definition is the pullback of* $\mathcal{O}_{\mathbb{P}^3}(1)$ *under the Segre embedding.*

1.4.2 Cartier divisors

Definition 1.174. *Let* X *be a scheme. A* ***Cartier divisor*** *is given by an open cover* $\{U_i\}$ *together with functions* $f_i \in \mathcal{O}_X(U_i)^\times$ *such that on overlaps*

$$f_i/f_j \in \mathcal{O}_X^\times(U_i \cap U_j).$$

Equivalently, a Cartier divisor is a global section of the sheaf of groups $K_X^\times/\mathcal{O}_X^\times$, where K_X is the sheaf of total quotient rings of $\mathcal{O}_X$. A Cartier divisor is ***principal*** *if it is given globally by a single rational function $f \in K(X)^\times$.*

Definition 1.175. *A Cartier divisor is* ***effective*** *if it can be represented by functions f_i that are regular and nonvanishing on U_i. In that case we define the associated subscheme of codimension 1, Y, to be the closed subscheme defined by the ideal sheaf $\mathcal{I}$ which is locally generated by f_i.*

This gives a correspondence between effective Cartier divisors and locally principal closed subschemes. The ideal sheaf $\mathcal{I}$ of the corresponding closed subscheme is $\mathcal{O}_X(-D)$ because we are thinking about functoins which vanish along D.

1.4.3 Relation between Weil and Cartier divisors

Proposition 1.176. *Let X be an integral, separated noetherian scheme, all of whose local rings are unique factorization domains (in which case we say X* ***is locally factorial****). Then the group $\operatorname{Div} X$ of Weil divisors on X is isomorphic to the group of Cartier divisors $\Gamma(X, \mathcal{K}^*/\mathcal{O}^*)$, and furthermore, the principal Weil divisors correspond to the principal Cartier divisors under this isomorphism.*

Proof. First note that X is normal, hence satisfies the conditions at the beginning of this section, since a UFD is integrally closed. Thus it makes sense to talk about Weil divisors. Since X is integral, the sheaf $\mathcal{K}$ is just the constant sheaf corresponding to the function field K of X. This is because the sheaf $\mathcal{K}$ is the sheaf which on an open set U invert the nonzerodivisors in $\mathcal{O}_X(U)$, and since X is integral, the nonzerodivisors are precisely the nonzero elements of K, so $\mathcal{K}(U) = K$ for every nonempty open set U. Note that on an integral scheme, if two rational functions agree on a nonempty open set, then they are equal.

Now let a Cartier divisor be given by $\{(U_i, f_i)\}$ where $\{U_i\}$ is an open cover of X, and $f_i \in \Gamma(U_i, \mathcal{K}^*) = K^*$. We define the associated Weil divisor as follows. For each prime divisor Y, take the coefficient of Y to be $\operatorname{ord}_Y(f_i)$, where i is any index for which $Y \cap U_i \neq \emptyset$. If j is another such index, then f_i/f_j is invertible on $U_i \cap U_j$, so $\operatorname{ord}_Y(f_i/f_j) = 0$ and $\operatorname{ord}_Y(f_i) = \operatorname{ord}_Y(f_j)$. Thus we obtain a well-defined Weil divisor

$$D = \sum \operatorname{ord}_Y(f_i)\, Y$$

on X. The sum is finite because X is noetherian.

Conversely, if D is a Weil divisor on X, let $x \in X$ be any point. Then D induces a Weil divisor D_x on the local scheme $\operatorname{Spec} \mathcal{O}_{X,x}$. Since $\mathcal{O}_{X,x}$ is a UFD, D_x is a principal divisor, so let $D_x = (f_x)$ for some $f_x \in K$. Now the principal divisor (f_x) on X has the same restriction to $\operatorname{Spec} \mathcal{O}_{X,x}$ as D, hence they differ only at prime divisors which do not pass through x. There are only finitely many of these which have a nonzero coefficient in D or (f_x), so there is an open neighborhood U_x of x such that D and (f_x) have the same restriction to U_x. Covering X with such open sets U_x, the functions f_x give a Cartier divisor on X.

Note that if f, f' give the same Weil divisor on an open set U, then f/f' has order zero at every prime divisor intersecting U, so f/f' is a unit in $\mathcal{O}_{X,\mathfrak{p}}$ for every $\mathfrak{p}$ of height 1 with $\mathfrak{p} \cap U \neq \emptyset$. Since X is normal, we have $\mathcal{O}_X(U) = \bigcap_{\substack{\mathfrak{p} \in U \\ \mathrm{ht}(\mathfrak{p})=1}} \mathcal{O}_{X,\mathfrak{p}} \subset K(X)$ and so we can conclude that $f/f' \in \mathcal{O}_X(U)^\times$. Thus the Cartier divisor we constructed from D is well-defined. These two constructions are inverse to each other, so we see that the groups of Weil divisors and Cartier divisors are isomorphic. Furthermore it is clear that the principal divisors correspond to each other. $\square$

Remark 1.177. *Since a regular local ring is a UFD, this proposition applies in particular to any regular integral separated noetherian scheme. A scheme is regular if all of its local rings are regular local rings.*

Remark 1.178. *If X is a normal scheme, which is not necessarily locally factorial, we can define a subgroup of* $\operatorname{Div} X$ *consisting of the locally principal Weil divisors: D is* locally principal *if X can be covered by open sets U such that $D|_U$ is principal for each U. Then the above proof shows that the Cartier divisors are the same as the locally principal Weil divisors.*

Example 1.179. *Let X be the affine quadric cone*

$$\operatorname{Spec} k[x, y, z]/(xy - z^2)$$

treated above. The ruling Y is a Weil divisor which is not locally principal in the neighborhood of the vertex of the cone. Indeed, our earlier proof shows that its prime ideal $\mathfrak{p}A_\mathfrak{m}$ is not a principal ideal even in the local ring $A_\mathfrak{m}$. Thus Y does not correspond to a Cartier divisor.

On the other hand $2Y$ is locally principal, and in fact principal. So in this case the group of Cartier divisors modulo principal divisors is 0, whereas

$$\operatorname{Cl} X \cong \mathbb{Z}/2\mathbb{Z}.$$

1.4.4 Divisors and line bundles

Definition 1.180. *To a Cartier divisor D, one associates an invertible sheaf $\mathcal{O}_X(D)$, defined by gluing the trivial line bundles on U_i via the transition functions f_i/f_j on overlaps. Explicitly, the sections of $\mathcal{O}_X(D)$ over an open set U are given by*

$$\mathcal{O}_X(D)(U) = \{s \in K(X) : \operatorname{div}(s|_{U_i \cap U}) + D|_{U_i \cap U} \geq 0 \text{ for all } i\}.$$

Note that if D is a principal Cartier divisor given by a global rational function f, then $\mathcal{O}_X(D) \cong \mathcal{O}_X$ via the isomorphism sending a section s to fs.

Proposition 1.181. *On any scheme X, the association $D \mapsto \mathcal{O}_X(D)$ gives an injective homomorphism from the group of Cartier divisors modulo principal divisors to the Picard group* $\operatorname{Pic}(X)$ *of isomorphism classes of line bundles on X. If X is integral, then this map is an isomorphism.*

Proof. The map $D \mapsto \mathcal{O}_X(D)$ maps into the invertible subsheaves of $\mathcal{K}$. Thus we have only to show that every invertible sheaf is isomorphic to a subsheaf of $\mathcal{K}$, which in this case is the constant sheaf K, where K is the function field of X.

So let $\mathcal{L}$ be any invertible sheaf, and consider the sheaf

$$\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{K}.$$

On any open set U where $\mathcal{L} \cong \mathcal{O}_X$, we have

$$\mathcal{L} \otimes \mathcal{K} \cong \mathcal{K},$$

so it is a constant sheaf on U. Now because X is irreducible, it follows that any sheaf whose restriction to each open set of a covering of X is constant is in fact a constant sheaf. Thus $\mathcal{L} \otimes \mathcal{K}$ is isomorphic to the constant sheaf $\mathcal{K}$, and the natural map

$$\mathcal{L} \to \mathcal{L} \otimes \mathcal{K} \cong \mathcal{K}$$

expresses $\mathcal{L}$ as a subsheaf of $\mathcal{K}$. $\square$

More precisely, from D one gets an invertible sheaf $\mathcal{O}(D)$ along with a meromorphic section s_D such that $(s_D) = D$. Effective Cartier divisors D are precisely those for which the meromorphic section s_D is regular.

One can think of s_D as the constant function 1. In particular, recall that $\mathcal{O}(D)$ is locally isomorphic to $\mathcal{O}_R$ by picking local defining equations η_α for D on an open cover U_α. Recall that on a smooth variety there is an equivalence between Cartier divisors and Weil divisors. Then the isomorphism

$$\mathcal{O}(D)|_{U_\alpha} \to \mathcal{O}_R|_{U_\alpha}$$

is given by multiplication by η_α. Then the canonical meromorphic section s_D, when restricted to U_α, is given by η_α which has divisor $D|_{U_\alpha}$.

Therefore, there is an isomorphism of abelian groups

$$\mathrm{Cl}(R) \to \text{subgroup of } \mathrm{Pic}(R) \text{ those invertible sheaves admitting meromorphic sections}$$

$$D \mapsto (\mathcal{O}(D), 1)$$

and the RHS is all of $\mathrm{Pic}(R)$ whenever every invertible sheaf admits a meromorphic section, which is the case for example when R is integral.

Remark 1.182. *Note that in general, we have flat pullback and proper pushforward for prime Weil divisors, and they are compatible with linear equivalence and intersection products, so we get flat pullback and proper pushforward for Weil divisor classes. We also have a pullback for Cartier divisors for any morphism, defined by precomposing with the morphism on the local equations. This is compatible with linear equivalence, for the reason that if $D - D' = (f)$ and g is a morphism, then $g^*D - g^*D' = (f \circ g)$.*

However, in general we do not have a pushforward for Cartier divisors, because the pushforward of a locally principal subscheme need not be locally principal.

Example 1.183. *An effective Cartier divisor D on an affine scheme* $\operatorname{Spec} A$ *need not be principal. Let A be a Dedekind domain with nontrivial class group, e.g.,*

$$A = \mathbb{Z}[\sqrt{-5}].$$

Let $\mathfrak{p} = (2, 1 + \sqrt{-5})$. Then

- *$\mathfrak{p}$ is a non-principal prime ideal.*
- *But $\mathfrak{p}$ is invertible (all nonzero ideals in a Dedekind domain are invertible).*

Then D is an effective Cartier divisor because $\mathfrak{p}$ is invertible and hence locally principal. If I is an invertible ideal then, $I \cdot J = A$ for some ideal J, so at any prime $\mathfrak{p}$ of A, $I_{\mathfrak{p}} \cdot J_{\mathfrak{p}} = A_{\mathfrak{p}}$, so $I_{\mathfrak{p}}$ is finitely generated projective of rank 1 over the local ring $A_{\mathfrak{p}}$. Projective modules over local rings are free and so therefore $I_{\mathfrak{p}}$ is generated by a single element, so I is locally principal.

However D is not principal because $\mathfrak{p}$ is not principal (note that on an affine scheme, principal Cartier divisors correspond precisely to principal ideals).

Example 1.184. *Let X be the cuspidal cubic curve $y^2 z = x^3$ in $\mathbb{P}^2_k$, with* $\operatorname{char} k \neq 2$. *In this case X is not regular in codimension 1, namely it has a singular point at $Z = (0, 0, 1)$ which is not a regular point of X. On the affine chart $z = 1$, the cuspidal cubic is*

$$X = \operatorname{Spec} A, \qquad A = k[x, y]/(y^2 - x^3),$$

and the cusp is the maximal ideal $\mathfrak{m} = (x, y)$. The local ring $A_{\mathfrak{m}}$ is not regular because $\dim A_{\mathfrak{m}} = 1$ *but* $\dim_k \mathfrak{m}/\mathfrak{m}^2 = 2$ *since $\mathfrak{m} = (x, y)$ and $\mathfrak{m}^2 = (x^2, xy, y^2)$, so the classes of x and y are linearly independent modulo $\mathfrak{m}^2$.*

Hence $A_{\mathfrak{m}}$ is not regular and therefore not a DVR, since DVRs are precisely the regular local rings of dimension 1. Note that A is also not integrally closed: the element $t = y/x \in \operatorname{Frac}(A)$ satisfies $t^2 = x \in A$ so it is integral over A, but $t \notin A$. Indeed, $A \cong k[t^2, t^3] \subset k(t)$ and $t \notin k[t^2, t^3]$.

Thus we cannot talk about Weil divisors on X. However, we can talk about the group $\operatorname{CaCl} X$ *of Cartier divisor classes modulo principal divisors. We will show:*

(a) there is a surjective degree homomorphism

$$\deg : \operatorname{CaCl} X \to \mathbb{Z};$$

(b) there is a one-to-one correspondence between the set of nonsingular closed points of X and the kernel of the degree map, which makes it into a group variety; and in fact

(c) there is a natural isomorphism of group varieties between $\operatorname{CaCl}^0 X$ *and the additive group G_a of the field k.*

To define the degree of a Cartier divisor on X, note that any Cartier divisor is linearly equivalent to one whose local function is invertible in some neighborhood of the singular

point $Z = (0,0,1)$. To see this, consider an open neighborhood V of Z on which the divisor is represented by a single function $g \in K(X)^\times$. Now subtract the principal divisor (g). On V, the new local equation becomes 1, which is invertible. Then this Cartier divisor corresponds to a Weil divisor

$$D = \sum n_i P_i$$

on $X - Z$, and we define the degree of the original divisor to be

$$\deg D = \sum n_i.$$

Suppose D and D' are linearly equivalent Cartier divisors, and both are represented by local equations invertible near Z. Then

$$D' - D = (f)$$

for some $f \in K(X)^\times$ and since D and D' are both represented by local equations invertible near Z, we have that f is invertible at Z. If $f \in K$ is invertible at Z, then the principal divisor (f) on $X - Z$ has degree 0. In particular, f has neither zero nor pole at Z and so the difference $D' - D$ is in fact supported on U, so it is enough to consider $D'|_U - D|_U = \operatorname{div}(f|_U)$, which we argue has degree 0.

Let $\nu : \widetilde{X} \to X$ be the normalization. For the cuspidal cubic,

$$\widetilde{X} \cong \mathbb{P}^1.$$

Also, ν is an isomorphism over $U = X - \{Z\}$. So the closed points of U correspond exactly to the points of $\widetilde{X}$ away from the point lying over Z.

Now pull f back to $\widetilde{X}$. Since $\widetilde{X}$ is a complete nonsingular curve,

$$\deg \operatorname{div}_{\widetilde{X}}(\nu^* f) = 0.$$

Because f is invertible at Z, the pullback $\nu^* f$ has no zero or pole at the point(s) of $\widetilde{X}$ over Z. So all zeros and poles of $\nu^* f$ occur over U. Hence

$$\deg(f)|_U = 0.$$

Thus the degree of a Cartier divisor on X is well-defined, and it passes to linear equivalence classes to give a surjective homomorphism

$$\deg : \operatorname{CaCl} X \to \mathbb{Z}.$$

Now let P_0 be the point $(0,1,0)$ as in the case of the nonsingular cubic curve. To each closed point $P \in X - Z$, we associate the Cartier divisor D_P which is 1 in a neighborhood of Z, and which corresponds to the Weil divisor $P - P_0$ on $X - Z$.

First note this map is injective: if $P \neq Q$ are two points in $X - Z$, and if $D_P \sim D_Q$, then there is an $f \in K^*$, which is invertible at Z, and such that

$$(f) = P - Q \quad \text{on } X - Z.$$

Then f gives a morphism of X to $\mathbb{P}^1$, which must be birational because it is degree 1. But then we would have a local map of local rings $\mathcal{O}_{\mathbb{P}^1,0} \to \mathcal{O}_{X,Z}$ which is dominant. Recall that we say a map of local rings $A \to B$ is dominant if it is local, injective, and A and B have the same fraction field. The injectivity and fraction field condition follow immediately from the birationality. The following proposition lemma shows that the local map is in fact an isomorphism, which is a contradiction since $\mathcal{O}_{\mathbb{P}^1,0}$ is regular and $\mathcal{O}_{X,Z}$ is not.

Proposition 1.185. *Let A be a DVR with fraction field K. If B is a local domain with fraction field K that dominates A, then $B = A$.*

Proof. Let $v : K^\times \to \mathbb{Z}$ be the valuation of A. Then

$$A = \{x \in K : v(x) \geq 0\}.$$

If B is local and dominates A, then for any $x \in B$, if $v(x) < 0$ then $x^{-1} \in A \subset B$, so x would be a unit of B, contradicting $v(x) < 0$. Hence $v(x) \geq 0$ for all $x \in B$, so $B \subset A$. Since $A \subset B$, we get $A = B$. $\square$

For the moment suppose that X is a nonsingular cubic curve. Then the kernel of the degree map is isomorphic to the group of closed points of X under the group law defined by the chord-tangent construction. Let $D \in \operatorname{Cl} X$ be in the kernel of the degree map.

Then we may write

$$D = \sum_i n_i(P_i - P_0)$$

for a fixed base point $P_0 \in X$. For any point $R \in X$, let the line P_0R meet X in a third point T, counted with multiplicity (if $R = P_0$, we take the tangent line at P_0, so $T = P_0$).

Then we have

$$P_0 + R + T \sim 3P_0,$$

so

$$R - P_0 \sim -(T - P_0).$$

If $n_i < 0$, take $P_i = R$. Replacing P_i by T, we obtain a linearly equivalent divisor in which the coefficient becomes $-n_i > 0$. Repeating this process, we may assume

$$D = \sum_i n_i(P_i - P_0), \quad n_i > 0 \text{ for all } i.$$

We prove by induction on $\sum_i n_i$ that $D \sim P - P_0$ for some point $P \in X$.

If $\sum_i n_i = 1$, there is nothing to prove.

Now suppose $\sum_i n_i \geq 2$. Choose two points P, Q among the P_i (possibly equal). Let the line PQ meet X in a third point R, and let the line P_0R meet X in a third point T.

Then

$$P + Q + R \sim 3P_0, \qquad P_0 + R + T \sim 3P_0,$$

so subtracting,

$$(P - P_0) + (Q - P_0) \sim (T - P_0).$$

Thus we may replace P and Q by T, obtaining a divisor linearly equivalent to D with total coefficient $\sum_i n_i - 1$. By induction,

$$D \sim P - P_0$$

for some point P.

The same is true for the cuspidal cubic curve, except that the group law is now given by the additive group structure on $\mathbb{A}^1$.

To show that every divisor in $\operatorname{CaCl}^0 X$ is linearly equivalent to D_P for some closed point $P \in X - Z$, we proceed exactly as in the case of the nonsingular cubic curve. The only difference is to note that the geometric constructions described there remain inside $X - Z$. Thus the group $\operatorname{CaCl}^0 X$ is in one-to-one correspondence with the set of closed points of $X - Z$, making it into a group variety.

In this case we are able to identify the group variety as G_a. Indeed, X is a rational curve, and

$$X - Z \cong \mathbb{A}^1_k.$$

More precisely, define a morphism

$$G_a = \operatorname{Spec} k[t] \to X - Z$$

by

$$t \mapsto (t, 1, t^3).$$

This is clearly an isomorphism of varieties and one can show that it is compatible with the group structures. Thus we obtain an isomorphism of group varieties

$$\operatorname{CaCl}^0 X \cong G_a.$$

Example 1.186. *Let X be the nonsingular cubic curve*

$$y^2 z = x^3 - xz^2 \subset \mathbb{P}^2_k,$$

Let $\mathcal{L} = \mathcal{O}(P_0)$ be the invertible sheaf associated to a point $P_0 \in X$.

Then $\mathcal{L}$ is ample, because

$$\mathcal{L}(3P_0) \cong \mathcal{O}_X(1),$$

and $\mathcal{O}_X(1)$ is very ample. On the other hand, $\mathcal{L}$ is not very ample, because $\mathcal{L}(P_0)$ is not generated by global sections. Indeed, if it were generated by global sections, then there is a section s of $\mathcal{L}(P_0)$ which does not vanish at P_0. Then the divisor of s must equal $Q - P_0$ for some point $Q \in X$, so P_0 would be linearly equivalent to some other point $Q \in X$, which is impossible since X is not rational. Thus an ample line bundle need not be very ample.

Example 1.187. *If D is a divisor on a complete nonsingular curve X, then $\mathcal{O}(D)$ is ample if and only if $\deg D > 0$. This is a consequence of the Riemann–Roch theorem.*

Proposition 1.188. *Let Y be a scheme and let E be a vector bundle on Y. Let $\mathbb{P}(E) = \mathrm{Proj}(\mathrm{Sym}^\bullet E^\vee)$ be the projectivization of a vector bundle E on a scheme Y parameterizing lines in the fibers of E. Then there is a natural isomorphism of abelian groups*

$$\mathrm{Pic}(\mathbf{P}(E)) \cong \mathrm{Pic}(Y) \oplus \mathbf{Z}\,[\mathcal{O}_{\mathbf{P}(E)}(1)]$$

where $\mathcal{O}_{\mathbf{P}(E)}(1)$ is the quotient line bundle on $\mathbf{P}(E)$, the dual of the tautological line bundle.

Proof. Define

$$\Phi : \mathrm{Pic}(Y) \oplus \mathbf{Z} \to \mathrm{Pic}(\mathbf{P}(E)), \qquad (\mathcal{L}, n) \mapsto \pi^*\mathcal{L} \otimes \mathcal{O}_{\mathbf{P}(E)}(n).$$

To prove surjectivity, let $\mathcal{M} \in \mathrm{Pic}(\mathbf{P}(E))$. For each $y \in Y$, the restriction

$$\mathcal{M}|_{\mathbf{P}(E_y)}$$

is a line bundle on $\mathbf{P}^r$, hence isomorphic to $\mathcal{O}_{\mathbf{P}^r}(n)$ for a unique integer n. This integer is locally constant in y, so after restricting to a connected component of Y, it is constant. Replacing $\mathcal{M}$ by $\mathcal{M}(-n)$, we may assume

$$\mathcal{M}|_{\mathbf{P}(E_y)} \cong \mathcal{O}_{\mathbf{P}(E_y)}$$

for all y. Then the seesaw theorem implies that $\mathcal{M}$ is pulled back from Y.

For injectivity, if

$$\pi^*\mathcal{L} \otimes \mathcal{O}(n) \cong \pi^*\mathcal{L}' \otimes \mathcal{O}(n'),$$

then restricting to any fiber $\mathbf{P}^r$ gives $\mathcal{O}_{\mathbf{P}^r}(n) \cong \mathcal{O}_{\mathbf{P}^r}(n')$, hence $n = n'$. Cancelling $\mathcal{O}(n)$ and pushing forward then gives $\mathcal{L} \cong \mathcal{L}'$. Therefore Φ is an isomorphism. □

Exercise 1.189. *Consider the compactification of the line bundle $O(n)$ over $\mathbb{P}^1$. What is its Picard group? What are its generators? What is the difference between the zero section and the "section at infinity" of this bundle?*

Solution 1.190. *Let*

$$X = \mathbf{P}\Big(\mathcal{O}_{\mathbf{P}^1} \oplus \mathcal{O}_{\mathbf{P}^1}(n)\Big)$$

with projection

$$\pi : X \to \mathbf{P}^1.$$

This is the Hirzebruch surface F_n, i.e. the compactification of the total space of $\mathcal{O}_{\mathbf{P}^1}(n)$ by adding the section at infinity. We will show that

$$\mathrm{Pic}(X) \cong \mathbf{Z} \oplus \mathbf{Z},$$

generated by a fiber f of π and by either the infinity section s_∞ or the zero section s_0. Note that all fibers are linearly equivalent because on $\mathbf{P}^1$ all points are linearly equivalent, and pulli back by π gives $\pi^*([p]) \sim \pi^*([q])$.

The two distinguished sections differ by

$$s_0 \sim s_\infty + nf$$

and can be distinguished by their self-intersection numbers

$$s_0^2 = n, \qquad s_\infty^2 = -n.$$

We are thinking about sections as morphisms

$$s : \mathbf{P}^1 \to F_n$$

such that $\pi \circ s = \mathrm{id}_{\mathbf{P}^1}$. *Equivalently, in the projective bundle*

$$F_n = \mathbf{P}(\mathcal{O} \oplus \mathcal{O}(n)),$$

sections correspond to line subbundles of $\mathcal{O} \oplus \mathcal{O}(n)$.

The two distinguished sections s_0 *and* s_∞ *come from the two natural inclusions of line bundles into* $\mathcal{O} \oplus \mathcal{O}(n)$. *These are the zero section and the section at infinity. There are, in general, many other sections. On the open subset identified with the total space of* $\mathcal{O}(n)$, *any global section*

$$\sigma \in H^0(\mathbf{P}^1, \mathcal{O}(n))$$

defines a section of π *whose image is disjoint from* s_∞; *geometrically, this is the graph of* σ. *Thus the "finite sections" are precisely the sections landing in the total-space chart, and for* $n \geq 0$ *they are parametrized by the vector space* $H^0(\mathbf{P}^1, \mathcal{O}(n))$.

All such finite sections are linearly equivalent to the zero section s_0. *By contrast,* s_∞ *is the unique section of self-intersection* $-n$ *when* $n > 0$, *which is why it is distinguished. A general section of* $F_n \to \mathbf{P}^1$ *need not stay in the finite part and may meet the section at infinity. In fact, the intersection product tells you how many times it meets the section at infinity:*

$$s \cdot s_\infty = (s_\infty + af) \cdot s_\infty = -n + a$$

Finally we check that the self intersections of the two distinguished sections are as claimed, by taking the degree of the normal bundle. Given a line subbundle $L \hookrightarrow E$ *corresponding to a section* s *and its normal bundle is*

$$N_{s/X} \cong \mathrm{Hom}(L, E/L)$$

Thus

$$N_{s_0/X} \cong \mathrm{Hom}(\mathcal{O}, \mathcal{O}(n)) \cong \mathcal{O}(n), \qquad N_{s_\infty/X} \cong \mathrm{Hom}(\mathcal{O}(n), \mathcal{O}) \cong \mathcal{O}(-n),$$

so

$$s_0^2 = n, \qquad s_\infty^2 = -n.$$

Consequently, if f *denotes the class of a fiber, then*

$$s_0 \sim s_\infty + nf.$$

1.5 Line bundles and maps to projective space

Let A be a ring and consider projective space

$$\mathbb{P}^n_A = \operatorname{Proj} A[x_0, \ldots, x_n].$$

On $\mathbb{P}^n_A$ we have the invertible sheaf $\mathcal{O}(1)$, and the homogeneous coordinates

$$x_0, \ldots, x_n \in \Gamma(\mathbb{P}^n_A, \mathcal{O}(1))$$

are global sections. These sections generate $\mathcal{O}(1)$, i.e. for each point $P \in \mathbb{P}^n_A$, their images generate the stalk $\mathcal{O}(1)_P$ as a module over $\mathcal{O}_P$.

Now let X be a scheme over A, and let

$$\varphi : X \to \mathbb{P}^n_A$$

be an A-morphism. Then

$$\mathcal{L} := \varphi^* \mathcal{O}(1)$$

is an invertible sheaf on X, and the global sections

$$s_i := \varphi^*(x_i) \in \Gamma(X, \mathcal{L}), \quad i = 0, \ldots, n,$$

generate $\mathcal{L}$. Generation of a line bundle by sections is preserved under pullback because it is a stalkwise statement and tensoring preserves generators. Conversely, $\mathcal{L}$ together with the sections s_i determine φ.

Theorem 1.191. *Let A be a ring and let X be a scheme over A.*

(a) If $\varphi : X \to \mathbb{P}^n_A$ is an A-morphism, then

$$\varphi^* \mathcal{O}(1)$$

is an invertible sheaf on X, generated by the global sections

$$s_i = \varphi^*(x_i), \quad i = 0, \ldots, n.$$

(b) Conversely, let $\mathcal{L}$ be an invertible sheaf on X, and suppose

$$s_0, \ldots, s_n \in \Gamma(X, \mathcal{L})$$

are global sections which generate $\mathcal{L}$. Then there exists a unique A-morphism

$$\varphi : X \to \mathbb{P}^n_A$$

such that

$$\mathcal{L} \cong \varphi^* \mathcal{O}(1) \quad \textit{and} \quad s_i = \varphi^*(x_i).$$

Proof. Part (a) is immediate from the discussion above. For (b), suppose given $\mathcal{L}$ and generating sections $s_0, \ldots, s_n$. For each i, define

$$X_i := \{\, P \in X \mid (s_i)_P \notin \mathfrak{m}_P \mathcal{L}_P \,\}.$$

Then each X_i is open, and since the s_i generate $\mathcal{L}$, the sets X_i cover X.

Recall that the standard open subset

$$U_i = \{x_i \neq 0\} \subset \mathbb{P}^n_A$$

is isomorphic to

$$U_i \cong \operatorname{Spec} A[y_0, \ldots, \widehat{y_i}, \ldots, y_n], \quad \text{where } y_j = x_j/x_i.$$

On X_i, we define a morphism

$$X_i \to U_i$$

by giving a ring homomorphism

$$A[y_0, \ldots, \widehat{y_i}, \ldots, y_n] \longrightarrow \Gamma(X_i, \mathcal{O}_X),$$

sending

$$y_j \longmapsto s_j/s_i.$$

This is well-defined because for $P \in X_i$, the section s_i is a unit in $\mathcal{L}_P$, so s_j/s_i is a well-defined section of $\mathcal{O}_X$ on X_i. These morphisms $X_i \to U_i$ agree on overlaps and hence glue to a morphism

$$\varphi : X \to \mathbb{P}^n_A.$$

By construction, $\mathcal{L} \cong \varphi^*\mathcal{O}(1)$ and $s_i = \varphi^*(x_i)$. Uniqueness is clear from the construction. □

In general, one can test very ampleness of a line bundle by checking the separation of points and tangent vectors by global sections.

Proposition 1.192. *Let k be an algebraically closed field, let X be a projective scheme over k, and let*

$$\varphi : X \to \mathbb{P}^n_k$$

be a morphism corresponding to an invertible sheaf $\mathcal{L}$ and sections

$$s_0, \ldots, s_n \in \Gamma(X, \mathcal{L}).$$

Let

$$V := \operatorname{span}_k\{s_0, \ldots, s_n\} \subseteq \Gamma(X, \mathcal{L}).$$

Then φ is a closed immersion if and only if:

*(1) (**Separation of points**) For any two distinct closed points $P, Q \in X$, there exists $s \in V$ such that*

$$s_P \in \mathfrak{m}_P\mathcal{L}_P \quad \text{and} \quad s_Q \notin \mathfrak{m}_Q\mathcal{L}_Q,$$

or vice versa.

*(2) (**Separation of tangent vectors**) For each closed point $P \in X$, the set*

$$\{\, s \in V \mid s_P \in \mathfrak{m}_P \mathcal{L}_P \,\}$$

spans the k-vector space

$$\mathfrak{m}_P \mathcal{L}_P / \mathfrak{m}_P^2 \mathcal{L}_P.$$

Example 1.193. *Let $X = \mathbb{P}^1_k$ and $\mathcal{L} = \mathcal{O}_{\mathbb{P}^1}(1)$. Then $\Gamma(X, \mathcal{L}) = \operatorname{span}\{x_0, x_1\}$.This gives the identity map $\varphi : \mathbb{P}^1 \to \mathbb{P}^1$. Let $P \neq Q \in \mathbb{P}^1$. Then at least one of x_0, x_1 vanishes at P but not at Q, since homogeneous coordinates distinguish points. Hence V separates points. Fix $P \in \mathbb{P}^1$. Choose affine coordinate $t = x_1/x_0$ near P (assuming $x_0(P) \neq 0$). Then $\mathcal{O}(1)|_{D_+(x_0)} \cong \mathcal{O}_{\mathbb{A}^1}$, and sections correspond to linear polynomials in t.*

The sections vanishing at P are multiples of $(t - t(P))$, and these generate $\mathfrak{m}_P/\mathfrak{m}_P^2$, which is 1-dimensional. Hence tangent vectors are separated. Therefore $\mathcal{O}_{\mathbb{P}^1}(1)$ is very ample.

1.5.1 Base locus and complete linear systems

Definition 1.194. *Let X be a scheme and $\mathcal{L}$ an invertible sheaf. Let*

$$V \subseteq \Gamma(X, \mathcal{L})$$

*be a finite-dimensional A-submodule. The **base locus** of V is*

$$\operatorname{Bs}(V) := \{\, x \in X \mid s_x \in \mathfrak{m}_x \mathcal{L}_x \text{ for all } s \in V \,\}.$$

Equivalently, it is the set of points where all sections in V vanish.

The choice of linear system V gives a morphism

$$X \setminus \operatorname{Bs}(V) \to \mathbb{P}(V)$$

and V is base point-free if and only if the sections in V globally generate $\mathcal{L}$.

Definition 1.195. *If D is a Cartier divisor on X, the **complete linear system** associated to D is*

$$|D| := \mathbb{P}\Big(\Gamma(X, \mathcal{O}_X(D))\Big).$$

Points of $|D|$ correspond to effective divisors linearly equivalent to D. A nonzero section $s \in \Gamma(X, \mathcal{O}_X(D))$ defines an effective divisor $(s) + D$.

Definition 1.196. *The **base locus** of a divisor D is*

$$\operatorname{Bs}(|D|) := \bigcap_{s \in \Gamma(X, \mathcal{O}_X(D))} \operatorname{Supp}((s) + D),$$

i.e. the common zero locus of all global sections.

1.5.2 Maps to projective space via $|D|$

Proposition 1.197. *Let X be a scheme and $\mathcal{L}$ an invertible sheaf. Let*

$$s_0, \ldots, s_n \in \Gamma(X, \mathcal{L})$$

be global sections. Then they define a morphism

$$\varphi : X \setminus \mathrm{Bs}(s_0, \ldots, s_n) \to \mathbb{P}^n_A.$$

Proof. On the open set where at least one s_i does not vanish, the sections generate $\mathcal{L}$, so the construction of the previous theorem applies locally and glues. □

Proposition 1.198. *Let D be a Cartier divisor on X. If $|D|$ is basepoint-free, i.e.*

$$\mathrm{Bs}(|D|) = \varnothing,$$

then the complete linear system defines a morphism

$$\varphi_{|D|} : X \to \mathbb{P}^n_A,$$

where $n + 1 = \dim \Gamma(X, \mathcal{O}_X(D))$.

Remark 1.199. *This is the canonical map associated to the linear system $|D|$. It depends only on the projectivization of the space of global sections.*

1.5.3 Very ampleness and embeddings

Definition 1.200. *An invertible sheaf $\mathcal{L}$ on X is **very ample** if there exists a closed immersion*

$$X \hookrightarrow \mathbb{P}^n_A$$

such that

$$\mathcal{L} \cong \mathcal{O}_X(1).$$

Definition 1.201. *An invertible sheaf $\mathcal{L}$ is **ample** if for every coherent sheaf $\mathcal{F}$, there exists n_0 such that for all $n \geq n_0$, the sheaf $\mathcal{F} \otimes \mathcal{L}^{\otimes n}$ is generated by global sections.*

Proposition 1.202. *If $\mathcal{L}$ is very ample, then it is basepoint-free.*

Proof. The embedding $X \hookrightarrow \mathbb{P}^n$ is given by global sections of $\mathcal{L}$, and at every point at least one coordinate is nonzero. □

Proposition 1.203. *Let $\mathcal{L}$ be very ample. Then the associated morphism*

$$\varphi : X \to \mathbb{P}^n_A$$

is a closed immersion.

Theorem 1.204 (Serre). *Let X be a projective scheme over a Noetherian ring A, and let $\mathcal{L}$ be ample. Then for $n \gg 0$, $\mathcal{L}^{\otimes n}$ is very ample.*

Example 1.205. *Let Q be the nonsingular quadric surface $xy = zw$ in $\mathbb{P}^3_k$ over a field k. We have seen that*

$$\mathrm{Pic}(Q) \cong \mathbb{Z} \oplus \mathbb{Z},$$

and so we speak of invertible sheaves of type (a, b), with $a, b \in \mathbb{Z}$.

Now $Q \cong \mathbb{P}^1 \times \mathbb{P}^1$. If $a, b > 0$, then we consider an a-uple embedding

$$\mathbb{P}^1 \to \mathbb{P}^{n_1}$$

and a b-uple embedding

$$\mathbb{P}^1 \to \mathbb{P}^{n_2}.$$

Taking their product and composing with the Segre embedding, we obtain a closed immersion

$$Q = \mathbb{P}^1 \times \mathbb{P}^1 \longrightarrow \mathbb{P}^{n_1} \times \mathbb{P}^{n_2} \longrightarrow \mathbb{P}^n,$$

which corresponds to an invertible sheaf of type (a, b) on Q. Thus for any $a, b > 0$, the corresponding invertible sheaf is very ample, and hence ample.

On the other hand, if $\mathcal{L}$ is of type (a, b) with either $a < 0$ or $b < 0$, then by restricting to a fiber of the product $\mathbb{P}^1 \times \mathbb{P}^1$, one sees that $\mathcal{L}$ is not generated by global sections. Hence if $a \leq 0$ or $b \leq 0$, $\mathcal{L}$ cannot be ample.

Therefore, on Q, an invertible sheaf $\mathcal{L}$ of type (a, b) is ample if and only if it is very ample, which holds if and only if $a, b > 0$.

1.5.4 The tautological bundles on projective space

Let V be a finite-dimensional vector space over a field k. We write

$$\mathbf{P}(V)$$

for the projective space of lines in V, i.e. 1-dimensional linear subspaces $\ell \subset V$.

There are two natural line bundles on $\mathbf{P}(V)$, dual to one another.

Definition 1.206. *The **tautological line bundle** on $\mathbf{P}(V)$ is the line subbundle*

$$\mathcal{O}_{\mathbf{P}(V)}(-1) \subset V \otimes_k \mathcal{O}_{\mathbf{P}(V)}$$

whose fiber at a point $[\ell] \in \mathbf{P}(V)$ is the line ℓ itself:

$$\mathcal{O}_{\mathbf{P}(V)}(-1)_{[\ell]} = \ell.$$

Its dual line bundle is denoted

$$\mathcal{O}_{\mathbf{P}(V)}(1) := \mathcal{O}_{\mathbf{P}(V)}(-1)^\vee.$$

Thus for each point $[\ell] \in \mathbf{P}(V)$, we have

$$\mathcal{O}_{\mathbf{P}(V)}(1)_{[\ell]} = \ell^\vee.$$

Equivalently, if one thinks of $\mathbf{P}(V)$ as parametrizing 1-dimensional quotient spaces of V, then $\mathcal{O}_{\mathbf{P}(V)}(1)$ is the universal quotient line bundle.

Proposition 1.207. *There is a natural isomorphism*

$$H^0(\mathbf{P}(V), \mathcal{O}_{\mathbf{P}(V)}(1)) \cong V^\vee.$$

Proof. Let $\varphi \in V^\vee$ be a linear functional. For each point $[\ell] \in \mathbf{P}(V)$, we may restrict φ to the line ℓ, obtaining

$$\varphi|_\ell \in \ell^\vee = \mathcal{O}_{\mathbf{P}(V)}(1)_{[\ell]}.$$

Thus φ defines a section s_φ of $\mathcal{O}_{\mathbf{P}(V)}(1)$, given fiberwise by

$$s_\varphi([\ell]) = \varphi|_\ell.$$

This construction is linear in φ, so it gives a linear map

$$V^\vee \longrightarrow H^0(\mathbf{P}(V), \mathcal{O}_{\mathbf{P}(V)}(1)).$$

To see that this is an isomorphism, choose a basis of V, so that $\mathbf{P}(V) \cong \mathbf{P}^n$. Then $\mathcal{O}_{\mathbf{P}(V)}(1)$ identifies with the usual hyperplane bundle $\mathcal{O}_{\mathbf{P}^n}(1)$, whose global sections are precisely the linear homogeneous polynomials. These are naturally identified with $V^\vee$. $\square$

The zero locus of the section s_φ is easy to describe. Indeed,

$$s_\varphi([\ell]) = 0 \quad \Longleftrightarrow \quad \varphi|_\ell = 0 \quad \Longleftrightarrow \quad \ell \subset \ker(\varphi).$$

Therefore

$$Z(s_\varphi) = \mathbf{P}(\ker \varphi),$$

which is a hyperplane in $\mathbf{P}(V)$. This is why $\mathcal{O}_{\mathbf{P}(V)}(1)$ is called the hyperplane bundle.

Remark 1.208. *The punctured affine cone*

$$V \setminus \{0\} \longrightarrow \mathbf{P}(V), \qquad v \longmapsto [kv],$$

is naturally the total space of $\mathcal{O}_{\mathbf{P}(V)}(-1)$ with the zero section removed. Indeed, over a point $[\ell]$, the fiber of $\mathcal{O}_{\mathbf{P}(V)}(-1)$ is the line ℓ, and removing the zero vector gives precisely the set of nonzero vectors in that line.

In general, if $\mathcal{E}$ is a vector bundle on X, then its total space is

$$\underline{\mathrm{Spec}}_X\Big(\mathrm{Sym}_{\mathcal{O}_X}(\mathcal{E}^\vee)\Big).$$

The reason is that if $\mathcal{E}|_U \cong \mathcal{O}_U^{\oplus r}$, then

$$\mathrm{Sym}(\mathcal{E}^\vee)|_U \cong \mathcal{O}_U[t_1, \ldots, t_r],$$

so

$$\underline{\mathrm{Spec}}_U\Big(\mathrm{Sym}(\mathcal{E}^\vee)\Big) \cong U \times \mathbf{A}^r.$$

These local affine spaces glue exactly as the vector bundle glues.

1.5.5 Relative Spec and affine morphisms

Let X be a scheme, and let $\mathcal{A}$ be a quasi-coherent sheaf of $\mathcal{O}_X$-algebras. We define the **relative spectrum** of $\mathcal{A}$ over X, denoted

$$\underline{\mathrm{Spec}}_X(\mathcal{A}),$$

to be the scheme obtained by gluing the affine schemes

$$\mathrm{Spec}\,\mathcal{A}(U)$$

over affine open subsets $U = \mathrm{Spec}\,R \subset X$.

Theorem 1.209. *The construction*

$$\mathcal{A} \longmapsto \underline{\mathrm{Spec}}_X(\mathcal{A})$$

defines an anti-equivalence between the category of quasi-coherent $\mathcal{O}_X$-algebras and the category of affine morphisms $f : Y \to X$. The inverse functor sends an affine morphism $f : Y \to X$ to the quasi-coherent $\mathcal{O}_X$-algebra

$$f_*\mathcal{O}_Y.$$

Remark 1.210. *This is the relative version of the affine correspondence $A \leftrightarrow \mathrm{Spec}\,A$. In particular, a morphism $f : Y \to X$ is affine if and only if*

$$Y \cong \underline{\mathrm{Spec}}_X(f_*\mathcal{O}_Y).$$

Example 1.211. *If $\mathcal{E}$ is a vector bundle on X, its total space is*

$$\mathrm{Tot}(\mathcal{E}) := \underline{\mathrm{Spec}}_X\Big(\mathrm{Sym}_{\mathcal{O}_X}(\mathcal{E}^\vee)\Big).$$

Indeed, on any open set $U \subset X$ where $\mathcal{E}|_U \cong \mathcal{O}_U^{\oplus r}$, we have

$$\mathrm{Sym}(\mathcal{E}^\vee)|_U \cong \mathcal{O}_U[t_1, \ldots, t_r],$$

so

$$\mathrm{Tot}(\mathcal{E})|_U \cong U \times \mathbb{A}^r.$$

Thus $\mathrm{Tot}(\mathcal{E}) \to X$ is an affine morphism which is locally the projection $U \times \mathbb{A}^r \to U$.

1.6 Differentials

1.6.1 Kähler differentials $\Omega_{X/Y}$

Definition 1.212. *Let $A \to B$ be a ring homomorphism. The* ***module of Kähler differentials*** *$\Omega_{B/A}$ is the B-module generated by symbols db for $b \in B$, subject to:*

$$\begin{aligned} d(b+b') &= db + db', \\ d(bb') &= b\,db' + b'\,db, \\ da &= 0 \quad \textit{for all } a \in A. \end{aligned}$$

There is a universal derivation

$$d : B \to \Omega_{B/A}.$$

Definition 1.213. *For a morphism of schemes $f : X \to Y$, the* ***sheaf of relative differentials*** *$\Omega_{X/Y}$ is the quasi-coherent sheaf obtained by gluing $\Omega_{B/A}$ on affine opens.*

Example 1.214. *If $B = A[x_1, \ldots, x_n]$, then*

$$\Omega_{B/A} \cong \bigoplus_{i=1}^{n} B\,dx_i.$$

1.6.2 Universal property of differentials

Definition 1.215. *An A-derivation from B to a B-module M is an A-linear map*

$$D : B \to M$$

satisfying the Leibniz rule:

$$D(bb') = bD(b') + b'D(b).$$

Proposition 1.216 (Universal property)**.** *For any B-module M, there is a natural isomorphism*

$$\mathrm{Der}_A(B, M) \cong \mathrm{Hom}_B(\Omega_{B/A}, M),$$

sending D to the unique B-linear map $\Omega_{B/A} \to M$ with $db \mapsto D(b)$.

1.6.3 Sheaf of differentials Ω_X

Definition 1.217. *If X is a scheme over $\operatorname{Spec} k$, define*

$$\Omega_X := \Omega_{X/k}.$$

Proposition 1.218. *$\Omega_{X/Y}$ is quasi-coherent, and coherent if $X \to Y$ is of finite type.*

Definition 1.219. *A morphism $f : X \to Y$ is* ***smooth of relative dimension*** *n if:*

- *f is locally of finite presentation,*

- *f is flat,*
- *$\Omega_{X/Y}$ is locally free of rank n.*

Example 1.220. *If X is smooth over k, then Ω_X is locally free and*

$$\omega_X := \wedge^{\dim X}\Omega_X$$

is the canonical bundle.

1.6.4 Exact sequences for differentials

Proposition 1.221. *Let $f : X \to Y$ be a morphism of schemes over S. Then there is a natural right exact sequence of sheaves which that absolute differentials decompose into "directions coming from the base" and "new directions along the fibers"*

$$f^*\Omega_{Y/S} \longrightarrow \Omega_{X/S} \longrightarrow \Omega_{X/Y} \longrightarrow 0$$

where working on an open $V \subseteq Y$, this is $\mathcal{O}_Y(V) \to \mathcal{O}_X(f^{-1}(V)) \to \Omega_{X/S}(f^{-1}(V))$, the structure map followed by the universal derivation $d : \mathcal{O}_X(f^{-1}(V)) \to \Omega_{X/S}(f^{-1}(V))$. The composition is S-linear and satisfies the Leibniz rule so it factors through $f^*\Omega_{Y/S}$, giving $\Omega_{Y/S} \to f_*\Omega_{X/S}$ and by adjunction $f^*\Omega_{Y/S} \to \Omega_{X/S}$.

For a sequence of ring maps $A \to B \to C$, this corresponds to

$$\Omega_{B/A} \otimes_B C \longrightarrow \Omega_{C/A} \longrightarrow \Omega_{C/B} \longrightarrow 0$$

where the maps are $b\,db' \otimes 1 \longmapsto b\,db$ the image of b in C, and $dc \otimes 1 \longmapsto dc$.

If f is smooth, or more generally flat and locally of finite presentation, then the first map is injective and we get a short exact sequence. This is because flatness implies that no nonzero element $\sum db_i \otimes c_i$ maps to zero.

Proposition 1.222 (Conormal sequence)**.** *Let $i : Z \hookrightarrow X$ be a closed immersion with ideal sheaf $\mathcal{I}$. Then there is an exact sequence*

$$\mathcal{I}/\mathcal{I}^2 \longrightarrow \Omega_X|_Z \longrightarrow \Omega_Z \longrightarrow 0.$$

where $\mathcal{I}/\mathcal{I}^2 = N^\vee_{Z/X}$ is the conormal bundle of Z in X.

If $B = A/I$, then this corresponds to:

$$I/I^2 \longrightarrow \Omega_A \otimes_A B \longrightarrow \Omega_B \longrightarrow 0,$$

where the map sends $f \mapsto df$. If

$$B = A[x_1, \ldots, x_n]/(f_1, \ldots, f_r),$$

then

$$\Omega_{B/A} \cong \frac{\bigoplus B\,dx_i}{\langle df_j \rangle}, \quad \text{where} \quad df_j = \sum_i \frac{\partial f_j}{\partial x_i} dx_i.$$

Proposition 1.223. *Let $D \hookrightarrow X$ be an effective Cartier divisor. Then $\mathcal{I}_D$ is locally principal and*

$$\mathcal{I}_D/\mathcal{I}_D^2 \cong \mathcal{O}_D(-D).$$

In this case the conormal sequence becomes short exact:

$$0 \longrightarrow \mathcal{O}_D(-D) \longrightarrow \Omega_X|_D \longrightarrow \Omega_D \longrightarrow 0.$$

The normal bundle is a line bundle, reflecting that a Cartier divisor has codimension 1. Dualizing gives

$$0 \longrightarrow T_D \longrightarrow T_X|_D \longrightarrow \mathcal{O}_D(D) \longrightarrow 0.$$

1.6.5 Tangent space via $\mathrm{Hom}(\mathfrak{m}/\mathfrak{m}^2, k)$

Definition 1.224. *Let X be a scheme, $x \in X$ with residue field $k(x)$. The **Zariski tangent space** is*

$$T_xX := \mathrm{Hom}_{k(x)}(\mathfrak{m}_x/\mathfrak{m}_x^2, k(x)).$$

Proposition 1.225. *There is a natural identification*

$$T_xX \cong \mathrm{Der}_{k(x)}(\mathcal{O}_{X,x}, k(x)).$$

Proposition 1.226. *If X is locally of finite type over k, then*

$$T_xX \cong \mathrm{Hom}_{k(x)}(\Omega_{X/k} \otimes k(x), k(x)).$$

Corollary 1.227. $\dim T_xX \geq \dim X$, *with equality iff X is smooth at x.*

1.6.6 Relative differentials and base change

Proposition 1.228 (Base change)**.** *Given a cartesian square*

$$\begin{tikzcd} X' \arrow[r] \arrow[d] & X \arrow[d] \\ Y' \arrow[r] & Y \end{tikzcd}$$

there is a natural isomorphism

$$\Omega_{X'/Y'} \cong \Omega_{X/Y} \otimes_{\mathcal{O}_X} \mathcal{O}_{X'}.$$

Proof. Let $X \to Y$ be modeled by a ring map $A \to B$, and consider the base change along $A \to A'$ so that $B' := B \otimes_A A'$. Then the local statement is that $\Omega_{B'/A'} \cong \Omega_{B/A} \otimes_B B'$. The point is relative differentiation kills base scalars, so base change only enlarges coefficients and does not create new differential directions. □

The stalkwise version states that $f : X \to Y$ and $x \in X$ mapping to $y \in Y$,

$$\Omega_{X/Y} \otimes k(x) \cong \Omega_{X_y/k(y)} \otimes_{k(y)} k(x).$$

In particular, the fiber of $\Omega_{X/Y}$ at x computes the tangent space of the fiber X_y.

Theorem 1.229 (Smoothness criterion). *Let $f : X \to Y$ be locally of finite type. Then f is smooth at x iff:*

- *$\Omega_{X/Y}$ is locally free near x,*
- $\dim_{k(x)}(\Omega_{X/Y} \otimes k(x)) = \dim_x(X_y)$.

1.6.7 More examples

Example 1.230. *Let $B = k[x, y]/(xy)$. Then*

$$d(xy) = x\,dy + y\,dx,$$

so

$$\Omega_{B/k} = \frac{B\,dx \oplus B\,dy}{\langle x\,dy + y\,dx\rangle}.$$

At the origin $(x, y) = (0, 0)$, the relation $x\,dy + y\,dx$ vanishes, so the dimension of the tangent space is 2. However, the curve has dimension 1, so the origin is a singular point.

Example 1.231. *Another example is the dual numbers $B = k[x]/(x^2)$. We compute $d(x^2) = 2x\,dx = 0 \Rightarrow x\,dx = 0$ so*

$$\Omega_{B/k} \cong B\,dx/(x\,dx)$$

At the closed point, $\dim T = 1$ but the scheme is 0-dimensional.

Example 1.232. *A field extension L/k is separable if and only if $\Omega_{L/k} = 0$. A basic nonexample in characteristic $p > 0$ is*

$$k(t^p) \subset k(t).$$

This extension is not separable, hence not étale. Indeed, if we set $u = t^p$, then

$$k(t) \cong k(u)[t]/(t^p - u).$$

Therefore $\Omega_{k(t)/k(u)}$ is generated by dt subject to the relation

$$d(t^p - u) = 0.$$

But in characteristic p,

$$d(t^p) = pt^{p-1}dt = 0,$$

and also $du = 0$ in relative differentials over $k(u)$. So this imposes no condition on dt. Thus

$$\Omega_{k(t)/k(t^p)} \cong k(t)\,dt \neq 0.$$

1.7 Cohomological Techniques

Theorem 1.233. *The category of sheaves of abelian groups on a topological space X has enough injectives, i.e. for any sheaf $\mathcal{F}$ on X, there exists a monomorphism $\mathcal{F} \to \mathcal{I}$ into an injective sheaf $\mathcal{I}$.*

This allows us to define sheaf cohomology as the right derived functors of the global sections functor $\Gamma(X, -)$. We will give the definition now. Take an injective resolution of a sheaf $\mathcal{F}$ on X:

$$0 \to \mathcal{F} \to \mathcal{I}^0 \to \mathcal{I}^1 \to \mathcal{I}^2 \to \cdots$$

Applying the global sections functor $\Gamma(X, -)$ gives a complex of abelian groups

$$0 \to \Gamma(X, \mathcal{I}^0) \to \Gamma(X, \mathcal{I}^1) \to \Gamma(X, \mathcal{I}^2) \to \cdots$$

Definition 1.234. *The **sheaf cohomology** groups $H^i(X, \mathcal{F})$ are defined as the cohomology groups of the complex of global sections of an injective resolution of $\mathcal{F}$:*

$$H^i(X, \mathcal{F}) = R^i\Gamma(X, \mathcal{F}) = H^i(\Gamma(X, \mathcal{I}^\bullet))$$

However, in practice it is quite difficult to work with injective resolutions of sheaves. Instead, we use other types of resolutions which are easier to construct.

Definition 1.235. *Let F be a left exact functor on $\mathrm{AbSh}(X)$. A sheaf $\mathcal{I}$ is F**-acyclic** if $R^i\mathcal{F}(\mathcal{I}) = 0$ for all $i > 0$.*

Definition 1.236. *A sheaf $\mathcal{F}$ on a topological space X is **flabby** if for every inclusion of open sets $V \subseteq U \subseteq X$, the restriction map*

$$\mathcal{F}(U) \to \mathcal{F}(V)$$

is surjective.

Definition 1.237. *A sheaf $\mathcal{F}$ on a topological space X is **soft** if for every closed set $K \subseteq X$, the restriction map is surjective.*

$$\mathcal{F}(X) \to \mathcal{F}(K)$$

Proposition 1.238. *Flabby sheaves are soft on any topological space.*

Theorem 1.239. *On a paracompact Hausdorff space X, soft sheaves are $\Gamma(X, -)$-acyclic.*

Definition 1.240. *A sheaf $\mathcal{F}$ on a topological space X is **fine** if for every locally finite open cover $\{U_i\}_{i \in I}$ of X, there exists a partition of unity subordinate to the cover, i.e. a collection of sheaf endomorphisms $\{\varphi_i : \mathcal{F} \to \mathcal{F}\}_{i \in I}$ such that*

- *For each $i \in I$, the support of φ_i is contained in U_i,*
- *The sum $\sum_{i \in I} \varphi_i$ is the identity endomorphism of $\mathcal{F}$.*

Theorem 1.241. *Fine sheaves are soft. Sheaves of modules over fine sheaves of rings are soft.*

Example 1.242. *Let $\mathcal{E}_X^k$ be the sheaf of smooth k-forms on a smooth manifold X. Then $\mathcal{E}_X^k$ is a fine sheaf because we can construct partitions of unity using smooth bump functions subordinate to any locally finite open cover of X.*

Example 1.243 (Complex of sheaves of $\mathbb{Z}$ singular cochains)**.** *Consider the presheaf $\mathcal{C}^k$ on a topological space X defined by*

$$\mathcal{C}^k(U) = C^k_{sing}(U, \mathbb{Z})$$

where $C^k_{sing}(U, \mathbb{Z})$ is the group of singular k-cochains on U with integer coefficients. The restriction maps are given by restricting cochains to smaller open sets. The sheafification $\mathcal{S}^k = \mathcal{C}^{k+}$ is called the sheaf of singular k-cochains. The complex of sheaves

$$\cdots \to \mathcal{S}^{k-1} \to \mathcal{S}^k \to \mathcal{S}^{k+1} \to \cdots$$

with the usual coboundary maps is a fine resolution of the constant sheaf $\mathbb{Z}_X$ on X. This is because partitions of unity can be constructed for singular cochains by using barycentric subdivisions of simplices.

The complex $(\mathcal{C}^\bullet(X), d)$ is exact on stalks except for when $k = 0$, where the cohomology is $\mathbb{Z}$. Thus this complex is a fine resolution of the constant sheaf $\mathbb{Z}_X$. This shows that sheaf cohomology with coefficients in $\mathbb{Z}_X$ can be computed using the complex of singular cochains, and hence agrees with singular cohomology.

1.7.1 Čech cohomology

Let X be a scheme and $\mathcal{F}$ a sheaf of abelian groups.

Definition 1.244. *Let $\mathfrak{U} = \{U_i\}$ be an open cover of X. The **Čech complex** is*

$$\check{C}^p(\mathfrak{U}, \mathcal{F}) = \prod_{i_0, \ldots, i_p} \mathcal{F}(U_{i_0} \cap \cdots \cap U_{i_p}),$$

with the usual alternating differential. The cohomology is denoted $\check{H}^p(\mathfrak{U}, \mathcal{F})$

There is a natural map

$$\check{H}^p(\mathfrak{U}, \mathcal{F}) \to H^p(X, \mathcal{F}),$$

where the RHS is derived functor cohomology.

Proposition 1.245. *If $\mathfrak{U}$ is an affine open cover of a separated scheme X and $\mathcal{F}$ is quasi-coherent, then*

$$\check{H}^p(\mathfrak{U}, \mathcal{F}) \cong H^p(X, \mathcal{F}).$$

Theorem 1.246. *If X is quasi-compact and separated, and $\mathcal{F}$ is quasi-coherent, then Čech cohomology with respect to any affine open cover computes sheaf cohomology.*

Theorem 1.247. *If $X = \operatorname{Spec} A$ is affine and $\mathcal{F}$ is quasi-coherent, then*

$$H^i(X, \mathcal{F}) = 0 \quad \text{for all } i > 0.$$

A relative version of the vanishing theorem holds for affine morphisms.

Theorem 1.248. *If $f : X \to Y$ is an affine morphism and $\mathcal{F}$ is a quasi-coherent sheaf on X, then*

$$R^i f_* \mathcal{F} = 0 \quad \text{for all } i > 0.$$

In particular,

$$H^i(X, \mathcal{F}) = H^i(Y, f_*\mathcal{F}) \quad \text{for all } i \geq 0$$

1.7.2 Cohomology of $\mathbb{P}^n$

Let k be a field and $X = \mathbb{P}^n_k$.

Theorem 1.249. *For all $m \in \mathbb{Z}$:*

1. *$H^0(\mathbb{P}^n, \mathcal{O}(m))$ is the space of homogeneous polynomials of degree m:*

$$H^0(\mathbb{P}^n, \mathcal{O}(m)) = \begin{cases} S_m & m \geq 0, \\ 0 & m < 0. \end{cases}$$

2. *$H^i(\mathbb{P}^n, \mathcal{O}(m)) = 0$ for $0 < i < n$.*

3. *Top cohomology comes exactly from monomials x^a with all exponents negative and with total degree m:*

$$H^n(\mathbb{P}^n, \mathcal{O}(m)) \cong \begin{cases} 0 & m > -n-1, \\ k^{\binom{-m-1}{n}} & m \leq -n-1. \end{cases}$$

Sketch of the computation. Let

$$X = \mathbb{P}^n_k = \operatorname{Proj} S, \qquad S = k[x_0, \ldots, x_n],$$

and cover X by the standard affine opens

$$\mathfrak{U} = \{U_i = D_+(x_i)\}_{i=0}^n.$$

For a subset $I = \{i_0, \ldots, i_p\}$, write $U_I = U_{i_0} \cap \cdots \cap U_{i_p}$. Then

$$\mathcal{O}_X(m)(U_I) = (S_{x_I})_m, \qquad x_I := x_{i_0} \cdots x_{i_p}.$$

Thus

$$\check{C}^p(\mathfrak{U}, \mathcal{O}_X(m)) = \bigoplus_{|I|=p+1} (S_{x_I})_m.$$

A basis for $(S_{x_I})_m$ is given by Laurent monomials

$$x^a = x_0^{a_0} \cdots x_n^{a_n}, \qquad a_i \in \mathbb{Z}, \qquad \sum_{i=0}^n a_i = m,$$

such that $a_j \geq 0$ for every $j \notin I$. Equivalently, the variables with negative exponent must all belong to I.

For such an exponent vector $a = (a_0, \ldots, a_n)$, set

$$N(a) := \{i \mid a_i < 0\}.$$

Then the monomial x^a appears in the summand indexed by I if and only if $N(a) \subseteq I$. Hence the Čech complex decomposes as a direct sum of monomial subcomplexes:

$$\check{C}^\bullet(\mathfrak{U}, \mathcal{O}_X(m)) = \bigoplus_{\substack{a \in \mathbb{Z}^{n+1} \\ a_0 + \cdots + a_n = m}} C^\bullet(a),$$

where $C^\bullet(a)$ is spanned by the single monomial x^a.

Fix such an a, and let $r = |N(a)|$. Then

$$C^p(a) = 0 \quad \text{if } p + 1 < r,$$

and for $p + 1 \geq r$ we have

$$C^p(a) = \bigoplus_{\substack{I \supseteq N(a) \\ |I| = p+1}} k \cdot x^a.$$

Thus $C^\bullet(a)$ is the simplicial cochain complex of a simplex on the remaining vertices $\{0, \ldots, n\} \setminus N(a)$, shifted by $r - 1$.

There are three cases.

Case 1: $N(a) = \varnothing$**.** Then all exponents are nonnegative, so x^a is an honest homogeneous polynomial of degree m. In this case $C^\bullet(a)$ is the ordinary Čech complex of the full simplex, which has the cohomology of a point:

$$H^q(C^\bullet(a)) = \begin{cases} k & q = 0, \\ 0 & q > 0. \end{cases}$$

Summing over all such monomials gives

$$H^0(\mathbb{P}^n, \mathcal{O}(m)) = \begin{cases} S_m & m \geq 0, \\ 0 & m < 0. \end{cases}$$

Case 2: $\varnothing \neq N(a) \neq \{0, \ldots, n\}$**.** Let $M = \{0, \ldots, n\} \setminus N(a)$ and write each index set as

$$I = N(a) \sqcup J, \qquad J \subseteq M.$$

Then

$$C^p(a) \cong \bigoplus_{\substack{J \subseteq M \\ |J| = p+1-|N(a)|}} k.$$

Thus $C^\bullet(a)$ is the augmented simplicial cochain complex of the simplex on the vertex set M, shifted by $|N(a)| - 1$. It is augmented because $J = \varnothing$ occurs when $p = r - 1$. So the complex starts with $C^{r-1}(a) \cong k$ then comes $C^r(a) \cong k^{|M|}$ and higher terms.

Since a simplex is contractible, its augmented cochain complex is exact. Therefore

$$H^q(C^\bullet(a)) = 0 \qquad \text{for all } q.$$

Case 3: $N(a) = \{0, \dots, n\}$**.** Then every exponent is negative. In this case the only subset I containing $N(a)$ is the full set $\{0, \dots, n\}$, so the complex is concentrated in degree n:

$$C^p(a) = 0 \quad (p < n), \qquad C^n(a) = k \cdot x^a.$$

Hence

$$H^q(C^\bullet(a)) = \begin{cases} k & q = n, \\ 0 & q \neq n. \end{cases}$$

Thus top cohomology is spanned by monomials x^a with all $a_i < 0$ and $\sum a_i = m$.

Writing

$$a_i = -b_i - 1, \qquad b_i \geq 0,$$

the condition $\sum a_i = m$ becomes

$$\sum_{i=0}^{n} b_i = -m - n - 1.$$

Therefore such monomials exist if and only if

$$m \leq -n - 1.$$

When this holds, a basis for $H^n(\mathbb{P}^n, \mathcal{O}(m))$ is given by the classes

$$\left[\frac{1}{x_0^{b_0+1} \cdots x_n^{b_n+1}}\right], \qquad b_i \geq 0, \qquad \sum_{i=0}^{n} b_i = -m - n - 1.$$

In fact this basis is more naturally interpreted in terms of rational top differential forms on $\mathbb{P}^n$. Let

$$\Omega = \sum_{i=0}^{n} (-1)^i x_i \, dx_0 \wedge \cdots \widehat{dx_i} \cdots \wedge dx_n$$

be the standard homogeneous $(n+1)$-form on $\mathbb{A}^{n+1} \setminus \{0\}$. Then one can check that Ω has weight $n+1$ with respect to the standard scaling action. Then a monomial x^a with all exponents negative corresponds to the rational differential form

$$\frac{\Omega}{x_0^{b_0+1} \cdots x_n^{b_n+1}},$$

which descends to a well-defined rational section of the canonical bundle $\omega_{\mathbb{P}^n}$. $\square$

Solution 1.250. *Let*

$$U = \mathbb{A}_k^2 \setminus \{(0,0)\} = D(x) \cup D(y).$$

Since $D(x)$, $D(y)$, and $D(x) \cap D(y) = D(xy)$ are affine, Čech cohomology with respect to this cover computes $H^1(U, \mathcal{O}_U)$. Thus

$$H^1(U, \mathcal{O}_U) \cong \frac{\mathcal{O}(D(xy))}{\mathcal{O}(D(x)) + \mathcal{O}(D(y))}.$$

Now

$$\mathcal{O}(D(x)) = k[x, x^{-1}, y], \qquad \mathcal{O}(D(y)) = k[x, y, y^{-1}],$$

and

$$\mathcal{O}(D(xy)) = k[x, x^{-1}, y, y^{-1}].$$

Therefore

$$H^1(U, \mathcal{O}_U) \cong \frac{k[x, x^{-1}, y, y^{-1}]}{k[x, x^{-1}, y] + k[x, y, y^{-1}]}.$$

A monomial $x^i y^j$ lies in $k[x, x^{-1}, y]$ if and only if $j \geq 0$, and it lies in $k[x, y, y^{-1}]$ if and only if $i \geq 0$. Hence the only monomials which survive in the quotient are those with

$$i < 0, \qquad j < 0.$$

It follows that

$$H^1(U, \mathcal{O}_U) \cong \bigoplus_{i,j<0} k \cdot x^i y^j,$$

so $H^1(U, \mathcal{O}_U)$ is isomorphic to the k-vector space spanned by

$$\{x^i y^j \mid i, j < 0\}.$$

In particular, it is infinite-dimensional over k.

Since $\mathcal{O}_U$ is quasi-coherent, if U were affine then we would have

$$H^1(U, \mathcal{O}_U) = 0$$

Thus U is not affine. This also shows that it is not proper since it has infinite-dimensional cohomology.

Theorem 1.251 (Serre duality and vanishing)**.** *Let X be a smooth projective scheme of dimension n over a field k, and let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module. Then there is a natural perfect pairing*

$$H^i(X, \mathcal{F}) \times \operatorname{Ext}^{n-i}_{\mathcal{O}_X}(\mathcal{F}, \omega_X) \longrightarrow k,$$

functorial in $\mathcal{F}$, which induces canonical isomorphisms

$$H^i(X, \mathcal{F})^\vee \cong \operatorname{Ext}^{n-i}_{\mathcal{O}_X}(\mathcal{F}, \omega_X).$$

Here $\omega_X := \Omega^n_{X/k}$ is the canonical (dualizing) sheaf.

Corollary 1.252. *If $\mathcal{F}$ is locally free, then*

$$H^i(X, \mathcal{F})^\vee \cong H^{n-i}(X, \mathcal{F}^\vee \otimes \omega_X).$$

Theorem 1.253 (Serre vanishing)**.** *Let X be a projective scheme over a Noetherian ring A, and let $\mathcal{L}$ be an ample line bundle. Then for every coherent sheaf $\mathcal{F}$, there exists n_0 such that for all $n \geq n_0$:*

$$H^i(X, \mathcal{F} \otimes \mathcal{L}^{\otimes n}) = 0 \quad \text{for all } i > 0.$$

Corollary 1.254. *For $n \gg 0$, the sheaf $\mathcal{F} \otimes \mathcal{L}^{\otimes n}$ is generated by global sections.*

1.7.3 Semicontinuity of cohomology, base change, and seesaw principle

We begin with a foundational result about the coherent pushforward of cohomology.

Theorem 1.255. *Let $f : X \to Y$ be a proper morphism of Noetherian schemes, and let $\mathcal{F}$ be a coherent sheaf on X. Then $R^i f_* \mathcal{F}$ is coherent on Y for all $i \geq 0$.*

Theorem 1.256 (Semicontinuity)**.** *Let $f : X \to Y$ be a projective morphism of Noetherian schemes, and let $\mathcal{F}$ be a coherent sheaf on X, flat over Y. Then for each $i \geq 0$, the function*

$$y \longmapsto h^i(y, \mathcal{F}) := \dim_{k(y)} H^i(X_y, \mathcal{F}_y)$$

is upper semicontinuous on Y. Moreover, the function

$$y \longmapsto \sum_{i=0}^{\infty} (-1)^i h^i(y, \mathcal{F})$$

is locally constant on Y.

In particular the dimension of sheaf cohomology can only jump up upon specialization, as is demonstrated by the following example.

Example 1.257. *Let $X = E \times E \to E$ be the trivial family of an elliptic curve E, and let $\mathcal{L}$ be a line bundle on X whose restriction $\mathcal{L}_y$ varies in $\operatorname{Pic}^0(E)$, with $\mathcal{L}_0 \cong \mathcal{O}_E$ and $\mathcal{L}_y \ncong \mathcal{O}_E$ for $y \neq 0$. For example, we can take $\mathcal{L}$ to the line bundle of the divisor $\Delta - (E \times \{0\})$, where Δ is the diagonal in $E \times E$.*

Then $\mathcal{L}$ is flat over Y. Note that because $\mathcal{L}$ is locally free, $\mathcal{L}_x \cong \mathcal{O}_{X_x}$ for every $x \in X$, and so it is enough to check that $X \to E$ is flat. The latter fact holds because the map $X \to E$ is just the projection from a product, which is flat. Locally on E, if E is covered by affine opens $\operatorname{Spec} A$, and Y by affine opens $\operatorname{Spec} R$, then

$$X|_{\operatorname{Spec} A \times \operatorname{Spec} R} = \operatorname{Spec}(A \otimes_k R),$$

and $A \otimes_k R$ is flat over R, because it is actually a free R-module once you choose a k-basis of A. Another way to see this is that $E \times_{\operatorname{Spec} k} E \to E$ is the base change of the flat morphism $E \to \operatorname{Spec} k$ and the structure map of a variety to $\operatorname{Spec} k$ is always flat, and flatness is preserved under base change.

Then we see that

$$h^0(E, \mathcal{L}_y) = \begin{cases} 1, & y = 0, \\ 0, & y \neq 0. \end{cases}$$

This is because a nontrivial degree 0 line bundle on an elliptic curve has no nonzero sections, while $\mathcal{O}_E$ has one. This is true on any smooth projective curve C for example. If a line bundle $\mathcal{L}$ is degree zero and has nonzero section s then s has an effective divisor of zeros D and $\mathcal{L} \cong \mathcal{O}_C(D)$ and $\deg D = 0$ and $D \geq 0$ implies that $D = 0$ and $\mathcal{L} \cong \mathcal{O}_C$.

This also shows that one cannot naively think of $\pi_* \mathcal{L}$ as obtained by "gluing together" the spaces $H^0(E, \mathcal{L}_p)$ Indeed, in this example one has

$$\pi_* \mathcal{L} = 0,$$

To see this, it is enough to check that for every open $U \subseteq X$ the stalk of $\pi_*\mathcal{L}$ at the generic point of U is zero (because $X = E \times E$ is integral). But the restriction of $\mathcal{L}$to the generic point of U is a nontrivial degree 0 line bundle since η_U is not equal to the closed point 0 of E, so it has no nonzero sections, and hence the stalk of $\pi_*\mathcal{L}$ at η_U is zero.

Thus this example shows cohomology does not commute with taking fibers everywhere. Let's closely examine the failure of cohomology and base change here.

The relevant base change map in degree 0 is

$$\varphi^0(p) : (\pi_*\mathcal{L}) \otimes k(p) \longrightarrow H^0(E, \mathcal{L}_p).$$

Since $\pi_*\mathcal{L} = 0$, the source is always 0. So

$$\varphi^0(p) : 0 \longrightarrow H^0(E, \mathcal{L}_p).$$

Hence:

- *if $p \neq p_0$, then $H^0(E, \mathcal{L}_p) = 0$, so $\varphi^0(p)$ is $0 \to 0$, an isomorphism;*
- *if $p = p_0$, then $H^0(E, \mathcal{L}_{p_0}) \cong k$, so $\varphi^0(p_0)$ is $0 \to k$, not surjective, hence not an isomorphism.*

So cohomology and base change fails exactly at the special point p_0.

The semicontinuity theorem also implies that the locus where line bundles are trivial is Zariski closed, which is made precise by the following result.

Proposition 1.258. *Suppose $X \to Y$ is a flat projective morphism with integral fibers. Let $\mathcal{L}$ be a line bundle on X, which is automatically flat over Y since it is locally free. Then the locus of $y \in Y$ where $\mathcal{L}_y$ is trivial on X_y is a closed set.*

Proof. The locus where

$$h^0(X_y, \mathcal{L}_y) \geq 1$$

is closed by the Semicontinuity Theorem, and the same holds for the locus where

$$h^0(X_y, \mathcal{L}_y^\vee) \geq 1.$$

Now invoke the fact that if $\mathcal{L}$ is an invertible sheaf on an integral projective scheme X such that both $\mathcal{L}$ and $\mathcal{L}^\vee$ have non-zero sections, then $\mathcal{L}$ is trivial. In the proof of this fact, we can remove projective if we know that the only global functions on X are the constants.

This is because the multiplication map if s and t are nonzero sections of $\mathcal{L}$ and $\mathcal{L}^\vee$, respectively, they are both nonzero in the stalk at the generic point. Otherwise they would be the zero section; this is where we use that X is integral.

Under the natural map

$$\mathcal{L} \otimes \mathcal{L}^\vee \longrightarrow \mathcal{O}_X,$$

the section $s \otimes t$ maps to st, which is also nonzero. But the only global functions, i.e. global sections of $\mathcal{O}_X$, are the constants, so st is a nonzero constant. It follows that s is nowhere vanishing, for otherwise st would vanish somewhere. Hence $\mathcal{L}$ has a nowhere vanishing section, and therefore $\mathcal{L} \cong \mathcal{O}_X$. $\square$

Theorem 1.259 (Cohomology and Base Change). *Let $f : X \to Y$ be a projective morphism of Noetherian schemes, and let $\mathcal{F}$ be a coherent sheaf on X, flat over Y. Let $y \in Y$. Then:*

1. *If the natural map*

$$\varphi^i(y) : (R^i f_* \mathcal{F}) \otimes_{\mathcal{O}_Y} k(y) \longrightarrow H^i(X_y, \mathcal{F}_y)$$

   is surjective, then it is an isomorphism, and the same holds for all y' in a suitable neighborhood of y.

2. *Assume that $\varphi^i(y)$ is surjective. Then the following are equivalent:*

   (a) $\varphi^{i-1}(y)$ is surjective;

   (b) $R^i f_* \mathcal{F}$ is locally free in a neighborhood of y.

Remark 1.260. *One naive thing one might try to do to understand the failure of cohomology and base change is to take a Čech complex $C^\bullet$ computing $H^i(X, \mathcal{F})$ and tensor it with $k(s)$. The problem is that tensoring with $k(s)$ is not exact in general, since $k(s)$ is usually not flat over A. Thus cohomology need not commute with tensor product:*

$$H^i(C^\bullet) \otimes_A k(s) \ncong H^i(C^\bullet \otimes_A k(s)).$$

Moreover, for a general family and a general sheaf, the terms of the Čech complex need not be free or even flat over A, and a given Čech cover need not compute the actual sheaf cohomology.

With this result in mind, we are ready to state the Seesaw principle, which is a powerful criterion for recognizing when a line bundle on a family of integral projective schemes is pulled back from the base.

Theorem 1.261 (Seesaw principle: recognition of pullbacks). *Let $\pi : X \to Y$ be a flat projective morphism of Noetherian schemes with geometrically integral fibers, and let $\mathcal{L}$ be an invertible sheaf on X. Assume that for every $y \in Y$,*

$$\mathcal{L}|_{X_y} \cong \mathcal{O}_{X_y}.$$

Then there exists an invertible sheaf $\mathcal{M}$ on Y such that

$$\mathcal{L} \cong \pi^* \mathcal{M}.$$

Moreover, $\mathcal{M}$ is unique up to unique isomorphism.

Discussion of the mechanism. The key input is cohomology and base change applied to $\mathcal{L}$. Since each fiber X_y is proper and geometrically integral over $k(y)$, we have

$$H^0(X_y, \mathcal{O}_{X_y}) \cong k(y).$$

To see this, fix the fiber

$$X_y = X \times_Y \operatorname{Spec} k(y),$$

and suppose it is proper and geometrically integral over $k(y)$. Set $K := H^0(X_y, \mathcal{O}_{X_y})$. Because X_y is proper over $k(y)$, K is a finite $k(y)$-algebra. Because X_y is integral, K is a domain, hence a field finite over $k(y)$. We need to see that this extension is trivial. Extend scalars from $k(y)$ to an algebraic closure $\overline{k(y)}$. Then

$$X_{\overline{k(y)}} := X_y \times_{k(y)} \operatorname{Spec} \overline{k(y)}.$$

Geometric integrality means exactly that this new scheme is still integral.

But global functions also base change in degree 0 here (the map $\operatorname{Spec} \overline{k(y)} \to \operatorname{Spec} k(y)$ is flat!), so

$$H^0(X_{\overline{k(y)}}, \mathcal{O}_{X_{\overline{k(y)}}}) \cong K \otimes_{k(y)} \overline{k(y)}.$$

So if $K \otimes_{k(y)} \overline{k(y)}$ is not a domain, then $X_{\overline{k(y)}}$ cannot be integral, because an integral scheme has its ring of global sections a domain.

If $K/k(y)$ separable, then over the algebraic closure, K splits:

$$K \otimes_{k(y)} \overline{k(y)} \cong \overline{k(y)}^{[K:k(y)]}.$$

So it is a product of fields and in particular not a domain. A product ring corresponds geometrically to several components, so the base change becomes reducible.

If $K/k(y)$ is purely inseparable, then after base change, nilpotents appear. Consider the simplest case where in characteristic p we have a degree p extension

$$k \subset k(\alpha), \qquad \alpha^p = a \in k,\ \alpha \notin k.$$

Over an algebraic closure $\bar{k}$,

$$K \otimes_k \bar{k} \cong \bar{k}[t]/(t-\beta)^p$$

for some $\beta \in \bar{k}$. This ring has nilpotents, so it is not reduced, hence not integral. So in this case the base change becomes nonreduced. Thus proper and geometrically integral implies that $H^0(X_y, \mathcal{O}_{X_y}) \cong k(y)$. By the hypothesis $\mathcal{L}|_{X_y} \cong \mathcal{O}_{X_y}$, it follows that

$$H^0(X_y, \mathcal{L}_y) \cong k(y),$$

so in particular

$$h^0(X_y, \mathcal{L}_y) = 1 \qquad \text{for all } y \in Y.$$

Now consider the base change map

$$\varphi^0(y) : (\pi_* \mathcal{L}) \otimes_{\mathcal{O}_Y} k(y) \longrightarrow H^0(X_y, \mathcal{L}_y).$$

Because the target has constant dimension 1, cohomology and base change implies that $\pi_*\mathcal{L}$ is locally free of rank 1 on Y, hence an invertible sheaf. Set

$$\mathcal{M} := \pi_*\mathcal{L}.$$

There is a natural adjunction morphism

$$\pi^*\pi_*\mathcal{L} \longrightarrow \mathcal{L},$$

that is,

$$\pi^*\mathcal{M} \longrightarrow \mathcal{L}.$$

To show that this is an isomorphism, it suffices to check on fibers. Restricting to X_y, the induced map is

$$H^0(X_y, \mathcal{L}_y) \otimes_{k(y)} \mathcal{O}_{X_y} \longrightarrow \mathcal{L}_y.$$

Since $\mathcal{L}_y \cong \mathcal{O}_{X_y}$ and $H^0(X_y, \mathcal{L}_y) \cong k(y)$, this is the canonical map

$$k(y) \otimes_{k(y)} \mathcal{O}_{X_y} \longrightarrow \mathcal{O}_{X_y},$$

hence an isomorphism for every $y \in Y$. Therefore

$$\pi^*\mathcal{M} \xrightarrow{\sim} \mathcal{L},$$

as claimed. Uniqueness is immediate: if $\pi^*\mathcal{M}_1 \cong \pi^*\mathcal{M}_2$, then pushing forward and using $\pi_*\mathcal{O}_X \cong \mathcal{O}_Y$ gives $\mathcal{M}_1 \cong \mathcal{M}_2$. $\square$

1.7.4 Higher direct image sheaves

If $f : X \to Y$ is a morphism of schemes and $\mathcal{F}$ is a sheaf on X, then the pushforward $f_*\mathcal{F}$ is a sheaf on Y. Therefore f_* is a functor from the category of sheaves on X to the category of sheaves on Y. This functor is left exact, but not necessarily right exact. The right derived functors of f_* are denoted by R^if_* and are called the **higher direct image functors**.

We explicilty construct R^if_* as we do $R\Gamma$. Let $\mathcal{I}^\bullet$ be an injective resolution of $\mathcal{F}$. Then $f_*\mathcal{I}^\bullet$ is a complex of sheaves on Y, and we define

$$R^if_*\mathcal{F} := \mathcal{H}^i(f_*\mathcal{I}^\bullet).$$

to be the i-th cohomology sheaf of this complex. The resulting R^if_* is independent of the choice of injective resolution, and it is a functor from the category of sheaves on X to the category of sheaves on Y.

It is true that $R^if_*\mathcal{F}$ is the sheafification of the presheaf

$$U \mapsto H^i(f^{-1}(U), \mathcal{F})$$

In particular, the stalk of $R^if_*\mathcal{F}$ at a point $y \in Y$ is given by

$$(R^if_*\mathcal{F})_y \cong \varinjlim_{y\in U} H^i(f^{-1}(U), \mathcal{F}).$$

Moreover it is also true that for affine open subsets $U \subseteq Y$, we have

$$H^0(U,\ R^q\pi_*\mathcal{F}) \cong H^q\Big(\pi^{-1}(U), \mathcal{F}\Big)$$

because of the degeneration of the Leray spectral sequence for the restriction $\pi_U : \pi^{-1}(U) \to U$

$$E_2^{p,q} = H^p(U,\ R^q\pi_*\mathcal{F}|_U) \ \Rightarrow\ H^{p+q}\Big(\pi^{-1}(U), \mathcal{F}\Big)$$

Now use the fact that U is affine to conclude that $H^p(U, -) = 0$ for $p > 0$.

Theorem 1.262. *Suppose $\pi : X \to Y$ is a quasi-compact separated morphism of schemes. Then:*

(a) For each $i \geq 0$, the higher direct image

$$R^i\pi_* : \mathrm{QCoh}_X \to \mathrm{QCoh}_Y$$

is a covariant functor.

(b) There is a natural isomorphism of functors

$$R^0\pi_* \cong \pi_*.$$

(c) If

$$0 \to \mathcal{F} \to \mathcal{G} \to \mathcal{H} \to 0$$

is a short exact sequence of sheaves on X, then there is an induced long exact sequence of higher direct image sheaves on Y:

$$0 \to R^0\pi_*\mathcal{F} \to R^0\pi_*\mathcal{G} \to R^0\pi_*\mathcal{H} \to R^1\pi_*\mathcal{F} \to R^1\pi_*\mathcal{G} \to R^1\pi_*\mathcal{H} \to \cdots.$$

(d) If π is projective, Y is locally Noetherian, and $\mathcal{F}$ is a coherent sheaf on X, then for all i,

$$R^i\pi_*\mathcal{F}$$

is a coherent sheaf on Y.

Example 1.263 (Computation for the projection $\mathbf{P}^1 \times Y \to Y$)**.** *Let Y be a scheme, and let*

$$p : \mathbf{P}^1 \times Y \to Y, \qquad q : \mathbf{P}^1 \times Y \to \mathbf{P}^1$$

be the two projections. We compute the higher direct images of $q^*\mathcal{O}_{\mathbf{P}^1}(n)$.

For every affine open $U = \operatorname{Spec} A \subseteq Y$, we have

$$p^{-1}(U) \cong \mathbf{P}^1_A, \qquad q^*\mathcal{O}_{\mathbf{P}^1}(n)\Big|_{p^{-1}(U)} \cong \mathcal{O}_{\mathbf{P}^1_A}(n).$$

Hence

$$(R^ip_*q^*\mathcal{O}_{\mathbf{P}^1}(n))(U) \cong H^i(\mathbf{P}^1_A, \mathcal{O}_{\mathbf{P}^1_A}(n)).$$

Since cohomology of $\mathcal{O}_{\mathbf{P}^1}(n)$ commutes with extension of scalars in this case (because all of the terms in the Čech complex for the standard affine cover are free A-modules), we have

$$H^i(\mathbf{P}^1_A, \mathcal{O}_{\mathbf{P}^1_A}(n)) \cong H^i(\mathbf{P}^1_k, \mathcal{O}_{\mathbf{P}^1}(n)) \otimes_k A.$$

Therefore

$$R^i p_* q^* \mathcal{O}_{\mathbf{P}^1}(n) \cong H^i(\mathbf{P}^1, \mathcal{O}_{\mathbf{P}^1}(n)) \otimes_k \mathcal{O}_Y.$$

Now recall the standard cohomology of line bundles on $\mathbf{P}^1$:

$$h^0(\mathbf{P}^1, \mathcal{O}(n)) = \begin{cases} n+1, & n \geq 0, \\ 0, & n < 0, \end{cases} \qquad h^1(\mathbf{P}^1, \mathcal{O}(n)) = \begin{cases} 0, & n \geq -1, \\ -n-1, & n \leq -2. \end{cases}$$

Thus

$$p_* q^* \mathcal{O}_{\mathbf{P}^1}(n) \cong \begin{cases} \mathcal{O}_Y^{\oplus(n+1)}, & n \geq 0, \\ 0, & n < 0, \end{cases}$$

and

$$R^1 p_* q^* \mathcal{O}_{\mathbf{P}^1}(n) \cong \begin{cases} 0, & n \geq -1, \\ \mathcal{O}_Y^{\oplus(-n-1)}, & n \leq -2. \end{cases}$$

Moreover,

$$R^i p_* q^* \mathcal{O}_{\mathbf{P}^1}(n) = 0 \qquad \text{for all } i \geq 2,$$

since the fibers of p are 1-dimensional.

1.8 Curves

1.8.1 Structure theory

Theorem 1.264 (Local normal form and ramification index)**.** *Let $f : R \to S$ be a nonconstant holomorphic map between Riemann surfaces, and let $p \in R$ with $q := f(p)$. Then there exist local holomorphic coordinates*

$$z : (R, p) \xrightarrow{\sim} (\mathbb{C}, 0), \qquad w : (S, q) \xrightarrow{\sim} (\mathbb{C}, 0),$$

and an integer $k \geq 1$ such that

$$w \circ f \circ z^{-1}(\zeta) \;=\; \zeta^k$$

for all ζ in a neighbourhood of 0. The integer k is uniquely determined and is called the ***ramification index*** *of f at p.*

Corollary 1.265. *Let $f : R \to S$ be a nonconstant holomorphic map between Riemann surfaces. Then f is an open map.*

Corollary 1.266. *Every holomorphic function defined everywhere on a compact connected Riemann surface is constant.*

Exercise 1.267. *Show that if R is a compact Riemann surface, $R - \{p\}$ is affine for every $p \in R$.*

Proof. By Riemann-Roch for sufficiently large n, there exists

$$f \in H^0(C, \mathcal{O}_C(np)) \setminus H^0(C, \mathcal{O}_C((n-1)p))$$

This means f has pole of order exactly n at p, and no poles elsewhere.

So f defines a nonconstant morphism $f : C \to \mathbf{P}^1$ whose only pole is at p. Therefore $f^{-1}(\infty) = \{p\}$. Restricting gives $f|_U : U = C \setminus \{p\} \to \mathbf{A}^1$.

Since any morphism from a projective curve to $\mathbf{P}^1$ is finite when nonconstant, f is finite, hence so is $f|_U$. Therefore U is finite over $\mathbf{A}^1$, so U is affine. □

1.8.2 Genus and degree

Theorem 1.268. *Let $f : R \to S$ be a nonconstant holomorphic map between compact Riemann surfaces. Pick $s \in S$ and define the quantity*

$$d := \sum_{p \in f^{-1}(s)} e_p$$

Then d is independent of the choice of $s \in S$ and is called the ***degree*** *of the map f.*

Theorem 1.269 (Riemann–Hurwitz)**.** *Let $f : R \to S$ be a nonconstant holomorphic map of compact Riemann surfaces of degree d. Write*

$$B \;:=\; \sum_{p \in R} (e_p - 1)\, p$$

for the ramification divisor of f. Then

$$2g_R - 2 \;=\; d(2g_S - 2) + \deg(B),$$

or equivalently,

$$\chi(R) \;=\; d\,\chi(S) - \sum_{p \in R} (e_p - 1).$$

Proof via Euler characteristic. If f were an unramified covering map, then R would be a d-fold cover of S and by the description of Euler characteristic as the alternating sum of cells in a cell decomposition, we would have

$$\chi(R) = d\,\chi(S).$$

However, ramification changes this count. We have to delete $e_p - 1$ points of R for each ramification point p, and each deletion drops the Euler characteristic by 1. Thus we obtain the desired formula. □

Proof via canonical bundles. Let K_R and K_S denote the canonical bundles of R and S. Choose local coordinates w near $p \in R$ and z near $f(p)$ so that the map is written $z = w^{e_p}$. Then

$$f^*(dz) = d(w^{e_p}) = e_p\, w^{e_p - 1}\, dw,$$

so $f^*(dz)$ has a zero of order $e_p - 1$ at p. Globalizing, one obtains an isomorphism of line bundles

$$K_R \cong f^* K_S \otimes \mathcal{O}_R(B),$$

where $B = \sum(e_p - 1)p$ is the ramification divisor.

Taking degrees on both sides, and using $\deg K_X = 2g_X - 2$, yields

$$2g_R - 2 = d(2g_S - 2) + \sum_p (e_p - 1),$$

the desired identity. □

Remark 1.270. *These two proofs are related by the fact a holomorphic* 1*–form* ω *on a compact Riemann surface satisfies* $\deg(\omega) = 2g - 2 = -\chi(X)$*. In particular, there is the isomorphism of line bundles* $K_R = f^* K_S + B$*. which refines the Riemann–Hurwitz formula.*

Theorem 1.271 (Adjunction formula for curves)**.** *If* X *is a smooth variety and* $D \subset X$ *is a smooth effective Cartier divisor, then*

$$\omega_D \cong (\omega_X \otimes \mathcal{O}_X(D))|_D.$$

Equivalently, at the divisor-class level,

$$K_D = (K_X + D)|_D.$$

Proof. The conormal exact sequence

$$0 \to \mathcal{I}_D/\mathcal{I}_D^2 \to \Omega_X^1|_D \to \Omega_D^1 \to 0$$

becomes, since D is Cartier,

$$0 \to \mathcal{O}_X(-D)|_D \to \Omega_X^1|_D \to \Omega_D^1 \to 0.$$

Taking determinants gives

$$\omega_D \cong \omega_X|_D \otimes \mathcal{O}_X(D)|_D.$$

□

Exercise 1.272 (POAC 2.2)**.** *If* $C \subset \mathbb{P}^r$ *is a 1-dimensional variety with normalization* $\varphi : \widetilde{C} \to C$*, and* $X \subset \mathbb{P}^r$ *is any subscheme that does not contain* C*, we define the multiplicity of intersection* $\mathrm{mult}_p(C, X)$ *at the point* $p \in X \cap C$ *to be the sum of the lengths of the finite scheme* $\varphi^{-1}(X)$ *at the points of* $\varphi^{-1}(p)$*. Show that* C *is singular at* p *if and only if* $\mathrm{mult}_p(C, X) \geq 2$ *for all* X *containing* p*. Show further that if* $H \subset \mathbb{P}^r$ *is a hyperplane then*

$$\deg C = \sum_{p \in C} \mathrm{mult}_p(C, H).$$

Solution 1.273.

1. *Let I_X be the ideal sheaf of X in C, that is it is a subsheaf of $\mathcal{O}_C$ such that for any open set $U \subseteq C$, $I_X(U)$ is the ideal of $\mathcal{O}_C(U)$ consisting of functions that vanish on $X \cap U$. By definition, we have the formula*

$$\mathrm{mult}_p(C, X) = \sum_{q \in \varphi^{-1}(p)} \ell(\mathcal{O}_{\phi^{-1}(X),q}) = \sum_{q \in \varphi^{-1}(p)} \ell(\mathcal{O}_{\widetilde{C},q}/I_X\mathcal{O}_{\widetilde{C},q}).$$

where the last equality follows from the definition of the scheme theoretic pullback.

Recall that if $f : X \to Y$ is a morphism of schemes and let $Z \hookrightarrow Y$ be a closed subscheme defined by an ideal sheaf $\mathcal{I}_Z \subset \mathcal{O}_Y$. Then the scheme-theoretic preimage is $f^{-1}(Z) := X \times_Y Z = \mathrm{Spec}_X\big(\mathcal{O}_X/\mathcal{I}_Z\mathcal{O}_X\big)$. So at the sheaf level, $\mathcal{O}_{f^{-1}(Z)} = \mathcal{O}_X/\mathcal{I}_Z\mathcal{O}_X$.

Recall that the local ring $\mathcal{O}_{\tilde{C},q}$ is a discrete valuation ring and every nonzero ideal in a DVR is principal, generated by a power of a uniformizer t.

Let $q_1, \ldots, q_s$ be the points of $\widetilde{C}$ over p (the branches). If $s \geq 2$ (multibranch singularity, e.g. a node), then for any $X \ni p$, $\phi^* I_X$ vanishes at each branch, so $\mathrm{ord}_{q_i}(\phi^* I_X) \geq 1$ for all i. Hence $\mathrm{mult}_p(C, X) = \sum_i \mathrm{ord}_{q_i}(\phi^* I_X) \geq s \geq 2$. If $s = 1$ (unibranch singularity, e.g. a cusp), let q be the unique point over p with parameter t. Locally we have $A = \mathcal{O}_{C,p}$ is a 1-dimensional Noetherian local ring that is not regular (since p is singular). $R = \mathcal{O}_{\widetilde{C},q}$ is a DVR, with uniformizer t. So $A \subset R \subset \mathrm{Frac}(A)$. If A has an element of order 1 in R, then $A = R$ is regular, contradicting the singularity of p. So $\mathrm{ord}_q(\phi^* I_X) \geq 2$ for all $X \ni p$, hence $\mathrm{mult}_p(C, X) = \mathrm{ord}_q(\phi^* I_X) \geq 2$.

Conversely, if p is nonsingular, then we can take $X = H$ the tangent hyperplane to C at p. Then $\phi^* I_H$ has order 1 at the unique point q over p, so $\mathrm{mult}_p(C, H) = 1$.

Let $L := \mathcal{O}_{\mathbb{P}^r}(1)|_C$. A hyperplane H corresponds to a section $s_H \in H^0(C, L)$. Pulling back, $\phi^* s_H \in H^0(\widetilde{C}, \phi^* L)$ and its divisor is exactly $\phi^{-1}(H)$. Therefore

$$\deg C = \deg L = \deg \phi^* L = \deg \mathrm{div}(\phi^* s_H) = \sum_{p \in C} \mathrm{mult}_p(C, H)$$

This holds for any hyperplane H not containing C.

Remark 1.274. *Note that if you take a nodal curve $C \subset \mathbb{P}^2 \subset \mathbb{P}^3$ and pick a $\mathbb{P}^1$ which passes through the node transversely, then the resulting intersection is in fact reduced and smooth. For example if you take a nodal curve $(w,\ xy)$ and $(x - \beta w,\ y - \alpha w)$ and $\alpha, \beta \in k$. Then $(w, xy) + (x - \beta w,\ y - \alpha w) = (w, x, y)$, I was just suprised that you can start with something singular, intersect with something else and get something smooth. This is the basis of Bertini-type theorems: a general enough hyperplane cut of a singular variety often avoids carrying the singularity forward, unless you cut right along the tangent cone.*

Exercise 1.275. *Let $Q = \mathbb{P}^1 \times \mathbb{P}^1$, and let $C \subset Q$ be a smooth curve of bidegree $(4, 3)$.*

1. *Compute the genus of C.*

2. *Let $\pi := p_1|_C : C \to \mathbb{P}^1$ be the restriction to C of the first projection Compute the degree of π, and determine the total ramification of π.*

3. *Let $L = \mathcal{O}_Q(4,3)$. Compute $p_{1*}L$.*

4. *Let $L_C = \mathcal{O}_C(C) = \mathcal{O}_Q(4,3)|_C$, the nomal bundle of C in Q. Describe $\pi_* L_C$.*

Note that something I learned in this exercise is that $\mathcal{O}_Q(-C)$ is the ideal sheaf of C in Q, and $\mathcal{O}_C(C)$ is the normal bundle of C in Q. The first makes sense because if one asks for functions vanishing along C, then these are of course the functions that cut out C as a subscheme. If f is a local defining equation for C in Q, then we are saying that $1/f$ is a local generator of the normal bundle $\mathcal{O}_Q(C)$, which makes sense because a local section looks like g/f with g a regular function on Q, and restricting to C corresponds to modding out by $\mathcal{O}_Q$ (recall the following exact sequence):

$$0 \to \mathcal{O}_Q \to \mathcal{O}_Q(C) \to \mathcal{O}_C(C) \to 0$$

so we remember only the first order data of g mod f.

Solution 1.276. *Write F_1 and F_2 for the divisor classes of the two rulings on $Q = \mathbb{P}^1 \times \mathbb{P}^1$. Then $C \sim 4F_1 + 3F_2$.*

(1) Genus. *Q is a product and for smooth varieties $K_{X\times Y} \cong p_X^* K_X \otimes p_Y^* K_Y$. Therefore, the canonical class on Q is $K_Q = -2F_1 - 2F_2$ By adjunction,*

$$2g(C) - 2 = C \cdot (C + K_Q).$$

Now

$$C + K_Q = (4F_1 + 3F_2) + (-2F_1 - 2F_2) = 2F_1 + F_2.$$

Using

$$F_1^2 = F_2^2 = 0, \qquad F_1 \cdot F_2 = 1,$$

we get

$$C \cdot (C + K_Q) = (4F_1 + 3F_2) \cdot (2F_1 + F_2) = 4 \cdot 1 + 3 \cdot 2 = 10$$

It follows that $g(C) = 6$. In general for a smooth curve of type (a, b) on $\mathbb{P}^1 \times \mathbb{P}^1$, one has

$$g(C) = (a-1)(b-1),$$

(2) Degree and ramification of π. *The degree of $\pi = p_1|_C$ is the number of intersection points of C with a fiber of p_1, counted with multiplicity. A fiber of p_1 has class F_1, so*

$$\deg(\pi) = C \cdot F_1 = (4F_1 + 3F_2) \cdot F_1 = 3.$$

Now apply the Riemann–Hurwitz formula:

$$2g(C) - 2 = \deg(\pi)\Big(2g(\mathbb{P}^1) - 2\Big) + \deg(R),$$

where R is the ramification divisor. Since $g(C) = 6$, $g(\mathbb{P}^1) = 0$, and $\deg(\pi) = 3$, we obtain

$$2 \cdot 6 - 2 = 3(-2) + \deg(R).$$

so $\deg(R) = 16$.

(3) The pushforward $p_{1*}L$***.*** *Since*

$$L = \mathcal{O}_Q(4,3) \cong p_1^*\mathcal{O}_{\mathbb{P}^1}(4) \otimes p_2^*\mathcal{O}_{\mathbb{P}^1}(3),$$

the projection formula gives

$$p_{1*}L \cong \mathcal{O}_{\mathbb{P}^1}(4) \otimes p_{1*}\Big(p_2^*\mathcal{O}_{\mathbb{P}^1}(3)\Big).$$

But

$$p_{1*}\Big(p_2^*\mathcal{O}_{\mathbb{P}^1}(3)\Big) \cong H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(3)) \otimes \mathcal{O}_{\mathbb{P}^1},$$

is the trivial vector bundle of rank 4 *on* $\mathbb{P}^1$*. This is because* $p_1 : \mathbb{P}^1 \times \mathbb{P}^1 \to \mathbb{P}^1$ *is the projection from a product, and the sheaf* $p_2^*\mathcal{O}_{\mathbb{P}^1}(3)$ *is constant in the first factor and equals* $\mathcal{O}_{\mathbb{P}^1}(3)$ *along each fiber of* p_1*. Therefore*

$$p_{1*}\mathcal{O}_Q(4,3) \cong \mathcal{O}_{\mathbb{P}^1}(4)^{\oplus 4}.$$

(4) The pushforward $\pi_* L_C$***.*** *Consider the exact sequence on* Q*:*

$$0 \to \mathcal{O}_Q \to \mathcal{O}_Q(4,3) \to \mathcal{O}_C(C) \to 0.$$

Pushing forward by p_1*, and using*

$$p_{1*}\mathcal{O}_Q \cong \mathcal{O}_{\mathbb{P}^1}, \qquad R^1p_{1*}\mathcal{O}_Q = 0, \qquad R^1p_{1*}\mathcal{O}_Q(4,3) = 0,$$

we obtain an exact sequence

$$0 \to \mathcal{O}_{\mathbb{P}^1} \to \mathcal{O}_{\mathbb{P}^1}(4)^{\oplus 4} \to \pi_*\mathcal{O}_C(C) \to 0.$$

Hence $\pi_*\mathcal{O}_C(C)$ *is a rank* 3 *vector bundle on* $\mathbb{P}^1$*, namely the cokernel of*

$$\mathcal{O}_{\mathbb{P}^1} \longrightarrow \mathcal{O}_{\mathbb{P}^1}(4)^{\oplus 4}.$$

Its degree is

$$\deg\Big(\pi_*\mathcal{O}_C(C)\Big) = 4 \cdot 4 = 16.$$

By the Birkhoff–Grothendieck theorem, $\pi_*\mathcal{O}_C(C)$ *splits as a direct sum of three line bundles on* $\mathbb{P}^1$ *whose degrees add up to* 16*, but the exact splitting type depends on the particular curve* C*.*

Exercise 1.277. *Reason about the following facts about a smooth plane curve* $C \subset \mathbb{P}^2$ *of degree* d*:*

1. $g(C) = \frac{(d-1)(d-2)}{2}$.
2. $\omega_C \cong \mathcal{O}_C(d-3)$.
3. $\deg \mathcal{O}_C(1) = d$.

Solution 1.278. *A smooth plane curve $C \subset \mathbb{P}^2$ of degree d fits into*

$$0 \to \mathcal{O}_{\mathbb{P}^2}(-d) \to \mathcal{O}_{\mathbb{P}^2} \to \mathcal{O}_C \to 0.$$

Twisting by $\mathcal{O}(n)$ and using the cohomology of $\mathbb{P}^2$ (recall that $\chi(\mathcal{O}_{\mathbb{P}^2}(n)) = \binom{n+2}{2}$) one computes

$$\chi(\mathcal{O}_C(n)) = \binom{n+2}{2} - \binom{n-d+2}{2} = dn - \frac{d(d-3)}{2}.$$

Note that $\chi(\mathcal{O}_C(n))$ agrees with the Hilbert function for large n by the vanishing of $H^1(\mathcal{O}_C(n))$ for large n. An application of Riemann–Roch gives

$$\chi(\mathcal{O}_C(n)) = \deg \mathcal{O}_C(n) + 1 - g = dn + 1 - g,$$

it follows that

$$g = \frac{(d-1)(d-2)}{2}.$$

Also adjunction gives

$$\omega_C \cong (K_{\mathbf{P}^2} \otimes \mathcal{O}_{\mathbf{P}^2}(C))|_C \cong \mathcal{O}_C(d-3).$$

For the third statement, $\deg \mathcal{O}_C(1) = C \cdot H = d$ where H is a hyperplane class in $\mathbb{P}^2$, since C has degree d by definition.

Exercise 1.279. *Give an example of a smooth curve of genus 2.*

Solution 1.280. *A standard example is the smooth projective completion of $y^2 = x^5 - 1$. More conceptually, any smooth double cover of $\mathbb{P}^1$ branched at 6 points has genus 2. Another realization is as a smooth divisor of type $(3, 2)$ on $\mathbb{P}^1 \times \mathbb{P}^1$, since such a curve has genus $(3-1)(2-1) = 2$.*

Exercise 1.281. *What is the genus of the projective completion of the affine curve defined by $x^n - y^n = 1$? Use the degree-genus formula for plane curves, and then use Riemann–Hurwitz to check your answer.*

Solution 1.282. *The projective closure is $X^n - Y^n + Z^n = 0 \subset \mathbb{P}^2$, which is smooth if $\operatorname{char} k \nmid n$. Since it is a smooth plane curve of degree n, its genus is*

$$\frac{(n-1)(n-2)}{2}.$$

Alternatively, project to the y-coordinate:

$$\pi : C \to \mathbb{P}^1.$$

This has degree n. It is branched exactly over the n points with $y^n = 1$, and over each such point there is one totally ramified point of index n. So the total ramification is $n(n-1)$. Riemann–Hurwitz gives

$$2g - 2 = -2n + n(n-1),$$

hence

$$g = \frac{(n-1)(n-2)}{2}.$$

1.8.3 Riemann–Roch theorem

Let R be a compact Riemann surface, and let L be a holomorphic line bundle on R. The Riemann–Roch theorem is the best possible answer to the question of finding the dimension of the space $\Gamma(R; L)$ of holomorphic sections of a line bundle by topological methods. This question does not have a purely topological answer, as seen from the following.

Example 1.283. *Let $g(R) > 0$, fix a point $y \in R$ and consider the family of divisors $D_x = x - y$ on R, parametrized by $x \in R$. For $x \neq y$, D_x is not principal, because a meromorphic function with a simple pole at x and no other poles would describe a degree one map $R \to \mathbb{P}^1$, which would have to be an isomorphism. However, $D_y = 0$ has a nontrivial holomorphic section: the function 1. So the dimension of the space of holomorphic sections can jump in a family. Note that all line bundles $\mathcal{O}(D_x)$ have degree zero, so they are topologically trivial. This means that no topological information will detect* $\dim \Gamma(R; L)$.

The topologically invariant quantity turns out to be

$$\chi(R; L) := \dim H^0(R; L) - \dim H^1(R; L).$$

Definition 1.284. *The space $\mathcal{P}_x(V)$ of **principal parts** at $x \in R$ with coefficients in a holomorphic vector bundle V is the quotient*

$$\mathcal{M}_x(V)/\mathcal{O}_x(V)$$

of the space of germs at x of meromorphic sections of V by the subspace of germs of holomorphic sections. The space $\mathcal{P}(R; V)$ of principal parts over R is the direct sum, over all points $x \in R$, of the $\mathcal{P}_x(V)$.

In a local coordinate centred at x, $\mathcal{P}_x(V)$ is the space of negative Laurent polynomials with values in V. For later use, we note that $\mathcal{P}(R; V)$ is the space of sections of a sheaf $\mathcal{P}(V)$ over R; however, in order to satisfy the sheaf condition for (possibly infinite) coverings, one must allow on a general open set U a distribution of principal parts over any discrete, but possibly infinite set. It is the compactness of R that forces the restriction to finitely many points, and limits us to the direct sum. There is a linear map induced by the morphism of sheaves $\mathcal{M}(V) \to \mathcal{P}(V)$

$$p : \Gamma(R; \mathcal{M}(V)) \longrightarrow \Gamma(R; \mathcal{P}(V))$$

which assigns to each meromorphic section its principal parts. The kernel of p consists of (the sheaf of) holomorphic sections of V.

Definition 1.285. *The **first cohomology group** $H^1(R; V)$ with coefficients in the (sheaf of holomorphic sections of the) vector bundle V is the ratio*

$$H^1(R; V) := \mathcal{P}(R; V)/\operatorname{Im}(p),$$

that is, principal parts modulo global meromorphic sections.

In other words, we have a four-term exact sequence

$$0 \longrightarrow H^0(R;V) \longrightarrow \Gamma(R;\mathcal{M}(V)) \xrightarrow{p} \mathcal{P}(R;V) \longrightarrow H^1(R;V) \longrightarrow 0 \qquad (2)$$

The 0 is justified by the fact that $H^1(\mathcal{M}(V)) = 0$ where $\mathcal{M}(V)$ is the sheaf of meromorphic sections of V, and $\mathcal{P} = \mathcal{M}/\mathcal{O}$ is the quotient of the sheaf of meromorphic sections by the sheaf of holomorphic sections, so the long exact sequence in cohomology gives the above short exact sequence. The vanishing of $H^1(\mathcal{M}(V))$ is a consequence of the following fact.

Exercise 1.286. *Show that, on a compact Hausdorff space X, sheaf cohomology commutes with filtered colimits: if $\mathcal{S} = \varinjlim \mathcal{S}_n$, then*

$$\varinjlim H^q(X;\mathcal{S}_n) = H^q(X;\mathcal{S}).$$

Use this to prove that the sheaf of principal parts of meromorphic sections of a vector bundle on a Riemann surface has no H^1 or higher cohomology.

Solution 1.287. *Let X compact Hausdorff and $\{\mathcal{S}_\alpha\}$ a filtered direct system of sheaves. Let $\mathcal{S} = \varinjlim_\alpha \mathcal{S}_\alpha$ be the colimit in sheaves.*

Take a finite open cover $\mathfrak{U} = \{U_i\}_{i=1}^m$ of X (exists since X is compact). For any p,

$$C^p(\mathfrak{U},\mathcal{S}) = \prod_{i_0<\cdots<i_p} \mathcal{S}(U_{i_0} \cap \cdots \cap U_{i_p}).$$

We will use a couple of lemmas and justify them at the end of the solution.

Lemma 1.288. *Filtered colimits are computed stalkwise and commute with finite products.*

The lemma implies that the Čech complex for $\mathcal{S}$ is the filtered colimit of the Čech complexes for $\mathcal{S}_\alpha$ since we have the following isomorphisms which commute with the differentials:

$$C^p(\mathfrak{U},\mathcal{S}) = \prod_{i_0<\cdots<i_p} \mathcal{S}(U_{i_0} \cap \cdots \cap U_{i_p}) = \prod_{i_0<\cdots<i_p} \varinjlim_\alpha \mathcal{S}_\alpha(U_{i_0} \cap \cdots \cap U_{i_p})$$

$$\cong \varinjlim_\alpha \prod_{i_0<\cdots<i_p} \mathcal{S}_\alpha(U_{i_0} \cap \cdots \cap U_{i_p}) = \varinjlim_\alpha C^p(\mathfrak{U},\mathcal{S}_\alpha).$$

Lemma 1.289. *Filtered colimits of abelian groups are exact, and in particular taking cohomology commutes with $\varinjlim$:*

$$\check{H}^q(\mathfrak{U};\mathcal{S}) \cong \varinjlim_\alpha \check{H}^q(\mathfrak{U};\mathcal{S}_\alpha).$$

On a compact Hausdorff space (in particular, on a Riemann surface) Čech cohomology with respect to a good cover computes sheaf cohomology. Hence

$$H^q(X;\mathcal{S}) \cong \varinjlim_\alpha H^q(X;\mathcal{S}_\alpha)$$

for all q.

For the second part, let R be a Riemann surface and V a holomorphic vector bundle. For each effective divisor D on R, let $\mathcal{P}_D(V)$ be the sheaf of principal parts of meromorphic sections of V with poles bounded by D.

Observe that $\mathcal{P}_D(V)$ is supported on the finite set $\mathrm{Supp}(D)$*, so it is a finite direct sum of skyscraper sheaves.*

Lemma 1.290. *A sheaf on R with finite support has no higher cohomology. In particular, for $q \geq 1$,*

$$H^q(R; \mathcal{P}_D(V)) = 0,$$

The full principal parts sheaf is the filtered colimit over all effective divisors:

$$\mathcal{P}(V) = \varinjlim_{D} \mathcal{P}_D(V),$$

directed by $D \leq D'$ if D divides D'. Applying the result from the first part:

$$H^q(R; \mathcal{P}(V)) \cong \varinjlim_{D} H^q(R; \mathcal{P}_D(V)).$$

But each $H^q(R; \mathcal{P}_D(V)) = 0$ for $q \geq 1$, so the colimit is 0. Therefore

$$H^q(R; \mathcal{P}(V)) = 0 \quad \textit{for all } q \geq 1.$$

Now we justify the lemmas used above.

Proof of Lemma 1.288. Define a presheaf $\mathcal{F}(U) := \varinjlim_{\alpha} \mathcal{S}_\alpha(U)$. We claim $\mathcal{F}$ is already a sheaf. Then $\mathcal{F}$ is the colimit in the sheaf category, so for any open U, we have

$$\mathcal{S} := \varinjlim_{\alpha} \mathcal{S}_\alpha \quad \text{satisfies} \quad \mathcal{S}(U) = \varinjlim_{\alpha} \mathcal{S}_\alpha(U)$$

To show $\mathcal{F}$ is a sheaf, take an open cover $U = \bigcup_i U_i$. For separatedness, suppose $s \in \mathcal{F}(U)$ restricts to zero in every $\mathcal{F}(U_i)$. Represent s by some $s_\alpha \in \mathcal{S}_\alpha(U)$. Its restriction to $\mathcal{F}(U_i)$ is the image of $s_\alpha|_{U_i}$. If that image is zero in the colimit, then for each i there is some stage $\beta_i \geq \alpha$ with $s_{\beta_i}|_{U_i} = 0$. Since the system is filtered, find γ dominating all β_i. Then $(s_\gamma)|_{U_i} = 0$ for all i, so by the sheaf property of $\mathcal{S}_\gamma$, $s_\gamma = 0$, hence $s = 0$ in the colimit.

For gluing, suppose we have $s_i \in \mathcal{F}(U_i)$ that agree on overlaps. Represent each s_i by some $s_{i,\alpha_i} \in \mathcal{S}_{\alpha_i}(U_i)$. Filteredness gives a stage β dominating all α_i. Push all s_{i,α_i} to $\mathcal{S}_\beta$; they still agree on overlaps, so they glue to some $s_\beta \in \mathcal{S}_\beta(U)$. Its class in $\mathcal{F}(U)$ glues the original s_i.

Therefore the presheaf colimit is a sheaf, and $\mathcal{S}(U) = \varinjlim_{\alpha} \mathcal{S}_\alpha(U)$. Thus filtered colimits are computed stalkwise, so it is enough to show that in the category of abelian groups filtered colimits commute with finite products. $\square$

Proof of Lemma 1.289. Take a short exact sequence of direct systems

$$0 \to A_\alpha \xrightarrow{f_\alpha} B_\alpha \xrightarrow{g_\alpha} C_\alpha \to 0,$$

with all maps compatible. We need to show that

$$0 \to \varinjlim A_\alpha \to \varinjlim B_\alpha \to \varinjlim C_\alpha \to 0$$

is exact.

Left-exactness (injectivity at $\varinjlim A_\alpha$) holds for any colimit: if an element becomes zero in $\varinjlim B_\alpha$, it is already zero after some stage, and then in A_α by exactness there.

Surjectivity at $\varinjlim C_\alpha$ is where filteredness is used. Let $[c_\alpha] \in \varinjlim C_\alpha$. Pick representative $c_\alpha \in C_\alpha$. Exactness at C_α gives $b_\alpha \in B_\alpha$ with $g_\alpha(b_\alpha) = c_\alpha$. Let $[b_\alpha]$ be its class in $\varinjlim B_\alpha$; its image is $[c_\alpha]$.

We need to check this does not depend on choices: if we change stage or lift, filteredness gives a common stage where the choices agree, because kernels and images are compatible. That is standard and uses only that the system is filtered.

Hence the right map is surjective and the colimit of a short exact sequence is exact. □

Proof of Lemma 1.290. Let $\mathcal{F}$ be a sheaf on R with finite support $\{p_1, \ldots, p_n\}$. Then

$$\mathcal{F} \cong \bigoplus_{i=1}^{n} (i_{p_i})_* \mathcal{F}_{p_i},$$

where $i_{p_i} : \{p_i\} \hookrightarrow R$ is the inclusion. Each $(i_{p_i})_*\mathcal{F}_{p_i}$ is a skyscraper sheaf supported at p_i.

Skyscraper sheaves are flasque. For $\mathcal{G} = (i_x)_* A$, where A is an abelian group and $x \in R$, we have for $V \subset U$, the restriction map $\mathcal{G}(U) \to \mathcal{G}(V)$ is either:

$$\begin{cases} \mathrm{id}_A & \text{if both contain } x \\ A \to 0 & \text{if } x \in U, x \notin V \\ 0 \to 0 & \text{otherwise} \end{cases}$$

In all cases it is surjective. Therefore every restriction map is surjective, so $\mathcal{G}$ is flasque.

Finite direct sums of flasque sheaves are flasque since surjectivity is preserved under finite products. Flasque sheaves have vanishing higher cohomology. Since cohomology commutes with finite direct sums, we conclude that $H^q(R; \mathcal{F}) = 0$ for $q \geq 1$. □

Example 1.291 (Cohomology of $\mathbb{P}^1$ via principal parts)**.** *We compute $H^1(\mathbb{P}^1, \mathcal{O}(-n \cdot 0))$ using principal parts. Write t for the coordinate on $\mathbb{A}^1 = \mathbb{P}^1 \setminus \{\infty\}$ and $s = 1/t$ for a local coordinate at ∞.*

Let $\{a_i\}$ be finitely many points in $\mathbb{A}^1$, and prescribe principal parts

$$\sum_{k=1}^{N_i} \frac{c_{i,k}}{(t - a_i)^k}$$

at each a_i. Then there exists a rational function

$$f_0(t) = \sum_i \sum_{k=1}^{N_i} \frac{c_{i,k}}{(t-a_i)^k}$$

realizing these principal parts. Thus principal parts away from ∞ always lift to global meromorphic functions. If f and g are two rational functions with the same principal parts at all finite points, then $f-g$ has no poles on $\mathbb{A}^1$, hence is a polynomial. Thus all such functions are of the form

$$f(t) = f_0(t) + P(t), \qquad P(t) \in k[t].$$

Now we consider the principal part at ∞. For $\mathcal{O}(-n) \cong \mathcal{O}(-n\infty)$, the local condition at ∞ is

$$\mathcal{O}(-n)_\infty = s^n \mathcal{O}_\infty,$$

so the space of principal parts at ∞ is

$$\mathcal{P}_\infty(\mathcal{O}(-n)) = \mathcal{M}_\infty / s^n \mathcal{O}_\infty.$$

Thus a Laurent expansion

$$f(1/s) = \sum_{m \gg -\infty} a_m s^m$$

is remembered modulo s^n, so the principal part at ∞ consists of the terms

$$a_{-1}s^{-1} + a_{-2}s^{-2} + \cdots + a_0 + a_1 s + \cdots + a_{n-1}s^{n-1}.$$

A polynomial

$$P(t) = a_0 + a_1 t + \cdots + a_d t^d$$

expands at ∞ as

$$P(1/s) = a_0 + a_1 s^{-1} + \cdots + a_d s^{-d}.$$

Thus adding a polynomial can change the coefficients of 1, s^{-1}, s^{-2}, $\ldots$ but cannot affect the coefficients of s, s^2, $\ldots$, s^{n-1}. Therefore, after fixing principal parts at finite points, the coefficients of s, s^2, $\ldots$, s^{n-1} at ∞ cannot be adjusted by any global meromorphic function. These give independent obstructions. Hence

$$\dim H^0(\mathbb{P}^1; \mathcal{O}(n)) = \begin{cases} n+1, & n \geq -1, \\ 0, & n < -1, \end{cases} \qquad \dim H^1((\mathbb{P}^1; \mathcal{O}(n)) = \begin{cases} 0, & n \geq -1, \\ -n-1, & n < -1. \end{cases}$$

Theorem 1.292 (Riemann–Roch theorem for vector bundles)**.** *Let R be a compact Riemann surface of genus g, and let V be a holomorphic vector bundle of rank r and degree d on R. Then the Euler characteristic of V is given by*

$$\chi(R, V) = \dim H^0(R, V) - \dim H^1(R, V) = d + r(1-g)$$

The following lemma has an elemntary proof, but it also immediately follows from the fact that proper pushforward preserves coherence, and any finitely generated module over a regular local ring is free.

Lemma 1.293. *Let $f : X \to Y$ be a branched cover of Riemann surfaces. If $\mathcal{F}$ is a locally free $\mathcal{O}_X$-module, then $f_*\mathcal{F}$ is a locally free $\mathcal{O}_Y$-module. The rank of $f_*\mathcal{F}$ is $n \cdot \operatorname{rank} \mathcal{F}$, where n is the degree of the cover.*

Proof. Suppose y is not a branch point. Then the fibre $f^{-1}(y)$ consists of n distinct points $x_1, \ldots, x_n$ and there are neighbourhoods U of y and V_i of x_i such that

$$f^{-1}(U) = \bigsqcup_{i=1}^{n} V_i \quad \text{and} \quad f|_{V_i} : V_i \to U$$

is a biholomorphism.

Then

$$(f_*\mathcal{F})(U) = \mathcal{F}(f^{-1}(U)) = \bigoplus_{i=1}^{n} \mathcal{F}(V_i)$$

Shrinking U if necessary, we may assume that $\mathcal{F}$ is trivial on each V_i, so that

$$\mathcal{F}(V_i) \cong \mathcal{O}_X(V_i)^{\oplus r}$$

where r is the rank of $\mathcal{F}$. Since $f|_{V_i}$ is a biholomorphism, we have

$$\mathcal{O}_X(V_i) \cong \mathcal{O}_Y(U)$$

Therefore,

$$(f_*\mathcal{F})(U) \cong \bigoplus_{i=1}^{n} \mathcal{O}_Y(U)^{\oplus r} \cong \mathcal{O}_Y(U)^{\oplus nr}$$

showing that $f_*\mathcal{F}$ is locally free of rank nr near y.

Now suppose that y is a branch point. To show $f_*\mathcal{F}$ is a vector bundle, we must show that its stalks are free $\mathcal{O}_{Y,y}$-modules. By definition, for any sheaf $\mathcal{F}$ on X we have

$$(f_*\mathcal{F})_y \;=\; \varinjlim_{y \in U} \mathcal{F}\Big(f^{-1}(U)\Big).$$

Given a finite map, the topology of the fiber decomposes the stalk into a direct sum over preimages:

$$(f_*\mathcal{F})_y \;\cong\; \bigoplus_{x_i \in f^{-1}(y)} \mathcal{F}_{x_i}$$

Note that $\mathcal{F}_x$ is naturally an $\mathcal{O}_{X,x}$-module, and $\mathcal{O}_{Y,y}$ acts on it via the pullback map

$$f^{\#} : \mathcal{O}_{Y,y} \longrightarrow \mathcal{O}_{X,x}.$$

Thus $(f_*\mathcal{F})_y$ is an $\mathcal{O}_{Y,y}$-module through $f^{\#}$, and thus we want to show that $\mathcal{O}_X$ near x is a free module of rank n over $\mathcal{O}_Y$ near y. We identify

$$\mathcal{O}_{Y,y} \;\cong\; \mathbb{C}\{z\}, \qquad \mathcal{O}_{X,x} \;\cong\; \mathbb{C}\{w\}$$

Take any holomorphic function in w.

$$g(w) = \sum_{k=0}^{\infty} a_k w^k.$$

Grouping the terms according to their residue modulo n, we have

$$g(w) = h_0(z) + h_1(z)\, w + \cdots + h_{n-1}(z)\, w^{n-1},$$

Thus $\mathbb{C}\{w\}$ is a free $\mathbb{C}\{z\}$-module of rank n with basis $1, w, \ldots, w^{n-1}$. In other words, $\mathcal{O}_{X,x}$ is a free $\mathcal{O}_{Y,y}$-module of rank n. This completes the proof. $\square$

Example 1.294 (Riemann-Roch on $\mathbb{P}^1$)**.** *We will directly verify the Riemann-Roch theorem for vector bundles on $\mathbb{P}^1$. Since every vector bundle on $\mathbb{P}^1$ splits as a direct sum of line bundles by Grothendieck's theorem, it suffices to verify the theorem for line bundles (χ, rank, and degree are additive under direct sum).*

For each $n \in \mathbb{Z}$, we have already computed

$$\dim H^0(\mathbb{P}^1, \mathcal{O}(n)) = \begin{cases} n+1, & n \geq -1, \\ 0, & n < -1, \end{cases} \qquad \dim H^1(\mathbb{P}^1, \mathcal{O}(n)) = \begin{cases} 0, & n \geq -1, \\ -n-1, & n < -1, \end{cases}$$

so in all cases

$$\chi\Big(\mathbb{P}^1, \mathcal{O}(n)\Big) := \dim H^0(\mathbb{P}^1, \mathcal{O}(n)) - \dim H^1(\mathbb{P}^1, \mathcal{O}(n)) = n + 1.$$

is precisely the statement of Riemann-Roch for $\mathbb{P}^1$.

Proof of Riemann-Roch for vector bundles. We realize our surface R as an n-sheeted branched cover $\pi : R \to \mathbb{P}^1$ for some n. We will prove the theorem on R by pushing V down to $\mathbb{P}^1$ and using Riemann-Roch there. For $\mathbb{P}^1$, the theorem follows from Grothendieck's classification of vector bundles and the explicit calculation for line bundles.

Let $B \in \operatorname{Div}(R)$ be the branch divisor of the cover, the sum of branch points repeated according to the branching index. The degree of B is the total branching index b.

Consider the direct image bundle $\pi_* V$ on $\mathbb{P}^1$, associated to the sheaf $\pi_* \mathcal{O}(V)$. I now make the following three claims:

1. $H^i(R; V) \cong H^i(\mathbb{P}^1; \pi_* V)$ for $i = 1, 2$;
2. $\operatorname{rank} \pi_* V = n \cdot \operatorname{rank} V$;
3. $\deg \pi_* V = \deg V - \frac{1}{2} b \cdot \operatorname{rank} V$,

where b is the total branching index.

Claim (ii) follows from the previous lemma. (i) is clear for H^0 but less so for H^1; (iii) seems obscure. Granting the claims for now and writing $r = \operatorname{rank} V$, we find

$$\chi(R; V) = \chi(\mathbb{P}^1; \pi_* V) = (\deg V - \tfrac{1}{2} b\, r) + n \cdot r = \deg V + (1 - g) \cdot r,$$

using Riemann-Hurwitz, thus proving the Riemann-Roch theorem.

It remains to prove the claims. Actually, claim (i) follows from the two observations that

$$\mathcal{M}(\mathbb{P}^1; \pi_* V) = \mathcal{M}(R; V) \qquad \text{and} \qquad \mathcal{P}(\mathbb{P}^1; \pi_* V) = \mathcal{P}(R; V).$$

Both statements actually hold at the level of sheaves, $\pi_*\mathcal{M}(V) = \mathcal{M}(\pi_*V)$ and similarly for $\mathcal{P}$, but we do not need that fact.

Where π is unramified, a holomorphic section is a tuple of r-vectors of holomorphic functions on each sheet, and a meromorphic section is a tuple of r-vectors of meromorphic functions on each sheet, so the equality of sheaves $\pi_*\mathcal{M}(V) = \mathcal{M}(\pi_*V)$ and $\pi_*\mathcal{O}(V) = \mathcal{O}(\pi_*V)$ is clear.

Likewise, at a branch point, in the local form $w \mapsto z = w^n$, we again have the decomposition of every holomorphic/meromorphic section, which after trivializing V, can be written as an r-vector of holomorphic/meromorphic functions in w. Grouping terms according to their residue modulo n again gives the desired equalities of sheaves. Thus again we have $\pi_*\mathcal{M}(V) = \mathcal{M}(\pi_*V)$ and $\pi_*\mathcal{O}(V) = \mathcal{O}(\pi_*V)$. Quotienting gives the equality for principal parts.

Now claim (i) follows from the two exact sequences

$$0 \to H^0(R, V) \to \Gamma(R, \mathcal{M}(V)) \to \Gamma(R, \mathcal{P}(V)) \to H^1(R, V) \to 0$$
$$0 \to H^0(\mathbb{P}^1, \pi_*V) \to \Gamma(\mathbb{P}^1, \mathcal{M}(\pi_*V)) \to \Gamma(\mathbb{P}^1, \mathcal{P}(\pi_*V)) \to H^1(\mathbb{P}^1, \pi_*V) \to 0.$$

since we have identified the middle two terms in each sequence. The more sophisticated argument is to argue that π is affine and so $R^i\pi_* = 0$ for $i > 0$, so that the Leray spectral sequence degenerates and gives the desired isomorphisms of cohomology groups.

The curious claim is (iii), which we now prove. For starters, observe that the fibre of π_*V at some $z \in \mathbb{P}^1$ where R is not branched is the sum of the fibres of V at the points in $\pi^{-1}(z)$.

Recall that an elementary transformation of a vector bundle V at some point $x \in R$ is

$$0 \to V' \to V \to S \to 0$$

where S is a skyscraper sheaf supported at x with fibre isomorphic to a quotient V_x/W for some subspace $W \subset V_x$. In particular the transformed bundle V' has the same rank as V but degree $c = \operatorname{codim} W$ less than that of V. The important fact to know that any vector bundle on a Riemann surface can be transformed to the trivial bundle $\mathbb{C}^r$ by a sequence of elementary transformations.

An elementary transformation on V at some $y \in \pi^{-1}(z)$ effects an elementary transformation on π_*V at z, with a subspace of the same codimension (the original S, plus the sum of the other fibres). So the degrees of the two bundles change by the same amount. Transforming all the way to the trivial bundle $\mathcal{O}^r$ gives the relation

$$\deg \pi_*V - \deg V = \deg \pi_*\mathcal{O}^r - \deg \mathcal{O}^r = r \cdot \deg \pi_*\mathcal{O}.$$

So we need to show that $\deg \pi_*\mathcal{O} = -b/2$. We now consider the dual vector bundle $(\pi_*\mathcal{O})^\vee$, which has degree opposite that of $\pi_*\mathcal{O}$; the desired equality is equivalent to

$$\deg\Big((\pi_*\mathcal{O})^\vee\Big) = \deg \pi_*\mathcal{O} + b.$$

This equality, and with it the theorem, will follow from taking degrees in the isomorphism of Lemma 1.295 below. In particular the RHS becomes

$$\deg \pi_*\Big(\mathcal{O}(B)\Big) = \deg \mathcal{O}(B) + \operatorname{rank}(\pi_*\mathcal{O}(B)) \cdot \deg \pi_*\mathcal{O} = b + 1 \cdot \deg \pi_*\mathcal{O}$$

as $\square$

Lemma 1.295. *With notation as in the proof above,*

$$(\pi_*\mathcal{O})^\vee \cong \pi_*\Big(\mathcal{O}(B)\Big).$$

More generally,

$$(\pi_*(V))^\vee \cong \pi_*\Big(V^\vee \otimes \mathcal{O}(B)\Big)$$

Proof. We first prove

$$(\pi_*\mathcal{O}_R)^\vee \cong \pi_*\mathcal{O}_R(B).$$

Let $\mathcal{M}_R$ and $\mathcal{M}_{\mathbb{P}^1}$ denote the sheaves of meromorphic functions. Since π is finite, there is a trace morphism

$$\mathrm{Tr}_\pi : \pi_*\mathcal{M}_R \to \mathcal{M}_{\mathbb{P}^1}.$$

We claim that

$$\mathrm{Tr}_\pi\Big(\pi_*\mathcal{O}_R(B)\Big) \subset \mathcal{O}_{\mathbb{P}^1}.$$

This is a local computation. Near a ramification point $x \in R$ of index n, with $z = \pi(x)$, choose coordinates w at x and t at z such that

$$t = w^n.$$

Then locally

$$B = (n-1)x, \qquad \mathcal{O}_R(B)_x = w^{-(n-1)}\mathbb{C}\{w\}.$$

Thus a local section of $\mathcal{O}_R(B)$ has an expansion

$$\psi(w) = \sum_{m \geq -(n-1)} a_m w^m.$$

Its trace is

$$\mathrm{Tr}_\pi(\psi) = \sum_{j=0}^{n-1} \psi(\zeta^j w) = \sum_{m \geq -(n-1)} a_m \Big(\sum_{j=0}^{n-1} \zeta^{jm}\Big) w^m,$$

where $\zeta = e^{2\pi i/n}$. Since

$$\sum_{j=0}^{n-1} \zeta^{jm} = \begin{cases} n, & n \mid m, \\ 0, & n \nmid m, \end{cases}$$

only the terms with $n \mid m$ survive. But the only multiple of n among the integers $m \geq -(n-1)$ and $m < 0$ is none, so no negative power survives. Hence $\mathrm{Tr}_\pi(\psi)$ is holomorphic in t.

Therefore we obtain an $\mathcal{O}_{\mathbb{P}^1}$-bilinear pairing

$$\pi_*\mathcal{O}_R \times \pi_*\mathcal{O}_R(B) \to \mathcal{O}_{\mathbb{P}^1}, \qquad (\varphi, \psi) \mapsto \mathrm{Tr}_\pi(\varphi\psi),$$

and hence a morphism

$$\pi_*\mathcal{O}_R(B) \to (\pi_*\mathcal{O}_R)^\vee.$$

To show that this is an isomorphism, it suffices to check stalkwise. At an unramified point, the map is obviously perfect. At a ramification point with local form $t = w^n$, the $\mathcal{O}_{\mathbb{P}^1,z}$-module $\mathcal{O}_{R,x}$ has basis

$$1, w, \ldots, w^{n-1},$$

while $\mathcal{O}_R(B)_x$ has basis

$$1, w^{-1}, \ldots, w^{-(n-1)}.$$

The pairing satisfies

$$\mathrm{Tr}_\pi(w^i \cdot w^{-j}) = \begin{cases} n, & i = j, \\ 0, & i \neq j, \end{cases}$$

so the matrix of the pairing is nI_n, hence invertible. Thus the pairing is perfect, and

$$(\pi_* \mathcal{O}_R)^\vee \cong \pi_* \mathcal{O}_R(B).$$

The same argument, after local trivialization of V, gives

$$(\pi_* V)^\vee \cong \pi_*(V^\vee \otimes \mathcal{O}_R(B)).$$

Indeed, the pairing is

$$\pi_* V \times \pi_*(V^\vee \otimes \mathcal{O}_R(B)) \to \mathcal{O}_{\mathbb{P}^1}, \qquad (s, \lambda) \mapsto \mathrm{Tr}_\pi(\lambda(s)),$$

and in a local frame it is the tensor product of the scalar trace pairing above with the natural evaluation pairing $V \times V^\vee \to \mathcal{O}_R$. $\square$

1.8.4 Serre duality on a curve

Let R be a compact Riemann surface, V vector bundle on R. We have the canonical bundle $K_R = \Omega^1_R$ of holomorphic one-forms on R.

Theorem 1.296 (Serre duality). *There is an isomorphism of vector spaces, functorial in R:*

$$H^0(R, V) \cong H^1(R, K_R \otimes V^\vee)^\vee$$
$$H^1(R, V) \cong H^0(R, K_R \otimes V^\vee)^\vee$$

Functorial means that for $f : R \to S$ a nonconstant map of compact Riemann surfaces, we have the following commutative diagram:

$$\begin{tikzcd} H^0(R, V) \arrow[r, "\cong"] \arrow[d, "f^*"] & H^1(R, K_R \otimes V^\vee)^\vee \arrow[d, "f^*"] \\ H^0(S, V) \arrow[r, "\cong"] & H^1(S, K_S \otimes V^\vee)^\vee \end{tikzcd}$$

We will say a little bit more about the map on H^1.

Example 1.297. *We check the isomorphism for $\mathcal{O}(n)$ on $\mathbb{P}^1$ for $n \geq -1$. Move to $\mathcal{O}(n\infty)$. Check the residue pairing at $z = 0$. May assume the principal parts are at 0 and that meromorphic functions have only poles at 0. In this case*

$$\begin{aligned} H^0(\mathbb{P}^1, \mathcal{O}(n\infty)) &= \operatorname{span}\{1, z, z^2, \dots, z^n\} \\ H^1(\mathbb{P}^1, K_{\mathbb{P}^1} \otimes \mathcal{O}(-n\infty)) &= \{\frac{dz}{z}, \frac{dz}{z^2}, \dots, \frac{dz}{z^{n+1}}\} \end{aligned}$$

because by the principal parts argument $dz/z^{n+2} = z^{-n}dz/z^n$ is holomorphic at ∞. The residue pairing is clear.

Proof. The global residue pairing

$$H^0(R, V) \times H^1(R, K_R \otimes V^*) \to \mathbb{C}$$

comes from a local residue pairing at a point $x \in R$.

$$\begin{aligned} \mathcal{O}(V)_x \otimes \mathcal{P}(K_R \otimes V^*)_x &\to \mathbb{C} \\ (\sigma, \tau) &\mapsto \operatorname{Res}_x(\langle \sigma, \tau \rangle) \end{aligned}$$

In a local coordinate z near x, the pairing takes the form

$$z^n u \otimes \frac{v}{z^m} dz \mapsto \operatorname{Res}_x\left(\frac{uv}{z^{m-n}} dz\right) = \begin{cases} u(x)v(x) & m = n+1 \\ 0 & \text{otherwise} \end{cases}$$

Note that the holomorphic part of a differential has no residue so the map is defined on $\mathcal{P}(K_R \otimes V^*)_x$. The pairing is nondegenerate. The global residue pairing then is simply the sum of the local pairings over all points $x \in R$.

We will reduce the proof to the case $R = \mathbb{P}^1$. In this case, we have already explicitly verified the isomorphism. Using the affineness of the map $f : R \to \mathbb{P}^1$, we have $H^i(R, V) = H^i(\mathbb{P}^1, f_*V)$ for $i = 0, 1$. We want to transport the residue pairing on $\mathbb{P}^1$ along this isomorphism. Serre duality on $\mathbb{P}^1$ for the bundle f_*V gives

$$H^0(\mathbb{P}^1, f_*V) \cong H^1\Big(\mathbb{P}^1, K_{\mathbb{P}^1} \otimes (f_*V)^\vee\Big)^*.$$

So to match this with the group $H^1(R, K_R \otimes V^\vee)^*$ we will construct an isomorphism of sheaves on $\mathbb{P}^1$

$$f_*\Big(K_R \otimes V^\vee\Big) \;\cong\; K_{\mathbb{P}^1} \otimes (f_*V)^\vee. \tag{3}$$

Tensoring both sides by $K_{\mathbb{P}^1}^{-1}$, this is equivalent to

$$K_{\mathbb{P}^1}^{-1} \otimes f_*(K_R \otimes V^\vee) \;\cong\; (f_*V)^\vee. \tag{4}$$

By the projection formula,

$$K_{\mathbb{P}^1}^{-1} \otimes f_*(K_R \otimes V^\vee) \;\cong\; f_*\Big(f^*K_{\mathbb{P}^1}^{-1} \otimes K_R \otimes V^\vee\Big),$$

so (4) is equivalent to the existence of an isomorphism

$$f_*\Big(f^*K_{\mathbb{P}^1}^{-1} \otimes K_R \otimes V^\vee\Big) \;\cong\; (f_*V)^\vee. \tag{5}$$

By Lemma 1.295, we have

$$(f_*V)^\vee \;\cong\; f_*\Big(V^\vee(B)\Big),$$

where B is the branch divisor of f. Hence (5) will follow provided we can identify the bundles upstairs:

$$f^*K_{\mathbb{P}^1}^{-1} \otimes K_R \otimes V^\vee \;\cong\; V^\vee(B). \tag{6}$$

Cancelling $V^\vee$, condition (6) is equivalent to the line-bundle identity

$$K_R \otimes f^*K_{\mathbb{P}^1}^{-1} \;\cong\; \mathcal{O}_R(B), \tag{7}$$

which is precisely the Riemann–Hurwitz formula in line-bundle form.

To check the compatibility of residue pairings on R and $\mathbb{P}^1$. We can use the isomorphisms

$$f_*\mathcal{P}_x(V) \;\cong\; \mathcal{P}_{f(x)}(f_*V),$$

and the discussion of differentials at branch points. Choose local coordinates in standard form and write the residue pairings in the Laurent basis. □

Remark 1.298. *Some other consequences of Riemann Roch and Serre duality.*

1. *The degree of the canonical bundle K_R is $2g-2$ and there are g independent holomorphic differentials on R.*

2. *For all R with genus $g \geq 1$, for all $p \in R$ there exists a nontrivial holomorphic differential not vanishing at p. If not, then all holomorphic differentials zeroes at p and so $h^0(R, K_R(-p)) = g$. This implies that $h^0(R, \mathcal{O}(p)) = 2$ so there exists a nonconstant meromorphic function with a single simple pole at p. This function gives an isomorphism to $\mathbb{P}^1$ and so $g = 0$, a contradiction.*

3. *There exists a map of degree $g+1$ from R to $\mathbb{P}^1$. This is because*

$$h^0(R, \mathcal{O}((g+1)p)) = h^1(R, K_R(-(g+1)p)) + (g+1) - g + 1 \geq 2$$

4. *Weierstrass gap theorem. For a point $p \in R$, there are exactly g positive integers n such that $h^0(R, \mathcal{O}(np)) = h^0(R, \mathcal{O}((n-1)p))$. These are called the Weierstrass gaps at p. The non-gap numbers are the pole orders of meromorphic functions with a single pole at p. The gap numbers are the pole orders of holomorphic differentials vanishing at p.*

1.8.5 Elliptic curves

1.8.6 Hyperelliptic curves

Definition 1.299. *A compact Riemann surface R is hyperelliptic if there exists a degree 2 holomorphic map $R \to \mathbb{P}^1$.*

A standard family is $y^2 = f(x)$, where $f(x)$ is squarefree of degree $2g+1$ or $2g+2$. Its smooth projective model is hyperelliptic of genus g.

Exercise 1.300. *Prove that every compact Riemann surface of genus 2 is hyperelliptic*

Solution 1.301. *Being genus 2 means* $\dim H^0(R, K_R) = 2$. *Choose a basis* ω_1, ω_2. *Since* $\deg K_R = 2g - 2 = 2$, *any nonzero holomorphic differential has zero divisor of degree 2.*

We claim ω_1, ω_2 *have no common zero. If both vanished at* $p \in R$, *then*

$$H^0\Big(R, K_R(-p)\Big) \subset H^0(R, K_R)$$

would still contain both ω_1, ω_2, *so* $\dim H^0(R, K_R(-p)) \geq 2$. *But* $\deg K_R(-p) = 1$, *and a line bundle of degree 1 on a genus 2 curve cannot have two independent global sections (otherwise it would give a degree 1 map to* $\mathbb{P}^1$, *impossible for genus 2). Contradiction.*

Hence $f := \omega_1/\omega_2$ *is a well-defined meromorphic function on* R. *Its poles are zeros of* ω_2, *and its zeros are zeros of* ω_1. *Each contributes total degree 2, so* f *has degree 2. Therefore* f *defines a holomorphic map*

$$R \to \mathbb{P}^1$$

of degree 2. Thus R *is hyperelliptic.*

If C is hyperelliptic, a degree-2 map

$$\pi : C \to \mathbb{P}^1$$

comes with a unique involution

$$\iota : C \to C$$

exchanging the two points in a general fiber. This is the **hyperelliptic involution**.

It is unique. The quotient is

$$C/\langle \iota \rangle \cong \mathbb{P}^1;$$

and the fixed points of ι are exactly the ramification points of π.

By Riemann–Hurwitz, if C has genus g and

$$\pi : C \to \mathbb{P}^1$$

has degree 2, then

$$2g - 2 = 2(-2) + \deg R,$$

so

$$\deg R = 2g + 2.$$

Hence there are $2g+2$ ramification points counted with multiplicity. Over an algebraically closed field of characteristic $\neq 2$, a smooth hyperelliptic curve is branched at $2g+2$ distinct points of $\mathbb{P}^1$. For a hyperelliptic curve, the ramification points of the double cover are the Weierstrass points. There are exactly $2g + 2$ of them.

2 Representation Theory

2.1 Lie Algebras

2.1.1 Definitions and Basic Examples

2.1.2 Solvable and Nilpotent Lie Algebras

2.1.3 Semisimple Lie Algebras

2.1.4 Killing Form and Complete Reducibility

Exercise 2.1 (Past Quals)**.** *Let G be a connected Lie group whose Killing form B on $\mathfrak{g}$ is negative definite. Show that G is compact.*

The general fact to know here is that a real Lie algebra with negative definite Killing form is the Lie algebra of a compact simply connected Lie group.

Solution 2.2. *The negative of B defines an inner product on $\mathfrak{g}$ which is positive definite and* Ad*-invariant. Hence there is a map $G \to O(\mathfrak{g})$ given by the adjoint representation. This map has kernel $Z(G)$, the center of G. The image of G is a subgroup of $O(\mathfrak{g})$, which is compact. Therefore, we must see that $Z(G)$ is finite and the image is closed.*

Observe that the Lie algebra center is zero. Indeed, if $X \in Z(\mathfrak{g})$, then $\mathrm{ad}_X = 0$, so for every $Y \in \mathfrak{g}$,

$$B(X, Y) = \mathrm{tr}(\mathrm{ad}_X \, \mathrm{ad}_Y) = 0.$$

Since B is nondegenerate, this implies $X = 0$. Hence $Z(\mathfrak{g}) = 0$ and so we see that $Z(G)$ is discrete. Also we immediately see that $\mathfrak{g}$ is semisimple since Cartan's criterion says that the Killing form is nondegenerate if and only if $\mathfrak{g}$ is semisimple if and only if the center of $\mathfrak{g}$ is zero.

Next we show that the image is closed. Let

$$H := \overline{\mathrm{Ad}(G)} \subset O(\mathfrak{g}).$$

Since $O(\mathfrak{g})$ is compact, H is compact.

Now compare Lie algebras. The Lie algebra of $\mathrm{Ad}(G)$ *is*

$$\mathrm{ad}(\mathfrak{g}) \subset \mathfrak{so}(\mathfrak{g})$$

because $d(\mathrm{Ad})_e = \mathrm{ad}$. Note the funny fact that the Lie algebra of $O(n)$ is $\mathfrak{so}(n)$, because $O(n)$ has two connected components, distinguished by $\det = \pm 1$. *$SO(n)$ is the identity component of $O(n)$.*

The closure H has the same Lie algebra, so

$$\mathrm{Lie}(H) = \mathrm{ad}(\mathfrak{g}) = \mathrm{Lie}(\mathrm{Ad}(G))$$

Now $\mathrm{Ad}(G)$ *is connected, because G is connected. In a Lie group, a connected Lie subgroup is determined by its Lie algebra. Since* $\mathrm{Ad}(G)$ *and H are connected Lie subgroups of*

$O(\mathfrak{g})$ with the same Lie algebra, they must coincide so $\mathrm{Ad}(G) = H$ is closed. Hence $G/Z(G) \cong \mathrm{Ad}(G)$ is compact. We now have a covering map $G \to G/Z(G) = \mathrm{Ad}(G)$ with compact base and discrete fiber. We need to know that the fiber is in fact finite. We can prove this by reasoning about π_1.

$\pi_1(G)$ is abelian so $\pi_1(G) \cong H_1(G, \mathbb{Z})$ by the Hurewicz map. Thus it is enough to show $H_1(G, \mathbb{Z})$ is finite. On a compact connected Lie group, every de Rham cohomology class has a left-invariant representative. So it suffices to compute closed left-invariant 1-forms.

A left-invariant 1-form is the same as an element $\alpha \in \mathfrak{g}^*$. For left-invariant vector fields X, Y,

$$d\alpha(X, Y) = -\alpha([X, Y]).$$

Thus α is closed iff

$$\alpha([X, Y]) = 0 \qquad \forall X, Y \in \mathfrak{g},$$

that is, iff α vanishes on $[\mathfrak{g}, \mathfrak{g}]$. But $\mathfrak{g}$ is semisimple, so $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$. Hence the only closed left-invariant 1-form is 0. Therefore

$$H^1(G, \mathbb{R}) = 0.$$

Since G is a compact manifold, $H_1(G, \mathbb{Z})$ is a finitely generated abelian group. Also

$$H^1(G, \mathbb{R}) \cong \mathrm{Hom}(H_1(G, \mathbb{Z}), \mathbb{R}).$$

Because $H^1(G, \mathbb{R}) = 0$, there are no nonzero homomorphisms

$$H_1(G, \mathbb{Z}) \to \mathbb{R}.$$

So $H_1(G, \mathbb{Z})$ has no free $\mathbb{Z}$-part. Being finitely generated abelian, it must therefore be finite. Thus $\pi_1(G) \cong H_1(G, \mathbb{Z})$ is finite.

2.1.5 Cartan Subalgebras

2.1.6 Root Systems and Root Space Decomposition

2.1.7 Weyl Group

2.1.8 Dynkin Diagrams and Classification of Simple Lie Algebras

2.1.9 Examples: $\mathfrak{sl}_n$, $\mathfrak{so}_n$, $\mathfrak{sp}_{2n}$, $\mathfrak{u}_n$

2.2 Lie Groups and Algebraic Groups

Recall the classification of compact Lie groups.

- Let G be a compact Lie group. Let G_0 be the identity component of G. Then there is a short exact sequence of Lie groups

  $$1 \to G_0 \to G \to \pi_0(G) \to 1.$$

  The component group $\pi_0(G)$ is finite.

- Let G be a compact, connected Lie group. Then G has a universal cover $\tilde{G}$, and there is a short exact sequence
$$1 \to \pi_1(G) \to \tilde{G} \to G \to 1.$$
The fundamental group $\pi_1(G)$ is an abelian group. If G is already compact semisimple then $\pi_1(G)$ is finite. If we have torus factors then $\pi_1(G)$ has free abelian parts. Moreover, $\tilde{G}$ is the product of a simply connected compact Lie group and a torus.

- Let G be a compact, connected, simply connected Lie group. Then G is the product of a simply connected simple Lie groups.

- Let G be a compact, connected, simply connected simple Lie group. Then G is determined up to isomorphism by its Lie algebra $\mathfrak{g}$. The simple compact Lie algebras are classified by the Dynkin diagrams A_n, B_n, C_n, D_n, E_6, E_7, E_8, F_4, and G_2.

2.2.1 Lie Group–Lie Algebra Correspondence

Proposition 2.3. *If G is a Lie group, then* $\mathrm{Lie}(G) = T_eG$ *with Lie bracket induced by left-invariant vector fields. In particular, every $v \in T_eG$ extends to a unique left-invariant vector field on G, and the bracket of two vector fields is defined by the action of first order differential operators. The bracket respects G-equivariance, so it induces a Lie algebra structure on T_eG. If $f : G \to H$ is a Lie group homomorphism, then the differential at the identity*
$$(df)_e : \mathrm{Lie}(G) \to \mathrm{Lie}(H)$$
is a Lie algebra homomorphism.

Proposition 2.4. *Let G be connected, and let $f_1, f_2 : G \to H$ be Lie group homomorphisms. If $(df_1)_e = (df_2)_e$ then $f_1 = f_2$.*

Proof. For any $X \in \mathrm{Lie}(G)$, the one-parameter subgroup $t \mapsto \exp_G(tX)$ is sent by f_i to
$$f_i(\exp_G(tX)) = \exp_H\Big(t\,(df_i)_e(X)\Big).$$
Since the differentials agree, these are equal for $i = 1, 2$. Hence
$$f_1(\exp_G(tX)) = f_2(\exp_G(tX))$$
for all t, so f_1 and f_2 agree on all one-parameter subgroups. Because G is connected, it is generated by any neighborhood of the identity, and in particular by exponentials from a small neighborhood in $\mathrm{Lie}(G)$. Therefore $f_1 = f_2$ on all of G. $\square$

Theorem 2.5. *Let G be a connected and simply connected Lie group with Lie algebra $\mathfrak{g}$, and let H be a Lie group with Lie algebra $\mathfrak{h}$. For every Lie algebra homomorphism $\varphi : \mathfrak{g} \to \mathfrak{h}$ there exists a unique Lie group homomorphism $\Phi : G \to H$ such that $d\Phi_e = \varphi$.*

Theorem 2.6 (Universal covering Lie group)**.** *If G is a connected Lie group, then its universal cover*
$$p : \widetilde{G} \to G$$
admits a unique Lie group structure such that p is a Lie group homomorphism. Moreover:

1. *$\widetilde{G}$ is connected and simply connected;*
2. $\mathrm{Lie}(\widetilde{G}) \cong \mathrm{Lie}(G)$*;*
3. $\ker(p)$ *is discrete and central in $\widetilde{G}$;*
4. $\ker(p) \cong \pi_1(G)$.

Proof. The nontrivial part of this theorem is that we can put a Lie group structure on the universal cover $\tilde{G}$ of G such that the covering map $p : \tilde{G} \to G$ is a homomorphism. Once we have this, all of the other statements follow from general covering space theory and the fact that p is a local diffeomorphism. In particular remark that the kernel will be discrete because p is locally injective. $\square$

Theorem 2.7 (Closed subgroup theorem)**.** *If $H \subset G$ is a closed subgroup of a Lie group, then H is an embedded Lie subgroup and*

$$\mathrm{Lie}(H) \subset \mathrm{Lie}(G)$$

is a Lie subalgebra.

Theorem 2.8 (Lie's third theorem)**.** *Every finite-dimensional real Lie algebra is the Lie algebra of a Lie group. Equivalently, every such Lie algebra integrates to a connected simply connected Lie group.*

Example 2.9. *We give an example of a Lie algebra homomorphism that does not integrate to a Lie group homomorphism. Consider the Lie algebras $\mathfrak{su}(2)$ and $\mathfrak{so}(3)$. The map $\mathrm{SU}(2) \to \mathrm{SO}(3)$ is the universal double cover, with kernel $\{\pm I\}$, so it induces an isomorphism of Lie algebras $\mathfrak{su}(2) \cong \mathfrak{so}(3)$.*

The inverse Lie algebra isomorphism $\mathfrak{so}(3) \to \mathfrak{su}(2)$ does not integrate to a Lie group homomorphism $S^3 = \mathrm{SO}(3) \to \mathrm{SU}(2) = \mathbb{RP}^3$, because if it did, it would split the covering $\mathrm{SU}(2) \to \mathrm{SO}(3)$. But the covering is nontrivial, so no splitting exists.

2.2.2 Fundamental Group and Center

2.2.3 Simply Connected and Adjoint Forms

2.2.4 Real, Compact, and Split Forms

2.2.5 Cartan Involution and Maximal Compact Subgroups

2.2.6 Borel, Parabolic, and Levi Subgroups

2.2.7 Cartan and Bruhat Decompositions

2.3 Representation Theory

2.3.1 Universal Enveloping Algebra and PBW Theorem

2.3.2 Harish-Chandra Homomorphism

2.3.3 Weights and Highest Weight Theory

2.3.4 Verma Modules and Highest Weight Modules

2.3.5 Finite-Dimensional Representations

Exercise 2.10. *Let V be a finite-dimensional vector space over $\mathbb{C}$ equipped with a non-degenerate symmetric bilinear form B. Show that $O(V, B)$ acts irreducibly on V.*

We present an elementary solution and then another solution using representation theory of Lie algebras.

Solution 2.11. *Take a nonzero $O(V, B)$-invariant subspace $0 \neq W \subseteq V$. Pick a nonzero vector $w = (a_1, \ldots, a_n) \in W$. Some coordinate a_i is nonzero.*

Now $O(V, B)$ contains the sign-change matrix

$$s_i = \operatorname{diag}(1, \ldots, 1, -1, 1, \ldots, 1),$$

with the -1 in the i-th place. Since $s_i \in O(V, B)$ and W is $O(V, B)$-stable, $s_i w \in W$. Therefore

$$w - s_i w = 2a_i e_i \in W.$$

Since $a_i \neq 0$, this gives $e_i \in W$. Next, $O(V, B)$ contains permutation matrices, so for any j, there is $p_{ij} \in O(V, B)$ with $p_{ij} e_i = e_j$. Because W is $O(V, B)$-stable and $e_i \in W$, we get $e_j \in W$ for all j. Hence all standard basis vectors lie in W, so $W = V$.

Solution 2.12. *It is enough to show that V is $SO(V, B)$-irreducible. This is equivalent to showing that V is $\mathfrak{so}(V, B)$-irreducible because $SO(V, B)$ is connected and the Lie algebra of $SO(V, B)$ is $\mathfrak{so}(V, B)$. Choose the standard Cartan subalgebra $\mathfrak{h} \subset \mathfrak{g}$ consisting of diagonal matrices of the form*

$$\operatorname{diag}(t_1, \ldots, t_m, -t_1, \ldots, -t_m)$$

in type D_m, and

$$\operatorname{diag}(t_1, \ldots, t_m, 0, -t_1, \ldots, -t_m)$$

in type B_m. Let $\varepsilon_i \in \mathfrak{h}^*$ be given by $\varepsilon_i(h) = t_i$.

With respect to the standard basis, the defining representation V has weight decomposition

$$V = \bigoplus_{\lambda} V_\lambda,$$

where the weights are

$$\{\pm\varepsilon_1, \ldots, \pm\varepsilon_m\} \quad \text{in type } D_m,$$

and

$$\{\pm\varepsilon_1, \ldots, \pm\varepsilon_m, 0\} \quad \text{in type } B_m.$$

Moreover, each weight space is one-dimensional.

For the standard choice of positive roots, V has a highest weight vector e_1 of weight ε_1. Let $W \subseteq V$ be the $\mathfrak{g}$-subrepresentation generated by e_1. Then W is a nonzero subrepresentation of V, so it suffices to show that $W = V$. For this, note that the negative root spaces move e_1 to all the other weight vectors in V:

- *root vectors for roots $-(\varepsilon_1 - \varepsilon_j)$ send e_1 to a nonzero multiple of e_j for $j > 1$;*
- *root vectors for roots $-(\varepsilon_1 + \varepsilon_j)$ send e_1 to a nonzero multiple of e_{-j};*
- *in type B_m, a root vector for the root $-\varepsilon_1$ sends e_1 to a nonzero multiple of e_0.*

General theory says that if a representation has a unique highest weight vector up to scaling, then the representation is irreducible. We have shown that e_1 is a highest weight vector and that the subrepresentation it generates contains all the other weight vectors, so it must be all of V. Hence V is irreducible.

2.3.6 Casimir Elements and Central Characters

2.3.7 Weyl Character Formula

2.3.8 Peter–Weyl Theorem

2.3.9 Flag Variety and Borel–Weil–Bott Theorem

3 Algebraic Topology

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